Committed to Education: School Fees, Vouchers, and the Timing of Cash Transfers

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Michelle Layvant* Gonzalo Ferres†‡

October 13, 2026

Abstract

Primary-school dropout remains widespread in Sub-Saharan Africa, and school fees are a major barrier to completion. In an experiment with 2,000 parents in eastern Uganda, we study whether time-inconsistent preferences help explain why fees go unpaid. We elicit parents' willingness to pay for two restrictions on a cash transfer, delaying it until fees are due and, holding timing fixed, earmarking it as a voucher paid directly to the school. Demand for both restrictions is substantial. Sixty-nine percent of parents are willing to pay for earmarking and 62 percent for delay. We then randomly assign parents to receive the same amount as cash six weeks before fees are due, as cash when fees are due, as a voucher, or no transfer. Both cash transfers modestly increase fee payment relative to no transfer, and delivering the cash when fees are due does little beyond delivering it earlier. In contrast, the voucher more than doubles fees paid and reduces within-term dropout by more than half, without detectably worse household outcomes than under cash. Fee-day cash and the voucher also raise school attendance. The effects of earmarking on fee payment and attendance are no larger among parents who pay for it than among those who prefer cash. This divergence partly reflects misprediction. Parents substantially underestimate the effect of earmarking on their own fee payment. In a model of quasi-hyperbolic discounting, these findings are consistent with partial naïveté, in which parents anticipate their own time inconsistency but underestimate its severity.

JEL Codes: C93, D91, I21, I25, O15
Keywords: commitment devices, present focus, school fees, cash transfers, in-kind transfers, education, Uganda

*University of California, Berkeley. Email: mlayvant@berkeley.edu.
†University of California, Berkeley. Email: gonzalo_ferres@berkeley.edu.
‡Conducted in partnership with the Uganda Ministry of Education and Sports and the Jinja District Education Office. Pre-registered at the AEA RCT Registry (AEARCTR-0016626). IRB: UC Berkeley (2024-09-17772).

1 Introduction

Following the abolition of primary-school tuition in many low-income countries, enrollment rose sharply.[1] Yet in Sub-Saharan Africa, about one in three children still do not complete primary school (United Nations, 2023; World Bank, 2024). One reason is that abolishing tuition has not eliminated the costs of schooling. Schools continue to levy fees for expenses not fully covered by government funding, and these costs remain an important barrier to participation.[2] Many of these fees recur at predictable points in the school year, yet they often go unpaid, putting children at risk of being suspended until their balances are cleared.

Why do families fail to make these payments? One explanation is that parents rationally choose not to pay school fees, trading off the marginal utility of other consumption against the benefits of schooling (Björkman-Nyqvist, 2013; Dillon, 2021). But paying school fees is also an intertemporal problem. A parent may plan to set aside money for the fee but spend it before the deadline, or redirect it to other needs when the fee comes due. Such reversals arise under time-inconsistent preferences (Strotz, 1955; Laibson, 1997; O'Donoghue and Rabin, 1999), because once the money is in hand, spending it on current needs offers an immediate benefit while the benefits of paying the fee lie in the future. Parents who anticipate failing to follow through may demand commitment devices that restrict their own future saving or spending decisions. Little is known about whether these self-control problems are an important source of unpaid school fees, or whether parents anticipate them and are willing to pay to prevent them.

We study these questions in a field experiment with 2,000 parents of children in 30 government primary schools in eastern Uganda, where schools charge fees each term for children to attend.[3] Specifically, we ask three questions. First, do parents anticipate self-control problems in paying school fees, and are they willing to pay to restrict their own future use of money? Second, are these self-control problems a barrier to fee payment and children's schooling? And third, do differences in parents' demand for commitment predict differences in its effects on their children's schooling?

Between December 2025 and May 2026, we introduced three transfers of UGX 35,000, approximately 85 percent of the average term fee: cash delivered within a day of the baseline interview (*Early Cash*), the same cash delivered six weeks later when fees were due (*Fee-Day Cash*), and a voucher paid directly to the school when the child re-enrolled (*Voucher*). The transfers therefore differ only in when the money becomes available and whether it can be spent on anything other than the fee.[4] Before random assignment, each parent chose between the Voucher and varying amounts of Fee-Day Cash, and between Early Cash and varying amounts of Fee-Day Cash. These choices measure parents' willingness to pay for two

[1] See Deininger (2003), Lucas and Mbiti (2012), and World Bank and UNICEF (2009).

[2] On fees charged after tuition abolition, see World Bank and UNICEF (2009), Nishimura et al. (2008), and Sakaue (2018). On schooling costs as a barrier to participation, see Schultz (2004), Baird et al. (2011), Duflo et al. (2015), and Duflo et al. (2021).

[3] Schools charge these fees despite the abolition of tuition in government primary schools under Universal Primary Education in 1997 (Deininger, 2003).

[4] The transfers were equated on transaction costs. The Voucher required a trip to the school, and each cash transfer included a small top-up to cover mobile-money withdrawal fees.

forms of commitment, earmarking the transfer for school fees and delaying it until fees are due. A randomly selected 2 percent of parents received the option implied by their choices at a randomly drawn price, making the choices incentive compatible. We then randomized the remaining 1,960 parents to Early Cash, Fee-Day Cash, the Voucher, or no transfer, and link these data to school fee ledgers, weekly unannounced attendance checks, and an endline survey in which we re-elicit demand for the Voucher.

Our first finding is that demand for both earmarking and delay is substantial. Sixty-nine percent of parents give up some cash to have the transfer paid directly to the school rather than receive the same amount in cash on the same day, and 62 percent give up cash to receive it when fees are due rather than immediately.[5] Mean willingness to pay is 35 percent of the transfer for earmarking and 24 percent for delay.[6] The earmarking premium, about UGX 12,000, is close to a day of median household consumption in the region. These choices are difficult to reconcile with standard models of time-consistent preferences. Cash weakly dominates an in-kind transfer of the same amount, since the household can always make the same purchase itself while retaining the option to spend on anything else (Moffitt, 1989; Currie and Gahvari, 2008). Similarly, absent constraints on saving, parents should prefer receiving cash earlier rather than paying to receive it later (Frederick et al., 2002).

A model of school-fee payment with quasi-hyperbolic discounting and uncertain income can explain both choices (Laibson, 1997; O'Donoghue and Rabin, 1999).[7] A parent receives income before the fee is due and decides how much to set aside for school fees. On the fee date, they decide whether to pay. Present focus can lead the parent to depart from their plan at either point. Fee-Day Cash removes the first opportunity by making the transfer unavailable before the fee date, and the Voucher removes both by also paying the school directly. A time-consistent parent would not pay for either restriction, so positive willingness to pay implies that the parent anticipates a self-control problem. Following O'Donoghue and Rabin (2001), we distinguish the parent's actual present focus, which governs their behavior, from their perceived present focus, which governs their forecasts of that behavior and hence their willingness to pay. A parent who anticipates the problem but underestimates its severity is partially naïve, a possibility we test below by comparing parents' forecasts of their own fee payment with what they actually pay.

Parents pay more for a restriction the more they expect it to change their own fee payment. In the model, a parent's willingness to pay for a restriction is the net return to paying the fee multiplied by the increase in the probability of paying that they expect the restriction to produce.[8] We measure the expected increase in payment directly. Before learning their

[5] An unplanned change to the school calendar provides a second measure of demand for delay. After the baseline survey, the government postponed the start of the term by eight days, from February 2 to February 10. We offered a random subset of 90 parents assigned to Fee-Day Cash a binding choice between receiving the transfer on February 2, as originally scheduled, or on February 10, when fees would now be due, and paid each parent on the date they chose. Sixty-three percent chose to wait.

[6] These means account for censoring at the bounds of the elicitation grid (Section 4.1).

[7] The baseline model assumes parents cannot borrow against future income. Appendix B.5 relaxes this assumption by allowing borrowing at a high interest rate.

[8] This expression relies on linear utility. With concave utility, willingness to pay also reflects the value of the flexibility that a restriction removes (Appendix B.6).

assignment, each parent reported the probability, from 0 to 100, that they would fully clear their child's fees with no transfer and under each of the three transfers. The Voucher forecast minus the Fee-Day Cash forecast gives the expected increase from earmarking, and the Fee-Day Cash forecast minus the Early Cash forecast gives the expected increase from delay. A 10-percentage-point larger expected increase is associated with a UGX 481 higher earmarking premium and a UGX 821 higher delay premium ($p < 0.001$ for both). The relationship is similar when we control for perceived returns to schooling and the child's baseline achievement, which proxy for the value of getting the fee paid.

Parents' open-ended explanations for their choices point to self-control problems. Many describe the temptation to spend the cash on other things before the fee is paid. As one parent explains, "I wouldn't want to be tempted to use the money to buy nice food like fish instead of paying it for school." Demand is not systematically related to household disagreement over school fees, redistributive pressures from kin, concern about theft, expected inflation, or social-desirability scores. These patterns suggest that parents pay for the restrictions primarily because they expect them to change their own allocation of resources, rather than because the restrictions protect them from other household members or outside claims.

Our second finding is that, although parents pay for both restrictions, only earmarking substantially increases fee payment. The Voucher more than doubles total fees paid and raises the share of the term fee paid by 60 percentage points, from a control mean of 37 percent ($p < 0.001$). Part of this effect is mechanical, since the Voucher is paid directly to the school once the child enrolls. Early Cash and Fee-Day Cash, in contrast, raise the share paid by only 11 to 12 percentage points, and their effects are statistically indistinguishable from each other ($p = 0.86$). Fee-Day Cash raises fees paid by less than UGX 5,000, so most of the UGX 35,000 transfer goes to other uses. Delivering cash on the day fees are due therefore does little beyond delivering it six weeks earlier.

The Voucher also reduces within-term dropout by more than half, from 5.1 percent in the control group to 2.4 percent ($p < 0.05$). The cash transfers reduce dropout by 1.1 and 1.5 percentage points, respectively. Neither effect is statistically significant, although we cannot reject that either equals the Voucher's effect. Because the transfers covered only one term, these effects may dissipate over time, particularly if children who dropped out in the control group subsequently re-enroll in future terms.

Fee-Day Cash and the Voucher also raise attendance, which we measure through weekly unannounced school visits. The effects are 3 and 5 percentage points, respectively, from a control mean of 77 percent ($p < 0.05$ and $p < 0.01$). The effect of Early Cash is smaller and not statistically significant. Most of these gains occur among children who remain enrolled throughout the term, rather than through reductions in dropout. This pattern is consistent with a reduction in the sanctions that schools impose for unpaid fees, since children whose fees are at least partly paid are less likely to be sent home from school. All three transfers reduce these sanctions ($p < 0.001$ for each). Relative to the amount transferred, the Voucher's effects on dropout and attendance are at the high end of estimates from randomized evaluations of programs that reduce schooling costs in low- and middle-income countries (Section 5.3). We detect no effect of any transfer on literacy or numeracy after one term.

On the household outcomes we observe, Voucher households fare no worse than cash households on average, even though the Voucher prevents parents from using the transfer for anything else. If parents had more valuable uses for the money, the Voucher's gains in fee payment and schooling would come at the expense of other needs. This is the standard argument for cash over in-kind transfers (Moffitt, 1989; Currie and Gahvari, 2008), and it is also the cost of commitment in models of self-control, in which a restriction that prevents spending on temptations also prevents spending on needs that arise unexpectedly (Amador et al., 2006). Food insecurity and disagreement over the child's fees fall by similar amounts in every arm, and livestock holdings and durable assets do not change detectably in any arm. Early Cash raises transfers to people outside the household, and both cash transfers raise borrowing, whereas the Voucher does neither. Because our follow-up covers a single term, we cannot detect costs or benefits that arise later, including longer-run investments financed by the additional borrowing.

If the Voucher had kept money from needs that parents considered more pressing, those who received it would have reason to pay less for it after experiencing it. Instead, experience with the Voucher raises demand for it. At endline, after one term with their assigned transfer, each parent again chose between a Voucher and Fee-Day Cash for a subsequent term. Seventy-seven percent of Voucher recipients pay at least as much for the Voucher at endline as at baseline, and their mean premium rises by more than half, from about UGX 6,400 to UGX 10,000. Relative to control parents, assignment to the Voucher raises the endline premium by UGX 4,000 ($p < 0.001$). Parents who received the Voucher therefore show no sign of regretting the restriction.

We next turn to the third question, whether parents' demand for commitment identifies the children whose schooling responds most to it. Even a restriction with little average effect on fee payment, such as delay, could have large effects for the parents who demand it. If demand tracks these effects, it could help target restrictions toward the parents for whom they are most effective, a rationale for self-selection in many programs (Roy, 1951; Alatas et al., 2016; Deshpande and Li, 2019; Finkelstein and Notowidigdo, 2019). Because demand was elicited before random assignment, we can test this directly.

We find no evidence of such selection for either restriction. Parents who expect larger effects of earmarking are willing to pay more for it, but its actual effects on fee payment are larger among those willing to pay less. Among parents who chose cash over a Voucher of the same amount, Fee-Day Cash has no effect on fees paid relative to the control group, whereas parents who chose the Voucher use part of the cash for fees. The Voucher's effect on fees paid relative to Fee-Day Cash is therefore larger among parents who chose cash, at UGX 23,900 compared with UGX 15,100 among parents who chose the Voucher ($p = 0.001$ for the difference). The effect on attendance follows the same pattern, at 5.3 compared with 0.1 percentage points ($p = 0.12$, or $p = 0.05$ when demand enters linearly). Demand for delay does not predict the effect of Fee-Day Cash relative to Early Cash.

Why does demand fail to predict treatment effects? One reason is that parents underestimate how much earmarking changes their own fee payment. In the model, willingness to pay depends both on how much parents value fee payment and on how much they expect a restriction to increase it. We examine the latter using parents' elicited forecasts. On average,

parents expect to clear their fees in full with a probability of about 74 percent under both Fee-Day Cash and the Voucher.[9] Yet 50 percent of Voucher recipients clear their fees, compared with only 15 percent of Fee-Day Cash recipients. Parents therefore underestimate the effect of earmarking on full clearing by 34 percentage points ($p < 0.001$). Because the two transfers deliver the same amount on the same day, forecast errors about future income or expenses, such as those studied by Augenblick et al. (2026), are common to both transfers and cancel in this comparison. The Voucher's advantage in full clearing also remains about 30 percentage points among households reporting no negative shock. Among parents who chose cash over the Voucher, only 10 percent clear their fees when assigned Fee-Day Cash, despite forecasting the same probability of clearing as those who chose the Voucher.

Through the lens of the model, these findings are consistent with partial naïveté (O'Donoghue and Rabin, 2001). Partially naïve parents anticipate self-control problems, as their willingness to pay implies, but underestimate their severity. In contrast to Bai et al. (2021), where partially naïve patients pay for commitment contracts that do not change their behavior, here partial naïveté leads parents to undervalue a restriction that does change their behavior, and some decline it.

Finally, we consider the implications of our findings for allocating financial assistance. Parental choice between Fee-Day Cash and the Voucher preserves flexibility for those who prefer cash, but forgoes much of the Voucher's effect on schooling. Using the randomized arms, we compare three equally costly rules: giving every parent Fee-Day Cash, giving every parent the Voucher, or allowing parents to choose between them. Allowing choice captures about 60 percent of the Voucher's increase in fees paid relative to cash, but none of its increase in attendance.

This paper contributes to three literatures. First, we contribute to the literature on schooling costs and participation by showing that parents pay a premium for a transfer paid directly to the school, rather than requiring a discount. Beyond fee reductions, in-kind and conditional transfers can increase participation (Kremer et al., 2009; Baird et al., 2011; Barrera-Osorio et al., 2011; Benhassine et al., 2015; Evans and Ngatia, 2021), as can loans for school fees (Jack et al., 2023; Gertler et al., 2024). Holding the amount fixed, we separate the timing of a transfer from its earmarking, and price each restriction. Existing studies typically assign a schooling subsidy and estimate how households respond to it, leaving unobserved recipients' preferences for one form over another, and the broader literature on cash versus in-kind transfers likewise infers the value of in-kind provision from take-up or cash-outs rather than eliciting it (Moffitt, 1989; Currie and Gahvari, 2008; Cunha et al., 2019; Lieber and Lockwood, 2019). In-kind transfers are often viewed as restricting households' choices to serve policymakers' objectives (Currie and Gahvari, 2008). Our results suggest that most parents choose this restriction themselves, and that paying the school directly can help them carry out educational investments they already plan to make.

Second, we contribute to the literature on self-control and commitment, which documents

[9]The forecast question asks about clearing the term fee in full. Because the Voucher covers only part of the fee, parents may expect it to raise the amount they pay without changing whether they clear the full balance. Their willingness to pay for the Voucher may therefore also reflect the benefits of partial payment, which can be enough to avoid sanctions such as being sent home (Section 5).

take-up of commitment devices for savings, smoking, work effort, farm inputs, and borrowing (Ashraf et al., 2006; Giné et al., 2010; Duflo et al., 2011; Dupas and Robinson, 2013; Kaur et al., 2015; Schilbach, 2019; Allcott et al., 2022; Carrera et al., 2022). We study commitment in education, where the person restricting their own choices is a parent investing on behalf of a child. In Bursztyn and Coffman (2012), parents pay to retain attendance conditions that discipline the child, whereas our restriction operates on the parent. Closest in setting, Karlan and Linden (2025) find that children in Uganda save less in a school savings account paid out in vouchers for school supplies than in one paid out in cash. Because our restriction applies to a transfer of fixed size, it cannot discourage deposits in this way. Demand for commitment in our setting is widespread, with 85 percent of parents willing to pay for at least one restriction on their transfer. Our results suggest that time inconsistency may contribute to underinvestment in education, even among parents who anticipate the problem. Because parents make these investments on behalf of their children, their self-control problems may have consequences for the next generation.

Third, we contribute to work on self-selection and targeting, which asks whether the individuals who take up or pay for a program are those whose behavior responds most to it (Ashraf et al., 2010; Cohen and Dupas, 2010; Jack, 2013; Alatas et al., 2016; Walters, 2018; Deshpande and Li, 2019; Finkelstein and Notowidigdo, 2019; Shapiro, 2020; Berry et al., 2020; Campos et al., 2025). The closest designs elicit willingness to pay and then randomize allocation to ask whether demand predicts use (Ashraf et al., 2010; Berry et al., 2020). We ask whether demand predicts treatment effects, and we use elicited beliefs to study why the two diverge. Within the commitment literature, Kaur et al. (2015) find that the workers with the largest payday effects choose commitment contracts more often, whereas Carrera et al. (2022) find little association between take-up and actual or perceived time inconsistency (see also John, 2020; Bai et al., 2021). Our results show that misprediction of one's own behavior can break the link between demand and effects, limiting what voluntary take-up can accomplish as a targeting device.

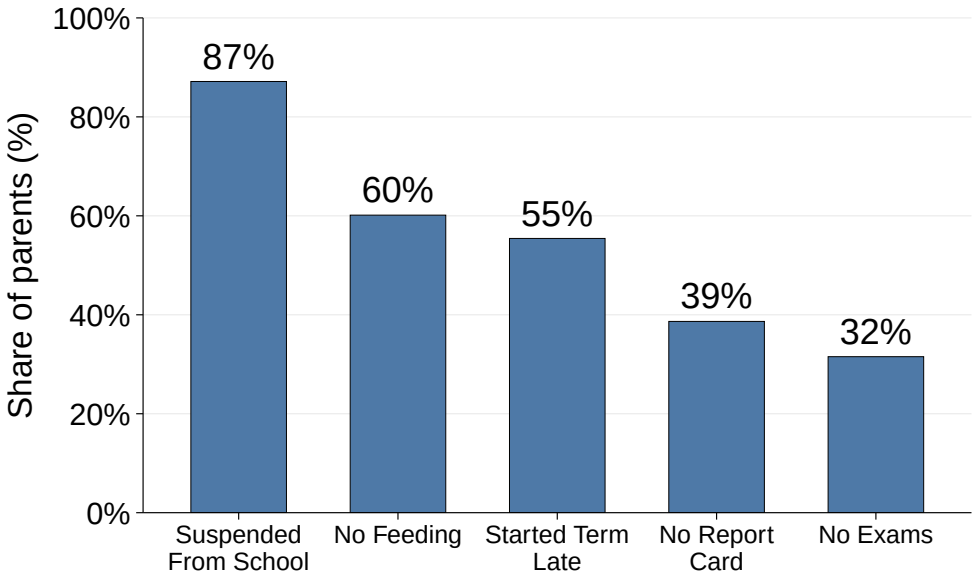
The remainder of the paper proceeds as follows. Section 2 describes the setting and experimental design. Section 3 presents the model. Section 4 examines demand for earmarking and delay, and Section 5 reports the effects of the transfers on fee payment, schooling, and the household. Section 6 asks whether demand identifies the children whose schooling responds most to the Voucher, compares anticipated with realized effects, and studies how experience changes demand. Section 7 concludes.

2 Setting and Experimental Design

2.1 School Fees in Uganda

Uganda introduced Universal Primary Education (UPE) in 1997, abolishing tuition fees in government (public) primary schools. Primary enrollment rose from about 3.1 million pupils in 1996 to 5.3 million in 1997 (UNESCO, 2026), and gender and income gaps in access narrowed (Deininger, 2003; Grogan, 2009). By 2025, Uganda had 9.1 million primary pupils, 57 percent of them in government schools (Uganda Bureau of Statistics, 2026). However,

Figure 1: Consequences of late or unpaid fees (baseline, prior 12 months)



Notes: Share of baseline parents reporting each consequence over the prior 12 months, excluding "don't know" responses; $N = 1{,}997$–$2{,}000$ per item.

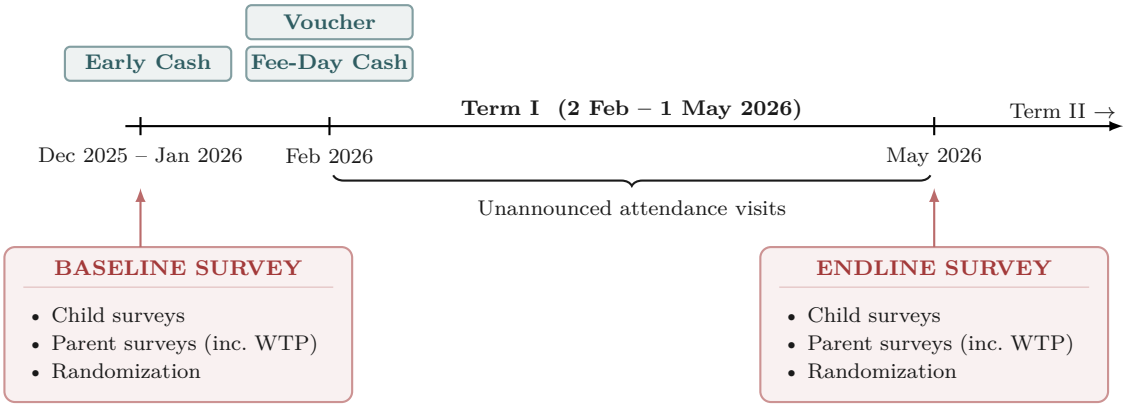
primary school completion remains low. Of the nearly 1.9 million children who entered Primary 1 in 2016, only about 832,000, or 44 percent, went on to sit for the final Primary 7 exam, known as the Primary Leaving Exam (Etukuri and Kwesiga, 2023).

One reason is that schooling remains costly despite the abolition of official tuition. The capitation grants that replaced tuition cover only part of what schools need to operate, and government schools commonly charge informal term fees for feeding, examinations, and support staff (Nishimura et al., 2008), on top of the scholastic materials and uniforms that parents must also buy. Fees are due as a lump sum at the start of each of the three school terms and are usually paid in cash, though the feeding component may instead be paid in kind, often in maize or beans valued at the prevailing market rate. In the schools in our study, the term fee excluding food averages UGX 48,000 and rises with grade (Appendix Figure A.1). In practice, many households pay late, and many do not pay at all.

Cost is also the most commonly cited reason children leave school. In the 2019/20 Uganda National Household Survey, lack of funding was the main reason given for leaving school among those aged 6 to 24, cited by 67 percent of boys and 62 percent of girls (UNICEF Uganda, 2024). Appendix Figure A.2 shows the same pattern for primary-age children.

Headteachers are responsible for collecting these fees, and unpaid fees carry direct consequences for children. In the baseline survey, 87 percent of parents report that their child had been sent home over unpaid fees in the previous 12 months, 60 percent that the child had been denied school feeding, 55 percent that the child had been made to start the term late or sit outside class, 39 percent that the child had been denied a report card, and 32 percent that the child had been barred from sitting examinations (Figure 1).

Figure 2: Study timeline



Notes: Term I was scheduled to begin on February 2, 2026. On January 16, after baseline data collection, the government postponed the start of term to February 10. Fee-Day Cash was delivered on February 2 as scheduled, except to the parents in the timing sub-experiment who chose to receive it on February 10 (Section 2.5). The Voucher was paid on February 10.

2.2 Sample

We conducted the study in Jinja District, a predominantly rural district in the Busoga sub-region of eastern Uganda.[10] The district has 59 government primary schools, of which 50 were eligible for the study, and we selected 30 at random.[11] Within each school, we drew a random sample of pupils enrolled in Primary 4 or Primary 6 in Term 3 of 2025, stratified by school, grade, and gender, oversampling girls, and selecting at most one child per household. The resulting sample included 2,000 pupils and their primary guardians. Following the pre-analysis plan, we drew pre-randomized replacement lists for households that could not be reached.[12] We surveyed both the sampled child and the child's primary guardian. We refer to the primary guardian as the parent throughout, although the guardian is not always the child's mother or father.

2.3 Transfers

Figure 2 shows the study timeline. We conducted the baseline survey between December 13, 2025 and January 12, 2026, during the two-month holiday between Term 3 of 2025 and Term 1 of 2026. The main harvest months are November–December and June–July, so the baseline survey took place at the end of a harvest. During the baseline interview, parents completed the willingness-to-pay elicitation described in Section 2.4 and were then informed of their assignment.

[10] In 2020 the urban core of the area was carved out as the separate Jinja City; the surrounding Jinja District, where this study takes place, remains predominantly rural, with about 281,000 residents across six sub-counties at the 2024 census (Uganda Bureau of Statistics, 2024).

[11] One school was excluded because it is a boarding school, and eight others were excluded for administrative reasons.

[12] Replacements were triggered when a household could not be located, the child had moved away, or the primary guardian declined to participate; each replacement was the next pre-randomized name in the same school–grade–gender cell.

The experiment centered on three transfers of equal monetary value, UGX 35,000, that differed in when the money arrived and whether it was earmarked for school fees. This amount is about US$26 at purchasing-power parity, on average 85 percent of a child's term fee excluding food, and about 8 percent of median monthly household consumption in the Busoga sub-region in 2019/20 (Uganda Bureau of Statistics, 2020). Of the 2,000 parents, 1,960 were randomly assigned to one of four arms, and the remaining 40 formed the Choice group described in Section 2.4. The *Control* group ($N = 495$) received no transfer. *Early Cash* (T_1; $N = 489$) was an unconditional mobile-money transfer to the parent within 24 hours of the baseline interview. *Fee-Day Cash* (T_2; $N = 490$) was the identical transfer delivered on February 2, 2026, the scheduled start of Term 1, when fees fell due; this was between three and seven weeks after the baseline interview, and five weeks on average. The *Voucher* (T_3; $N = 486$) paid the same amount at the start of term directly to the school the child attended in 2025, conditional on the child re-enrolling there, which we confirmed with the school before the fee was paid.

The transfers therefore provided the same resources while varying when those resources became available and whether they remained fungible. Comparing each transfer with the Control group gives the effect of receiving UGX 35,000 in that form relative to receiving no transfer. By design, the Voucher and Fee-Day Cash were both scheduled for the start of Term 1, so the difference between them isolates earmarking. Fee-Day Cash and Early Cash were both unrestricted, so the difference between them isolates timing. The Voucher was both delayed and earmarked and Early Cash was neither, so the difference between them gives the combined effect of the two restrictions, comparing the most restricted transfer with the least restricted. The start of term was later postponed by eight days (Section 2.5).

2.4 Eliciting Willingness to Pay and Beliefs

During the baseline interview, before assignment was revealed, each parent completed two incentive-compatible choice tasks, presented in random order. The *earmarking* task offered a choice between the UGX 35,000 Voucher and an amount of Fee-Day Cash, and the *timing* task a choice between UGX 35,000 in Early Cash and an amount of Fee-Day Cash. In each task, the amount of Fee-Day Cash started at UGX 35,000 and adjusted over four rounds to the parent's previous choices, in progressively finer steps of UGX 10,000, 5,000, and 2,500, bounded between UGX 17,500 and UGX 52,500. The four choices locate the amount of Fee-Day Cash at which the parent was indifferent between the two options to within UGX 2,500 (Appendix Figure C.1). After the first round of each task, parents gave a short open-ended explanation for their choice.[13]

These indifference points measure how parents value two restrictions on a transfer: earmarking it for school fees, and delaying it until fees fall due. The *earmarking premium* is the indifference point in the earmarking task minus UGX 35,000, the value of having resources earmarked and paid directly to the school rather than received as flexible cash at the same time. The *delay premium* is UGX 35,000 minus the indifference point in the timing task,

[13]Each parent then completed one of two additional exploratory timing tasks, assigned at random: Early Cash versus cash in one week, or Early Cash versus cash three weeks after the start of term. We do not use these tasks in this paper.

the value of receiving liquidity when fees are due rather than immediately. A premium is negative when the parent prefers the flexible or the earlier transfer at equal amounts.

We designed the options so that transaction costs would not influence parents' choices. Each cash transfer included a UGX 1,000 top-up to offset the mobile-money withdrawal fee. Fees are usually paid in cash at the school, so to avoid saving Voucher parents this trip, we required them to travel to the school on a separate day to enroll the child. We explained both requirements before parents made any choices. Differences in valuations and in treatment effects therefore reflect the timing of liquidity and the value of earmarking, not differences in value or hassle.

To keep the elicitation incentive-compatible, a randomly selected 2 percent of parents, the *Choice group*, received a live payout matching the option they revealed (following Allcott et al., 2022). Their payouts were set by a Becker–DeGroot–Marschak procedure: we drew one task and one price at random, and the parent received the option their choices implied at that price.[14] Before the tasks, we told parents that their answers might be used in this way and that it was therefore in their interest to answer each question honestly. We also told them that any cash they received could be spent as they chose and did not have to go toward school fees.

On a 0–100 scale, parents also reported the probability that they would fully clear their child's Term 1 fees without assistance and under each of the three transfers. We use these forecasts to measure how much each parent expected earmarking and delay to raise the probability of clearing fees. In the model in Section 3, a parent's willingness to pay for earmarking and for delay is proportional to these expected changes.

We re-elicited preferences at endline to test whether experiencing a transfer shifts demand for earmarking. After Term 1 ended, each parent completed a second incentive-compatible task that again compared a UGX 35,000 school-fee voucher with an equal-value unconditional cash transfer, now for an upcoming term: parents of pupils in their final year of primary school were asked about Term 3 of 2026, and the rest were randomly assigned to Term 3 of 2026 or Term 1 of 2027 with equal probability. The endline task mirrored the baseline design, with the same adaptive rounds and the same return-trip requirement for the Voucher. A randomly selected 5 percent of households received a future voucher or cash award, and for half of these, 2.5 percent of the sample, the award was a live payout of their choices, as at baseline. We use these valuations to study preference updating (Section 6.3).

2.5 Randomization and Delivery

We assigned treatment at the individual level within randomization strata defined by school, grade, and gender. Assignment was drawn by computer in advance and revealed only after the parent completed the preference elicitation and belief questions, so enumerators remained blind to treatment during the interview. The Choice group is excluded from the treatment-effect sample, which consists of the 1,960 parents in the four arms.

[14] The price was drawn from the full grid, UGX 17,500 to 52,500 in increments of 2,500, so it need not equal a price shown during the task; the payout was always consistent with the parent's revealed choices.

Both cash transfers were delivered by mobile money to the number and provider the parent confirmed at the close of the baseline interview. Early Cash transfers were initiated within 24 hours of the interview, and Fee-Day Cash transfers were preceded by a verification call, with in-person follow-up when a transfer failed.[15] For the Voucher, we verified re-enrollment against class registers, paid the school by mobile money together with a written list of beneficiary pupils, and collected the receipts and fee-ledger entries that schools issued in return. A voucher for a child who did not re-enroll was not disbursed: the unused voucher was forfeited, whereas unspent cash was kept.

On January 16, 2026, after all baseline data had been collected, the government postponed the start of Term 1 from February 2 to February 10 over election-related safety concerns. Fee-Day Cash remained scheduled for February 2, and the Voucher was paid on February 10, when the term began. The postponement opened a gap between the scheduled transfer date and the new start of term, which we used for a small, pre-registered timing sub-experiment. In the week before February 2, field staff offered a stratified random subset of 90 of the 490 Fee-Day Cash parents a binding choice between receiving the transfer on February 2, the original date, and receiving it on February 10, the new start of term. Each parent received the transfer on the date they chose; the remaining Fee-Day Cash parents received it on February 2 as planned. This choice provides a second measure of demand for delay (Section 4.2), and the random offer provides an instrument for the date on which Fee-Day Cash was received.

2.6 Data

We measure fee payment using administrative records from the 30 study schools. Headteachers maintain fee-tracking sheets that record each payment toward a child's term fees, in cash or in kind, and we value in-kind payments at the unit prices posted by the school. We combine these payments with each school's fee schedule, which sets the amount charged by grade, to compute the share of the term fee paid. We also collected scores on the examinations that schools administer each term, which are missing for children who were barred from sitting them over unpaid fees (Figure 1).

Our main measure of attendance comes from unannounced spot checks by enumerators. Throughout Term 1, enumerators visited each school once a week, on a randomly chosen day and time, and recorded which study children were present, which gives 12 observations per child. Schools also keep daily class registers, but we found them incomplete, so we use them only to examine spillovers onto classmates outside the study sample.

We surveyed parents and children at baseline and again at endline, between May 2 and June 7, 2026, after Term 1 ended on May 1. The surveys collected information on household demographics and economic outcomes and on the child's schooling, including, at endline, the child's enrollment status and the sanctions the child faced over unpaid fees. In both rounds, enumerators also tested each child at home using literacy and numeracy assessments

[15] When repeated attempts failed because a line was inactive or an account was blocked, field staff visited the household to update the number or, where mobile money was not feasible, disbursed in person (Appendix D).

built on the Uwezo and Beyond Basics instruments (Uwezo, 2021; Jones et al., 2014), which follow the citizen-led, household-based approach pioneered by India's ASER. The Uwezo tasks test English literacy and numeracy at the Primary 2 level, from letter recognition to comprehension of a short story and from counting to division, and the Beyond Basics tasks extend them for children who complete these. We standardize scores within grade using the control-group mean and standard deviation and average the two subjects into a learning index. Because the assessments were administered to children directly, they measure learning for every child, including those barred from school examinations or no longer enrolled.

2.7 Sample Characteristics and Balance

Table 1 reports baseline characteristics of the sample and balance across the four arms. The parent is the child's mother in 50 percent of households and the father in 17 percent, with the remainder mostly grandparents or other relatives. Most parents (77 percent) are women, with an average age of 44 years. Households have 6.7 members on average and are largely agricultural, with 64 percent reporting agriculture as their main source of income. The children are 13 years old on average; 57 percent are girls, reflecting the oversample, and 53 percent were in Primary 4 in 2025, with the rest in Primary 6.

The four arms are similar on parent, household, and child characteristics and on the baseline willingness-to-pay measures, and the joint orthogonality tests do not reject balance. One of the twenty per-variable tests, on the food-security index, rejects at five percent, which is the number of rejections chance produces at this many comparisons. There is no endline attrition: every study household completed the endline survey. Fee records and attendance checks, however, cover only the 1,776 children who did not transfer to another school during the term, and we report Lee (2009) bounds for those outcomes.

Table 1: Baseline Balance Across Treatment Arms

	Control Mean	**Early Cash (T1)**	**Fee-Day Cash (T2)**	**Voucher (T3)**	**Joint** *p*-value
Child characteristics					
Child age (years)	13.251	-0.062	-0.048	0.045	0.706
		(0.095)	(0.092)	(0.099)	
Child in Primary 4 (2025)	0.525	0.000	0.000	0.000	.
		(0.000)	(0.000)	(0.000)	
Child is female	0.578	0.000	0.000	0.000	.
		(0.000)	(0.000)	(0.000)	
Learning index (z-score)	0.000	0.024	0.092	0.081	0.356
		(0.061)	(0.060)	(0.061)	
Literacy index (z-score)	0.000	0.049	0.068	0.046	0.696
		(0.060)	(0.059)	(0.060)	
Numeracy index (z-score)	0.000	-0.006	0.094	0.098	0.167
		(0.062)	(0.062)	(0.063)	
Commute time to school (minutes)	34.8	0.1	-0.0	0.9	0.896
		(1.4)	(1.4)	(1.5)	
Parent characteristics					
Parent is female	0.792	-0.032	-0.045*	-0.012	0.344
		(0.027)	(0.027)	(0.026)	
Parent age (years)	43.670	-0.293	0.668	0.760	0.511
		(0.815)	(0.837)	(0.854)	
Parent is the child's mother or father	0.671	0.036	0.010	-0.024	0.250
		(0.029)	(0.030)	(0.030)	
Parent completed primary (P7+)	0.426	0.032	0.032	0.041	0.589
		(0.031)	(0.032)	(0.032)	
Parent is the household head	0.503	-0.005	0.004	-0.043	0.430
		(0.032)	(0.032)	(0.032)	
Household characteristics					
Adults in household	2.626	0.117	0.062	0.078	0.610
		(0.088)	(0.088)	(0.088)	
Children in household (<18)	3.984	0.107	0.075	-0.010	0.658
		(0.111)	(0.108)	(0.108)	
Single-adult household	0.188	-0.040*	-0.045*	-0.034	0.230
		(0.024)	(0.024)	(0.024)	
Wealth index (z-score)	0.000	0.075	0.080	0.085	0.469
		(0.063)	(0.063)	(0.062)	
Household income mainly agricultural	0.636	-0.007	-0.001	-0.003	0.996
		(0.029)	(0.029)	(0.029)	
Per capita HH income (past 14 days)	12754	-186	-491	1732	0.254
		(1096)	(1092)	(1178)	
Any household savings (past year)	0.771	-0.023	-0.015	0.020	0.386
		(0.027)	(0.027)	(0.026)	
Food security index (z-score)	-0.000	0.106*	-0.045	-0.043	0.035
		(0.060)	(0.062)	(0.062)	
Preference measures					
Earmarking premium (UGX)	5907	377	-606	492	0.532
		(808)	(818)	(799)	
Delay premium (UGX)	4152	87	1025	824	0.498
		(824)	(815)	(808)	
Joint orthogonality test (*p*-value)		0.836	0.808	0.779	
Observations	495	489	490	486	

Notes: The sample is the 1,960 study children, excluding the 40 in the Choice group. Column 1 reports control-group means. Columns 2–4 report each arm's difference from control, from regressions with randomization-strata fixed effects and robust standard errors in parentheses; the Joint column is the *p*-value that the three differences are jointly zero. Gender, grade, and school are balanced by construction. Monetary values are in UGX; variables labeled (z-score) are standardized; the preference rows are the earmarking and delay premia from the baseline elicitations. Household size in the text counts all members, reported here as adults and children separately. The final row reports, for each arm, a joint orthogonality test of all characteristics against control. $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

3 A Conceptual Framework for School-Fee Payment and Self-Control

In this section we present a model of school-fee payment with quasi-hyperbolic discounting (Laibson, 1997; O'Donoghue and Rabin, 1999) and uncertain income on fee day. We use the model to derive willingness to pay for each restriction, to generate predictions about the effects of each transfer, and to show how the comparison between the two identifies whether parents correctly forecast their own present focus. Formal statements of the decision rules, the derivation of the value functions, all proofs, and four extensions are collected in Appendix B.

3.1 Setup

Timing. There are four dates, $t = 0, 1, 2, 3$. Date $t = 0$ is the baseline interview, at which the parent chooses among transfers and we elicit willingness to pay. Date $t = 1$ is the period before the term begins: the parent receives income y_1 and chooses how much to save, $s \geq 0$. Date $t = 2$ is *fee day*, the first day of term: the parent receives income y_2 and either pays the fee F or does not. At $t = 3$ the parent receives the return to schooling, R, if and only if the fee was paid. In practice a parent can pay part of the fee or pay it late (Section 2.1). We treat payment as a single decision on fee day, which corresponds to clearing the fee in full, the outcome that parents forecast at baseline (Section 2.4).

Fee-day income. Income on fee day takes two values: $y_2 = y_H \geq F$ with probability $1 - q$, so income alone covers the fee, and $y_2 = y_L < F$ with probability q, a fee-day shortfall. The low-income state represents a term in which the household has too little cash on hand to pay the fee when it falls due.

Transfers. The program delivers a transfer of common cost $C = F$ through one of three channels, against a no-transfer control ($d = 0$): fungible *Early Cash* ($d = 1$, paid at $t = 1$); fungible *Fee-Day Cash* ($d = 2$, paid at $t = 2$); and a *Voucher* ($d = V$) paid directly to the school toward the fee at $t = 2$, contingent on enrollment. All three deliver the same amount F and differ only in when it arrives and in whether it can be spent on anything else.

Preferences and beliefs. The parent is a quasi-hyperbolic discounter with $\delta = 1$ and linear utility, so each self discounts any future period by $\beta \leq 1$. Writing c_t for consumption, the self active at each date maximizes

$$U_2 = c_2 + \beta c_3, \qquad U_1 = c_1 + \beta\,\mathbb{E}[c_2 + c_3], \qquad U_0 = \beta\,\mathbb{E}[c_1 + c_2 + c_3],$$

the last because the $t = 0$ self holds no consumption of their own. A parent with $\beta = 1$ is the time-consistent benchmark, for whom the design of the transfer is irrelevant.

We set $\delta = 1$ because the experiment spans about six weeks, following O'Donoghue and Rabin (1999, 2001), and we assume linear utility.[16] Appendices B.4 and B.6 relax each assumption.

[16]The longer delay between fee payment and the schooling return is common across all three transfers, and we interpret R as the value of that return as of fee day, thereby folding longer-run discounting into its

In both, a parent who believes they are time consistent pays for neither restriction.[17]

When $\beta < 1$ the parent need not be aware of it: following O'Donoghue and Rabin (2001), their true parameter is β but they believe their later selves will act with $\tilde{\beta}$, so that $\tilde{\beta} = \beta$ is *sophisticated*, $\tilde{\beta} = 1$ fully *naive*, and $\beta < \tilde{\beta} < 1$ partial naïveté. We do not impose $\tilde{\beta} \geq \beta$: a parent may instead overestimate their own present focus, and the data can tell the two cases apart (Proposition 2). The belief $\tilde{\beta}$ governs their forecast of their own saving and payment, and hence the perceived probabilities of paying the fee, $\tilde{\mu}_d(\tilde{\beta})$, and what they pay for commitment; the true β governs what their later selves do, and hence the treatment effects. The difference between $\tilde{\beta}$ and β is the parent's misperception of their own present focus.

Assumption 1. (i) Paying the fee is worthwhile when the return is not discounted, $R > F > 0$. (ii) For simplicity, parents cannot borrow: $s \geq 0$, and the fee can be paid only out of cash on hand.[18] (iii) The transfer covers the fee, $C = F$.[19]

Choices and payoffs. The parent chooses the transfer d at $t = 0$, saving $s \geq 0$ at $t = 1$, and whether to pay the fee, $a \in \{0, 1\}$, on fee day. They consume $c_1 = y_1 + C\,\mathbf{1}\{d = 1\} - s$ at $t = 1$. On fee day, paying ($a = 1$) costs F and yields the return, so under the cash arms $c_2 = \ell - aF$ and $c_3 = aR$, where ℓ is cash on hand, while under the Voucher the fee is paid directly to the school at no out-of-pocket cost. At $t = 0$ they choose the transfer that maximizes $V_0^{(d)}$, the value of U_0 under transfer d.

3.2 Saving and Payment Decisions

We solve the model backward from fee day. Holding cash ℓ at $t = 2$, the fee-day self compares paying with not paying: paying yields $(\ell - F) + \tilde{\beta}R$ and not paying yields ℓ, so the fee-day self pays if and only if paying is feasible, $\ell \geq F$, and worthwhile, $\tilde{\beta}R \geq F$. The first condition requires enough cash on hand; the second, the *payment condition*, is met when present focus is mild enough that the discounted return exceeds the fee. The $t = 0$ self weighs the same payment differently. For the $t = 0$ self the fee and the return both lie in the future, so the fee is worth paying whenever cash allows, because $R > F$. The fee-day self bears the fee today and discounts the return, and may therefore leave the fee unpaid with cash in hand. Under the Voucher neither condition applies: the fee is paid directly to the school.

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level.

[17] With $\delta < 1$ the characterization is unchanged provided the fee is worth paying from the ex ante perspective, $\delta R > F$, and parents who value neither restriction strictly prefer earlier cash, generating a negative delay premium. With concave utility a parent can pay for both restrictions. A parent with little cash on fee day may also prefer cash to a paid fee, so the restrictions carry the cost of lost flexibility studied by Amador et al. (2006) and Laibson (2015). With linear utility and $\delta = 1$ a restriction costs the parent nothing, so neither premium is negative.

[18] Duflo et al. (2011) and Casaburi and Macchiavello (2019) also rule out borrowing. Appendix B.5 lets parents cover a fee-day shortfall by borrowing at interest. The earmarking premium is unchanged, and the delay premium falls with the cost of credit and is zero when credit is free.

[19] In the experiment the transfer covered 85 percent of the term fee on average. Appendix B.7 sets $C < F$: the fee is then paid in full under the Voucher only if the parent pays the balance, and the most present-focused parents value it below cash.

Saving matters only under Early Cash, since Fee-Day Cash and the Voucher deliver F on fee day. With linear utility the period-1 self saves either nothing or $F - y_L$, the amount needed to pay the fee when fee-day income is low (Lemma B.2). Saving reduces consumption at $t = 1$ for certain and allows the fee to be paid only if income is low. A parent who meets the payment condition therefore saves $F - y_L$ if and only if

$$\tilde{\beta}\, q\,(R - F) \;\geq\; (1 - \tilde{\beta})\,(F - y_L) \quad \Longleftrightarrow \quad \tilde{\beta} \geq \bar{\beta}_S \equiv \frac{F - y_L}{(F - y_L) + q(R - F)}.$$

The cutoff $\bar{\beta}_S$ is higher when the shortfall $F - y_L$ is larger, and lower when a shortfall is more likely or the net return $R - F$ is larger. Combining the payment and saving rules gives the probability, as forecast at $t = 0$, that the parent pays the fee under each transfer:

$$\tilde{\mu}_V \equiv 1, \qquad \tilde{\mu}_2 = \mathbf{1}\{\tilde{\beta} R \geq F\}, \qquad \tilde{\mu}_1 = \mathbf{1}\{\tilde{\beta} R \geq F\}\big(1 - q\,\mathbf{1}\{\tilde{\beta} < \bar{\beta}_S\}\big). \tag{1}$$

These probabilities sort parents, by how present-focused they believe themselves to be, into three contiguous regions. When present focus is severe ($\tilde{\beta} R < F$, the *under-payment* region) the parent expects not to pay the fee under either cash transfer, even with the cash on hand, and expects the fee to be paid only under the Voucher. For intermediate present focus ($F/R \leq \tilde{\beta} < \bar{\beta}_S$, the *under-saving* region) the parent would pay but will not save, so they expect to pay the fee with certainty under Fee-Day Cash and the Voucher, and under Early Cash only when fee-day income is high. When present focus is mild ($\tilde{\beta} \geq \bar{\beta}_S$) the parent saves and pays on their own, and expects to pay the fee under every transfer. Early Cash leaves both decisions with the parent: the cash must be saved until fee day and then paid toward the fee.

3.3 Willingness to Pay for Commitment

We define willingness to pay for one transfer over another as the amount of Fee-Day Cash that leaves the $t = 0$ self indifferent between them, as in the elicitation (Section 2.4). With linear utility these amounts are (Appendix B.2)

$$\begin{aligned} \text{WTP}_{\text{voucher}} &= (R - F)\,(\tilde{\mu}_V - \tilde{\mu}_2), \\ \text{WTP}_{\text{delay}} &= (R - F)\,(\tilde{\mu}_2 - \tilde{\mu}_1). \end{aligned} \tag{2}$$

Each premium equals the net return to paying the fee, $R - F$, multiplied by the increase in the probability of paying that the parent expects from the restriction. Because the three transfers deliver the same amount, a parent who pays for a restriction pays for a change in their own later behavior, as in Carrera et al. (2022) and Allcott et al. (2022). Strictly, the premia require only that the parent expects cash to be spent before it becomes fees; their own present focus is the leading interpretation of that expectation, and Section 4.4 weighs it against claims on their cash by others. Substituting the probabilities in (1) into (2) gives Proposition 1.

Proposition 1 (Demand for commitment)**.** *For every* $\tilde{\beta} \in (0, 1]$,

$$\text{WTP}_{\text{voucher}}(\tilde{\beta}) = (R - F)\,\mathbf{1}\{\tilde{\beta}R < F\}, \qquad \text{WTP}_{\text{delay}}(\tilde{\beta}) = q\,(R - F)\,\mathbf{1}\{F/R \leq \tilde{\beta} < \bar{\beta}_S\}.$$

A parent pays for earmarking if they expect not to pay the fee out of cash on fee day, and pays for delay if they expect to pay on fee day but not to save beforehand. A parent who expects to save and pay values neither restriction. The range of $\tilde{\beta}$ *in which delay is valued is non-empty when* $y_L < F(1 - q)$.

Appendix Figure B.1 displays the partition. A parent who expects to pay the fee in any case values the Voucher exactly as they value cash, which is the usual result for an in-kind transfer of a good the recipient would have bought anyway (Moffitt, 1989; Currie and Gahvari, 2008). The Voucher is worth more than cash only to a parent who expects that they would otherwise leave the fee unpaid. Paying for either restriction requires $\tilde{\beta} < 1$: a parent restricts their own transfer only if they perceive a self-control problem to begin with (Bryan et al., 2010). In Proposition 1 no parent pays for both restrictions, which is specific to linear utility. With concave utility the same parent can pay for both (Appendix B.6), as many parents in our data do (Section 4.1).

Prediction 1 (Demand for restrictions). A parent pays for a restriction only if they believe they are present-focused, so the share of parents with a positive premium is a lower bound on the share with $\tilde{\beta} < 1$. Each premium is increasing in the parent's own perceived behavior change: $\text{WTP}_{\text{voucher}}$ in $\tilde{\mu}_V - \tilde{\mu}_2$ and $\text{WTP}_{\text{delay}}$ in $\tilde{\mu}_2 - \tilde{\mu}_1$.

Prediction 2 (Who demands commitment). Demand for delay requires a positive probability of a fee-day shortfall. On its support the delay premium increases in that probability, and the support widens with the size of the shortfall, $F - y_L$. In so far as poorer households face more likely and larger shortfalls, demand for delay is concentrated among them.

3.4 Treatment Effects and Naivete

Willingness to pay is governed by the forecast $\tilde{\beta}$; realized behavior is governed by the true β. Replacing $\tilde{\beta}$ with β in equation (1), the fee-day self pays if and only if $\beta R \geq F$, and the period-1 self saves if and only if $\beta \geq \bar{\beta}_S$ and they expect the fee-day self to pay. Under the Voucher the fee is paid regardless of β, because the fee-day self makes no payment decision. Under either cash transfer the fee is paid only if that self in fact chooses to pay, and, under Early Cash, only if the period-1 self in fact saved.

Prediction 3 (Effects). Fee payment is weakly ordered across transfers: it is highest under the Voucher, followed by Fee-Day Cash and then Early Cash. The Voucher's advantage over Fee-Day Cash is the share of parents with $\beta R < F$. Fee-Day Cash raises payment over Early Cash only among parents who pay on fee day but do not save beforehand, so the two cash transfers have similar effects if few parents are of this kind. Because schools send children home over unpaid fees (Section 2.1), we expect the same ranking for dropout.

Comparing forecasts with realized behavior identifies whether parents anticipate their own present focus (Acland and Levy, 2015; Augenblick and Rabin, 2019; Allcott et al., 2022;

Carrera et al., 2022). Let μ_d denote the realized probability that the parent pays the fee under transfer d, and for each cash transfer $d \in \{1, 2\}$ let $e_d \equiv (\tilde{\mu}_V - \tilde{\mu}_d) - (\mu_V - \mu_d)$ denote the forecast error in the gain in that probability from the Voucher over that transfer, the forecast gain minus the realized gain.

Proposition 2 (Forecast errors and naïveté)**.** *For $d \in \{1, 2\}$, $e_d = \mu_d - \tilde{\mu}_d$. A sophisticated parent ($\tilde{\beta} = \beta$) has $e_d = 0$. The error is negative only if the parent underestimates their present focus ($\beta < \tilde{\beta}$) and positive only if they overestimate it ($\tilde{\beta} < \beta$). The corresponding error for timing, $(\tilde{\mu}_2 - \tilde{\mu}_1) - (\mu_2 - \mu_1) = e_1 - e_2$, can take either sign in both cases.*

Under the Voucher the fee is paid regardless of the present-focus parameter, so an error in the forecast gain from the Voucher over cash is an error about the parent's own behavior under cash; forecast errors that do not depend on the transfer, such as errors about income or fees, difference out. The probability of paying the fee with cash is weakly increasing in the present-focus parameter. A parent who underestimates their present focus overestimates the probability that they pay the fee with cash, and therefore underestimates the gain from the Voucher. A parent who overestimates their present focus makes the opposite error. The sign of e_d reveals the direction of the misperception, and sophistication implies $e_d = 0$, which the data can reject. The timing error carries no such restriction, because the under-saving region lies in the middle of the parameter range: a parent can move into it or out of it as β moves away from $\tilde{\beta}$.

Prediction 4 (Naïveté). If parents underestimate their present focus, the realized gain in fee payment from the Voucher over cash exceeds the gain they forecast; if they overestimate it, the realized gain falls short of the forecast.

The distinction between $\tilde{\beta}$ and β also determines whether the parents who pay for a restriction are the parents it affects. The Voucher raises a parent's fee payment over Fee-Day Cash if and only if $\beta R < F$, and the parent pays for earmarking if and only if $\tilde{\beta} R < F$. For sophisticated parents the two conditions coincide, so the parents who pay for earmarking are those whose payment it changes, as under selection on gains (Roy, 1951). A parent who underestimates their present focus may pay nothing for the Voucher although it would raise their payment, and a fully naive parent pays for neither restriction whatever their β. A parent who overestimates their present focus may pay for the Voucher and yet pay the fee when given cash. If no parent overestimates their present focus, every parent who pays for earmarking has their payment raised by it, so the Voucher's effect is at least as large among parents who pay for it as among those who do not. The reverse ordering requires that some parents who pay for earmarking would have paid the fee with cash; we examine which holds in Section 6.2. In the same way, a parent can pay for delay ($\tilde{\mu}_2 > \tilde{\mu}_1$) yet find their payment unchanged ($\mu_2 = \mu_1$). We return to these points when parents re-value the transfers after experiencing them (Section 6.3).

4 Demand for Commitment

This section asks whether parents pay to restrict their own transfer and, if so, why. A parent who is time consistent would pay for neither restriction, and Section 4.1 compares parents'

choices with this benchmark. Section 4.2 asks whether the premia reflect the change in their own fee payment that parents expect from each restriction, as Prediction 1 implies, and Section 4.3 asks which parents pay the most. Section 4.4 examines motives other than self-control.

4.1 Willingness to Pay for Earmarking and Delay

In the model, a parent who believes they are time consistent, $\tilde{\beta} = 1$, pays for neither restriction (Proposition 1). Such a parent can always use cash to pay the fee and can save cash received early until fees are due, so cash is worth at least as much as the Voucher, and early cash at least as much as later cash (Moffitt, 1989; Currie and Gahvari, 2008; Frederick et al., 2002; Andreoni and Sprenger, 2012). This benchmark does not depend on the functional forms of our stylized model. It holds for any increasing utility function and discount factor, provided that cash can be saved at no cost and that paying the fee in cash costs the same as the Voucher, which our design was meant to ensure (Section 2.4; Proposition 3). A positive premium therefore indicates that the parent anticipates a self-control problem, unless holding cash until fees are due is costly for another reason or the choice reflects motives other than consumption, which we examine in Section 4.4.

A majority of parents nonetheless choose each restriction (Figure 3). Sixty-nine percent choose to have the transfer paid directly to the school rather than receive the same amount in cash on the same day, and 62 percent choose to receive the transfer when fees are due rather than the same amount several weeks earlier.[20] Many parents choose the restriction at every price offered. Forty-two percent choose the Voucher over every amount of Fee-Day Cash, and 35 percent choose Fee-Day Cash, at every amount offered, over UGX 35,000 of Early Cash. Both groups have premia of at least UGX 17,500, or half the transfer.

The premia are large relative to the transfer (Table 2). The median premium is UGX 11,250 for earmarking and UGX 6,250 for delay. Because many parents choose the restriction at every price, the premia are censored at the bounds of the elicitation grid, and the observed means of UGX 6,000 and UGX 4,569 understate the mean premia. An intercept-only two-limit Tobit model, which extrapolates beyond the bounds, gives mean premia of UGX 12,245 for earmarking and UGX 8,517 for delay, or 35 and 24 percent of the transfer. The mean earmarking premium is close to a day of median household consumption in the region. The medians and the shares choosing a restriction at every price do not depend on the Tobit model.

Most parents pay for at least one restriction, and many pay for both. Eighty-five percent of parents have a positive premium for at least one restriction, and 46 percent have a positive premium for both.[21] In the linear-utility model a parent pays for earmarking or for delay

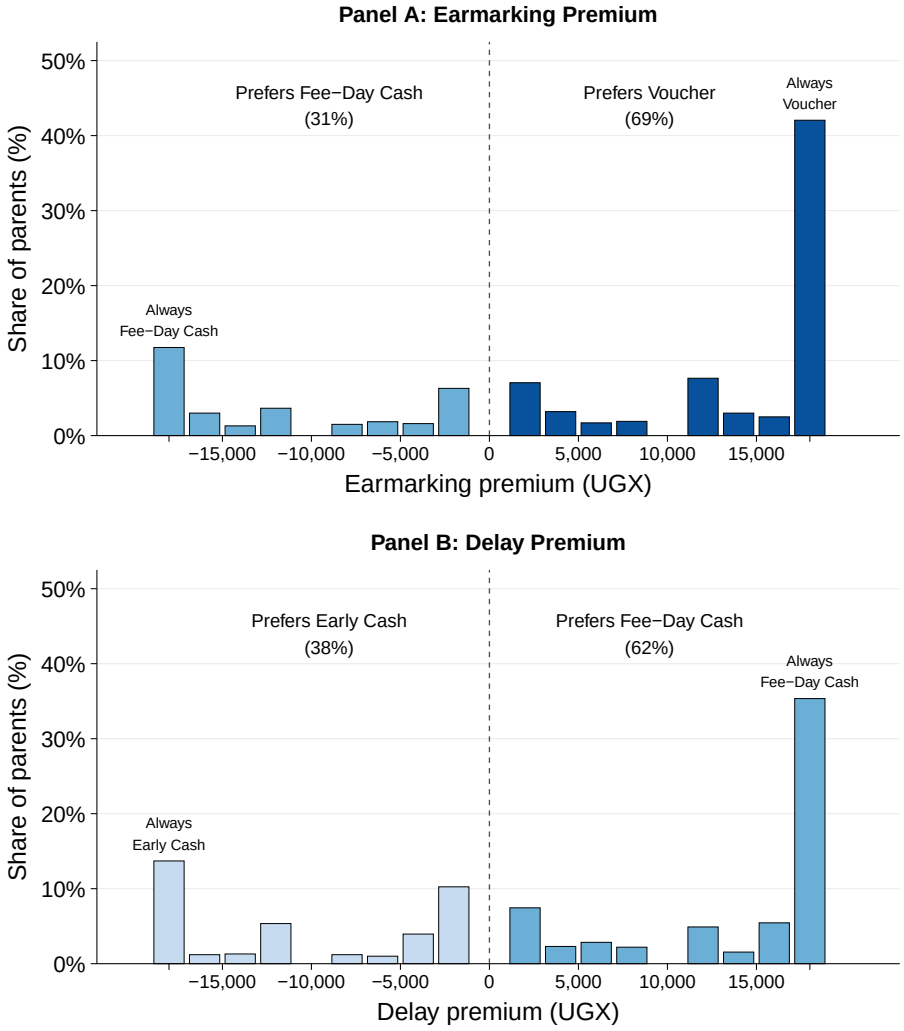
[20] Indifference points are recorded at midpoints of the UGX 2,500 grid, so no parent has a premium of exactly zero. A parent who chooses the restriction when both options offer the same amount is recorded with a premium of at least UGX 1,250, and a parent who chooses the unrestricted option with a premium of at most −UGX 1,250. Parents who are close to indifferent at equal amounts therefore appear among both the positive and the negative premia.

[21] Both tasks were incentivized. A randomly selected 2 percent of parents received the transfer implied by their choices at a randomly drawn price (Section 2.4). The tasks were presented in random order, and the

but not both. With concave utility, the same parent can pay for both (Appendix B.6).

The remaining parents choose the unrestricted option. Thirty-one percent prefer cash to the Voucher, and 38 percent prefer Early Cash to Fee-Day Cash, when both options offer the same amount. Some of these parents may be close to indifferent, and others may value the flexibility of unrestricted cash, for example to meet other needs as they arise (Amador et al., 2006).

Figure 3: Distribution of the Earmarking and Delay Premia



Notes: $N = 2{,}000$ parents. Premia are defined as in Table 2. Positive values mean that the parent gives up Fee-Day Cash to receive the Voucher (Panel A) or accepts less than UGX 35,000 on fee day rather than UGX 35,000 in Early Cash (Panel B). The outermost bars are parents who chose the same transfer at every amount offered, whose premia are censored at ±17,500.

A binding choice made outside the baseline interview provides a second measure of demand

order has no effect on either premium ($p = 0.22$ for earmarking and $p = 0.27$ for delay). The adaptive grid also prevents inconsistent choices within a task.

Table 2: Willingness to Pay for Earmarking and Delay

	Earmarking *(Voucher vs. Fee-Day Cash)*		**Delay** *(Fee-Day Cash vs. Early Cash)*	
	Indifference point	Premium	Indifference point	Premium
Mean	41,000	6,000	30,431	4,569
Mean, Tobit	47,245	12,245	26,483	8,517
Standard deviation	(13,064)	(13,064)	(13,062)	(13,062)
25th percentile	33,750	−1,250	17,500	−3,750
Median	46,250	11,250	28,750	6,250
75th percentile	52,500	17,500	38,750	17,500
Share with positive premium	69.0%		62.0%	

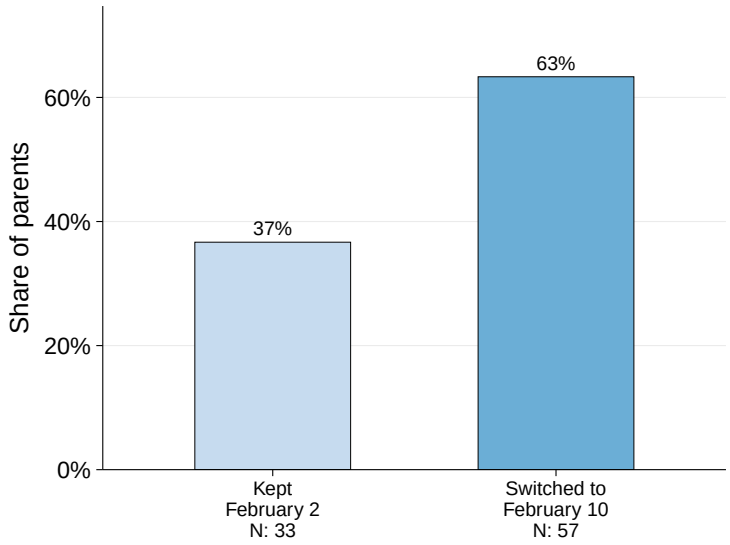
Notes: $N = 2{,}000$ parents. In each task, the indifference point is the amount of Fee-Day Cash, in UGX, at which the parent is indifferent between that amount and the UGX 35,000 Voucher (earmarking task) or UGX 35,000 in Early Cash (timing task). The earmarking premium is the indifference point minus 35,000, and the delay premium is 35,000 minus the indifference point, so positive values indicate demand for the restriction. Indifference points are recorded at midpoints of the 2,500-UGX grid from 17,500 to 52,500 and are censored at its bounds. "Mean, Tobit" is the latent mean from an intercept-only two-limit Tobit model (Appendix C). The last row is the share of parents with a strictly positive premium.

for delay. After the baseline interviews, the government postponed the start of Term 1 by eight days, from February 2 to February 10. We offered 90 randomly selected parents assigned to Fee-Day Cash a choice between receiving the transfer on February 2, as originally scheduled, or on February 10, when fees would now be due, and paid each parent on the date they chose (Section 2.5). Standard discounting predicts a weak preference for the earlier date. Instead, 63 percent chose to wait (Figure 4).

Demand of this extent is unusual in field studies of commitment. In most settings, a minority of participants take up commitment devices, and demand falls sharply once commitment has a price (Ashraf et al., 2006; Giné et al., 2010; Bryan et al., 2010; Kaur et al., 2015; Augenblick et al., 2015; Laibson, 2015; John, 2020; Bai et al., 2021). In Augenblick et al. (2015), for example, 59 percent of participants choose commitment when it is free, but only 9 percent do so when it costs $0.25.

Our estimates are closer to the few studies that find substantial costly demand. Eighty-six percent of the dairy farmers in Casaburi and Macchiavello (2019) give up a 15 percent higher price to be paid monthly rather than daily, and about a third of the rickshaw drivers in Schilbach (2019) choose sobriety incentives over a strictly higher unconditional payment. Payday borrowers in Allcott et al. (2022) value an incentive to avoid future borrowing about 30 percent above its value to a risk-neutral borrower who believes they are time consistent. In education, parents in Bursztyn and Coffman (2012) give up part of a transfer to retain an attendance condition for their children. Unlike most of these studies, which recruit participants for a particular behavior or savings product, our sample is drawn from pupils

Figure 4: Incentivized Choice Between February 2 and February 10 Cash



Notes: $N = 90$ parents. In the week before the originally scheduled transfer date, a stratified random subset of parents assigned to Fee-Day Cash chose, bindingly, between receiving the UGX 35,000 transfer on February 2, the original date, and February 10, the revised first day of term. Each parent was paid on the date they chose.

in randomly selected government schools.[22]

Demand may be especially high for school fees. Once the money is in hand, spending it on current needs offers an immediate benefit, while most of the benefits of paying the fee lie years in the future, the structure in which present focus has the largest effect on behavior (O'Donoghue and Rabin, 1999). Fees are also lumpy payments due by a fixed date each term, and a child whose fee is unpaid may be sent home (Section 5.3), so a single lapse has an immediate cost. The stakes are high as well. In the model, each premium scales with the net return to paying the fee, $R - F$, and returns to schooling in Uganda are high by international standards,[23] so even a modest expected change in fee payment can justify a large premium. One parent who chose the Voucher explained that "the child's education is a priority, and I might divert money to my farming business if given to me."

The form of the restrictions may also contribute. Commitment savings products typically require the user to deposit money they already hold, which a present-focused saver may keep

[22]The two restrictions also resemble other ways households set aside resources for planned expenses. Labeled transfers and savings devices have raised educational investment and health savings (Benhassine et al., 2015; Dupas and Robinson, 2013). Workers in Malawi voluntarily delay wage payments without interest (Brune et al., 2021), taxpayers overwithhold to defer access to income (Thaler, 1994), and offering farmers the chance to buy fertilizer right after harvest raises its use (Duflo et al., 2011).

[23]Using the Uganda National Panel Survey 2019/20, we regress the log hourly wage of wage earners aged 25 to 60 on years of schooling, potential experience and its square, and indicators for female and urban residence, using survey weights. An additional year of schooling is associated with a 16.3 percent higher hourly wage (standard error 1.6, clustered by district). Montenegro and Patrinos (2014) report a return of 15.9 percent for Uganda in 2010, compared with an average of 9.7 percent across 139 economies. These estimates are associations and need not reflect the causal effect of schooling.

postponing (Casaburi and Macchiavello, 2019). Both restrictions instead apply to a transfer that the parent has not yet received. They also restrict only the transfer, not the household's other income, so a parent who chooses them keeps other resources for unexpected needs. This limits the cost of lost flexibility, which Laibson (2015) argues keeps demand for commitment low in many settings.

4.2 Demand and Anticipated Fee Payment

Why do parents pay to restrict their own transfer? In the model, a parent pays for a restriction because they expect it to raise the probability that they pay the fee. By equation (2), the premium equals this expected increase, the *forecast gain*, multiplied by the net return to paying, $R - F$. The model therefore suggests that the earmarking premium rises with the forecast gain from earmarking, $\tilde{\mu}_V - \tilde{\mu}_2$, and the delay premium with the forecast gain from delay, $\tilde{\mu}_2 - \tilde{\mu}_1$ (Prediction 1).

We measure these forecasts directly. Before treatment assignment, each parent reported the probability that they would fully clear their child's Term 1 fees with no transfer and under each of the three transfers, the counterparts of $\tilde{\mu}_1$, $\tilde{\mu}_2$, and $\tilde{\mu}_V$ (Section 2.4).[24] On average, parents expect to fully clear their fees with a probability of 43 percent with no transfer, 60 percent with Early Cash, and 73 percent under both Fee-Day Cash and the Voucher. Parents therefore expect delay to raise their chance of clearing the fee, but on average they expect earmarking to leave it unchanged.

Parents pay more for a restriction the more they expect it to change their own fee payment. Figure 5 plots each premium against the parent's own forecast gain. A 10-percentage-point larger forecast gain from earmarking is associated with a UGX 481 higher earmarking premium, and a 10-percentage-point larger forecast gain from delay with a UGX 821 higher delay premium ($p < 0.001$ for both). Gym members in Carrera et al. (2022) similarly pay more for attendance incentives the more they expect to attend, and payday borrowers in Allcott et al. (2022) pay more for an incentive to avoid borrowing when they expect it to reduce their own borrowing.

The relationship could also arise if parents who forecast larger gains have more at stake, that is, a larger $R - F$. The estimates are similar, however, when we control for the child's grade and achievement, household wealth, parents' expected returns to schooling, and the fee, and when we compare parents within randomization strata, at UGX 425 for earmarking and UGX 796 for delay ($p < 0.001$ for both; Appendix Table A.2). Proxies for the value of schooling also do not predict either forecast gain, so the forecasts capture the change in behavior that parents expect rather than how much they value schooling.

The forecast gain from delay also predicts the earmarking premium. Controlling for both forecast gains, a 10-percentage-point larger forecast gain from delay is associated with a UGX 424 higher earmarking premium ($p < 0.001$), whereas the forecast gain from earmarking does not predict the delay premium ($p = 0.87$; Appendix Table A.2, column 4). Because

[24] Enumerators elicited each probability on a ruler from 0 to 100, explaining that 0 means impossible and 100 means certain. Parents first practiced with the chance that it would rain the next day, and enumerators re-explained the scale until the parent could use it.

both forecast gains contain the forecast under Fee-Day Cash, noise in that forecast could link them mechanically, but such noise would also tend to make the forecast gain from earmarking predict the delay premium. The positive relationship is instead consistent with a component of demand common to both restrictions, such as a parent's general awareness of their own present focus. The forecast gain from delay may capture this component better than the forecast gain from earmarking, which is harder to interpret because the Voucher covers only part of the fee.

The forecasts predict differences in the earmarking premium across parents but not its level. On average, parents expect the Voucher to leave their chance of clearing the fee unchanged, yet 69 percent choose it over the same amount of cash. One reason is that the Voucher covers about 85 percent of the fee on average, while the forecasts concern clearing the fee in full. Partial payment can be enough to avoid sanctions such as being sent home (Section 5.3), so a parent may pay for the Voucher to have most of the fee paid even if they do not expect it to change whether they clear the full balance. Section 6 shows that parents in fact substantially underestimate the effect of earmarking on their own full clearing.

More than a third of parents report a negative forecast gain from earmarking, expecting to be less likely to clear their fees under the Voucher than under Fee-Day Cash (Figure 5, Panel A). In the model, a negative forecast gain from earmarking cannot occur, since the Voucher leaves only the balance for the parent to pay. One explanation is noise in reported probabilities, since respondents find it difficult to express expectations about their own behavior as probabilities and round to focal values (Manski, 2004; Delavande et al., 2011; Allcott et al., 2022). In our data, 77 percent of forecasts are multiples of 10, and 32 percent are exactly 0, 50, or 100. Noise of this kind produces negative forecast gains even for parents who expect the Voucher to help and, if it is classical, attenuates the relationship between forecasts and premia.

4.3 Demand and Self-Control

Parents' own explanations describe the self-control problem in the model, in which spending decisions made once the money is in hand depart from the plan to pay the fee. Asked why they preferred the Voucher, many describe the risk that cash will go to other uses before the fee is paid. One parent explains that "if it is sent on the phone, it might get other expenses, and I fail to pay fees," and another that "I have a lot of problems, so I might use it for other business, and at the end fail to clear school fees when the term resumes."[25] Among parents who chose the Voucher when both options offered the same amount, 54 percent say the cash would otherwise have gone to immediate household needs such as food, medical care, and essentials, fewer than 8 percent say they would have saved it, and 0.6 percent name alcohol.

The competing uses are rarely temptation goods. In some models of self-control, only particular goods such as alcohol tempt (Banerjee and Mullainathan, 2010). Parents' answers instead point to a preference for current consumption in general, as in our model. Food, medical care, and other immediate needs displace a planned investment in their child's schooling.

[25] Open-ended explanations were recorded by enumerators in English, sometimes in the third person; quotations are lightly edited for spelling and punctuation.

Such spending need not be frivolous to reflect a self-control problem. A time-consistent parent who expected urgent needs to arise would want to keep the option of meeting them, not pay to give it up (Amador et al., 2006). Paying to remove that option implies that parents expect their fee-day selves to weigh immediate needs more heavily than they do when choosing the transfer.

Parents' explanations for the February choice (Section 4.1) reflect the same trade-off between commitment and flexibility. Parents who waited referred to the risk of spending the money before fees were due. One explained, "I don't want to stay with money because money is so tempting," and another that "when I get money before the term starts I may misuse it for other purposes." Parents who chose the earlier date planned to buy uniforms and scholastic materials before the term began, the kind of need that gives early cash its value.

By equation (2), a parent may pay more for a restriction because they expect it to change their fee payment by more or because they have more at stake, a larger $R - F$. Saving and access to credit each predict one premium but not the other. The child's numeracy and household wealth predict both premia but neither forecast gain, which suggests differences in the stakes.

Parents who already save pay less for earmarking but not for delay. Parents who saved money in the past month pay UGX 2,584 less for earmarking ($p < 0.001$), and parents with a bank account or in a savings group, which typically requires regular contributions (Gugerty, 2007), pay UGX 1,944 less ($p < 0.001$; Appendix Table A.3). Neither group pays less for delay ($p = 0.42$ and $p = 0.44$). Demand for delay therefore does not appear to reflect a lack of places to save.

Access to credit has the reverse pattern. A one-standard-deviation increase in the reported ease of borrowing UGX 50,000 to 200,000 lowers the delay premium by UGX 726 ($p = 0.01$) but does not predict the earmarking premium ($p = 0.23$). A parent who can borrow to cover a shortfall on fee day has less need to keep the transfer until then, but borrowing does not help a parent who would rather spend the money than pay the fee (Appendix B.5).

The child's achievement appears to operate through the stakes. A child who is learning more stands to gain more from staying in school, and both premia rise with the child's numeracy. Controlling for grade and the asset index, a one-standard-deviation increase in the child's numeracy score is associated with a UGX 571 higher earmarking premium and a UGX 810 higher delay premium ($p = 0.04$ and $p = 0.005$; Table 3, Panel B), while literacy predicts neither premium. Numeracy does not predict either forecast gain ($p = 0.84$ and $p = 0.19$). This is consistent with equation (2), in which the return to paying the fee scales the premium without changing the gain in payment the parent expects.

Parents' expectations of the return to schooling, a more direct measure of the stakes, predict neither premium. Following Attanasio and Kaufmann (2014), each parent reports the child's expected earnings at age 25 if the child completes only Primary 7 and if the child completes Senior 4. Parents expect lower secondary schooling to roughly double earnings, an annualized return of 26 to 30 percent (Appendix Figure A.4). These expectations capture only part of the return that parents may attach to schooling, however, and earnings many years ahead

are difficult to forecast.[26]

Poorer and less liquid households pay more for both restrictions (Table 3, Panel A). A one-standard-deviation increase in the asset index is associated with a UGX 816 lower earmarking premium and a UGX 685 lower delay premium ($p = 0.003$ and $p = 0.01$), and a one-standard-deviation increase in the liquidity index lowers the two premia by UGX 834 and UGX 675. Parents who expect to be less likely to clear their fees with no transfer also pay more for both restrictions.

The gradient for delay is in line with Prediction 2, which links demand for delay to the risk of a shortfall on fee day. Poorer parents do not, however, forecast larger gains from either restriction ($p = 0.55$ and $p = 0.49$; Appendix Figure A.5), nor do they expect higher returns to schooling (Appendix Figure A.4). Their higher premia may instead reflect higher stakes. Having the fee paid may be worth more to poorer households, who may find an unpaid balance harder to clear later. The forecasts concern full clearing, however, and do not capture expected gains from partial payment. Allocating the restrictions by demand would direct them toward poorer households, and Section 6 asks whether these are also the households whose fee payment responds most.

Table 3: Heterogeneity in Willingness to Pay

		Earmarking Premium *(Voucher vs. Fee-Day Cash)*		**Delay Premium** *(Fee-Day Cash vs. Early Cash)*	
	Mean	Marg. effect	(s.e.)	Marg. effect	(s.e.)
Panel A: Wealth and Liquidity					
Asset index (z-score)	0.06	−816***	(272)	−685**	(277)
Liquidity index (z-score)	0.00	−834***	(283)	−675**	(288)
Forecast: clears fees with no transfer (0–100)	43.04	−26.6**	(11.5)	−23.9**	(11.7)
Panel B: Child Achievement and Expected Returns					
Learning index (z-score)	0.05	443	(281)	521*	(288)
Numeracy score (z-score)	0.05	571**	(281)	810***	(287)
Literacy score (z-score)	0.04	240	(281)	107	(288)
Expected return to schooling (z-score)	0.00	9	(283)	48	(293)

Notes: $N = 2{,}000$ parents. Each row reports the average marginal effect of the row variable on each premium, in UGX, from a separate two-limit Tobit model censored at ±17,500 (Appendix C). Continuous variables other than the forecast are standardized. The asset index is the first principal component of dwelling, water and sanitation, land, durable-goods, and livestock measures. The liquidity index averages five standardized measures of access to funds. The forecast is the parent's reported probability of fully clearing Term 1 fees with no transfer. The learning index averages the child's baseline literacy and numeracy scores, standardized within grade, and the learning rows control for grade and the asset index. The expected return to schooling follows Attanasio and Kaufmann (2014). Additional measures and extensive-margin estimates appear in Appendix Tables A.3, A.4, and A.5. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.4 Alternative Explanations

The evidence so far indicates that parents pay to protect the transfer from their own later choices. Other motives could also generate demand for one or both restrictions. Holding

[26] Beliefs about returns affect schooling decisions in other settings (Jensen, 2010; Attanasio and Kaufmann, 2014).

cash until fees are due may be costly because others can claim it or because it can be stolen, and parents may also choose a restriction because of how they label money or because it seems the more responsible answer. Each of these motives predicts which parents should pay more, and for which restriction, and Table 4 tests these predictions. These motives likely matter for some parents, but none appears to drive demand.[27]

The leading alternative is that the restrictions protect the transfer from other people. Individuals hide income from spouses and choose savings devices that limit their spouses' access (Anderson and Baland, 2002; Ashraf, 2009), and they forgo returns to keep income hidden from kin (Jakiela and Ozier, 2016). A Voucher paid to the school cannot be claimed, lent, or given away, and delay shortens the weeks during which cash sits in the household where others can ask for it, so this motive would raise both premia. It predicts higher premia in households that disagree over school fees, among parents with less control over household resources, and among parents who face more requests from relatives, and lower premia in single-adult households, where no other adult can claim the cash.

The data show none of these patterns. Parents in households that disagreed over school fees in the past year pay no more for either restriction ($p = 0.83$ for earmarking and $p = 0.28$ for delay), and neither premium varies with whether the parent is female or heads the household, with an index of pressure from relatives, or with net transfers out of the household (Table 4, Panels A and B). Parents in single-adult households demand the restrictions as much as other parents. Among the 312 such households, 72 percent pay for earmarking and 63 percent pay for delay, and the median premia equal those in the full sample. Asked how the cash would otherwise have been used, only 0.2 percent of parents name giving it to others outside the household.

Mental accounting is closely related to self-control and harder to separate from it. Households often treat money as non-fungible across mental accounts (Shefrin and Thaler, 1988; Thaler, 1990), and labeled transfers are spent disproportionately on the labeled good (Benhassine et al., 2015; Hastings and Shapiro, 2018). We described both cash transfers as fully fungible, so the Voucher differed from Fee-Day Cash in its label as well as its restriction, and our design cannot separate the value of the two. A parent can label cash themselves, however. A parent who expected a self-imposed account for school fees to hold would expect to pay the fee under either transfer and would gain nothing from the Voucher. The label is therefore worth paying for only if the parent expects to break such an account, which is the self-control problem that mental accounts address (Thaler and Shefrin, 1981).

Mental accounting without self-control also has difficulty explaining demand for delay. Early Cash and Fee-Day Cash were described identically, and in the February choice both options were the same unrestricted transfer. A parent who could set the money aside would weakly prefer to receive it on February 2, yet 63 percent chose to wait, and those who waited cited the risk of spending the money before fees were due.

[27] The restrictions also differ from the attendance conditions that parents use to monitor their children (Bursztyn and Coffman, 2012; De Walque and Valente, 2023). Saving the transfer and paying the fee are the parent's own decisions, and delaying the transfer reveals nothing about the child. The Voucher could still affect the parent's dealings with the school, which we examine through school pressure and stigma in Sections 5.3 and 5.5.

The risks of holding cash bear on the timing of the transfer rather than its use (Table 4, Panel C). Cash kept at home can be lost or stolen (Karlan et al., 2014), which would raise the delay premium, whereas a parent who expects prices to rise gains from receiving cash early, which would lower it. Concern about theft instead predicts a lower delay premium ($p = 0.05$), the opposite of what the theft motive implies, and expected inflation in the prices of maize and soap predicts neither premium.

Parents may also choose the restricted transfers because they appear the more responsible answer, which would raise both premia. Both tasks were incentivized, however, and the February choice, made outside the baseline interview, shows a similar preference for delay. Scores on the 13-item short form of the Marlowe–Crowne Social Desirability Scale, which measures the tendency to give socially desirable answers (Crowne and Marlowe, 1960; Reynolds, 1982), predict neither premium.[28]

Table 4: Alternative Explanations for Willingness to Pay

		Earmarking Premium *(Voucher vs. Fee-Day Cash)*		**Delay Premium** *(Fee-Day Cash vs. Early Cash)*	
	Mean	Marg. effect	(s.e.)	Marg. effect	(s.e.)
Panel A: Household Disagreement and Bargaining Position					
Disagreement over fees in past year (0/1)	0.25	140	(652)	707	(660)
Parent is female (0/1)	0.77	−386	(669)	76	(686)
Parent is household head (0/1)	0.49	−695	(569)	−678	(578)
Single-adult household (0/1)	0.16	663	(773)	−633	(801)
Panel B: Claims from Relatives and Others					
Kin pressure index (z-score)	0.00	270	(285)	−421	(290)
Net transfers out of household (1,000 UGX)	4.44	−1.81	(1.80)	−0.43	(1.81)
Panel C: Theft, Inflation, and Social Desirability					
Concern about theft (z-score)	0.00	−145	(284)	−563*	(288)
Expected inflation, maize	0.23	1,552	(1,524)	1,232	(1,540)
Expected inflation, soap	0.10	−1,007	(2,192)	1,556	(2,216)
Social desirability score (z-score)	0.00	249	(285)	−102	(290)

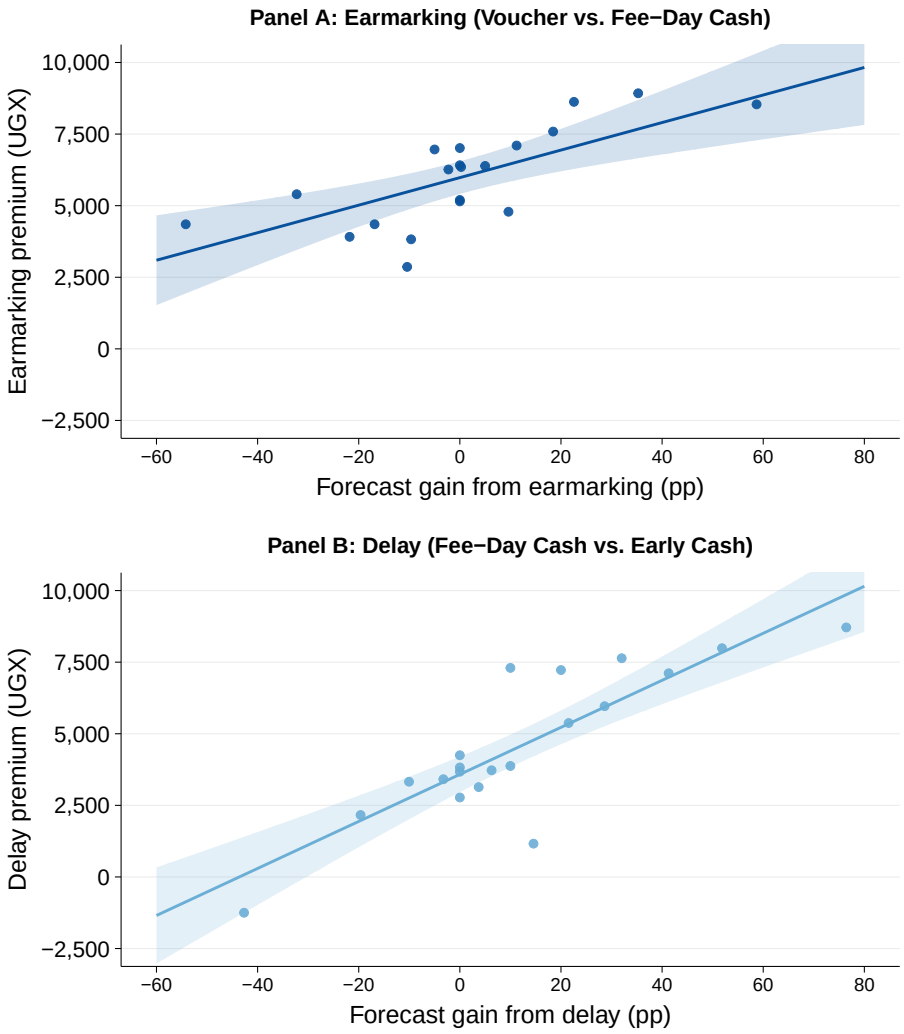
Notes: $N = 2{,}000$ parents. Average marginal effects from separate two-limit Tobit models, as in Table 3. For binary variables, the effect is the difference between one and zero. Panel A uses whether household members disagreed over school fees in the past 12 months and the parent's position in the household. Panel B uses an index of kin pressure, which combines how often the parent is asked for money, how difficult they find it to refuse, and whether they lent or gave money outside the household in the past year, and net transfers out of the household. Panel C uses concern about theft of cash kept at home, expected inflation in the prices of maize and soap, and the 13-item short form of the Marlowe–Crowne Social Desirability Scale (Reynolds, 1982). Standard errors in parentheses. $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

Taken together, the evidence points to anticipated self-control problems as the main source of demand. Parents pay for restrictions that a time-consistent parent would not pay for, and they pay more when they expect a restriction to change their own fee payment. The alternative motives, in contrast, do not predict who pays. If the premia reflect present focus, Prediction 1 implies that most of the 85 percent of parents who pay for at least one restriction perceive themselves as present-focused. Demand reflects this perceived present

[28]The scale asks whether statements such as "I'm always willing to admit it when I make a mistake" and "There have been occasions when I took advantage of someone" describe the respondent. A higher score indicates a greater tendency to present oneself favorably.

focus, $\tilde{\beta}$, rather than parents' actual present focus, β, which determines what the restrictions do. Section 5 estimates those effects, and Section 6 compares them with parents' forecasts.

Figure 5: Willingness to Pay and Forecast Gains



Notes: $N = 2{,}000$ parents. The horizontal axis is the parent's forecast gain, the difference between two baseline forecasts of the probability of fully clearing Term 1 fees (Section 2.4): the Voucher minus Fee-Day Cash in Panel A, and Fee-Day Cash minus Early Cash in Panel B. The vertical axis is the corresponding premium, defined as in Table 2. Points are means within 20 equal-sized bins of the forecast gain. The line and band are the OLS fit and its 95% confidence interval, estimated on the parent-level data. A 10-percentage-point larger forecast gain is associated with a UGX 481 (robust s.e. 122) higher earmarking premium and a UGX 821 (113) higher delay premium.

5 Effects of the Transfers

This section asks what the three transfers do to fee payment and schooling, and whether earmarking or timing accounts for the differences between them. Because the transfers have the same value, the Voucher and Fee-Day Cash differ in earmarking, and Fee-Day Cash and Early Cash differ only in timing. The Voucher is also paid only if the child re-enrolls at the baseline school and is otherwise forfeited, whereas cash is kept whatever the child does. Section 5.2 reports effects on fee payment, and Section 5.3 effects on enrollment and attendance, including the sanctions that schools impose for unpaid fees. Section 5.4 reports effects on learning, and Section 5.5 asks whether the Voucher's effects come at a cost to the rest of the household.

5.1 Empirical Specification

We estimate intention-to-treat effects with the specification

$$Y_{is} = \beta_1 T_{1,is} + \beta_2 T_{2,is} + \beta_3 T_{3,is} + X'_{is}\gamma + \delta_s + \varepsilon_{is}, \tag{3}$$

where $T_{1,is}$, $T_{2,is}$, and $T_{3,is}$ indicate assignment to Early Cash, Fee-Day Cash, and the Voucher, and δ_s are randomization-strata fixed effects. The vector X_{is} contains the baseline value of the outcome where one exists (ANCOVA) and, for enrollment outcomes, the gender of the responding parent. Standard errors are heteroskedasticity-robust. We estimate attendance at the child-week level with week fixed effects and standard errors clustered by child. The difference $\beta_3 - \beta_2$ measures the effect of earmarking, including the Voucher's re-enrollment requirement, $\beta_2 - \beta_1$ the effect of timing, and $\beta_3 - \beta_1$ the effect of both restrictions together.

The five primary outcomes are the share of the term fee paid, dropout, moving to another school, attendance, and the learning index.[29] We report Anderson (2008) sharpened q-values across these five outcomes and the three arms, and we do not adjust the estimates for other outcomes.

Fee records and attendance checks are collected at the baseline school, so fee payment is observed only for the 1,776 children who did not move to another school during the term. The share who move is 2.8 to 3.5 percentage points lower in every treatment arm (Table 5, column 5), so comparisons among the children who stayed may reflect selection as well as treatment. Appendix Table A.6 reports Lee (2009) bounds for fee payment and for attendance among these children. The bounds trim each treated arm's outcome distribution by its excess share of children who stayed, between 3.4 and 3.9 percent, and assume that no child moved schools because of treatment.

[29] Dropout and moving to another school together make up the pre-analysis plan's outcome of enrollment at the baseline school.

5.2 Fee Payment

The Voucher more than doubles the fees paid toward the term fee (Table 5). It raises total fees paid by UGX 22,800, an increase of 113 percent over the control mean of UGX 20,123 (column 1). The share of the term fee paid rises by 59.5 percentage points, from a control mean of 37 percent (column 2). Much of this effect is mechanical, because the Voucher is paid to the school when the child re-enrolls. The Voucher also raises the share of children who clear the fee, excluding the food charge, by 38 percentage points, from 12 percent in the control group (Appendix Table A.7).

Cash raises fee payment by much less, whether it arrives several weeks before fees are due or when fees are due. Early Cash raises fees paid by UGX 4,649 and Fee-Day Cash by UGX 4,762, increases of 23 and 24 percent, and each raises the share paid by 11 to 12 percentage points. The two effects are statistically indistinguishable ($p = 0.94$ for the amount and $p = 0.86$ for the share). Fee-Day Cash parents receive UGX 35,000 shortly before fees are due, yet they pay less than UGX 5,000 more toward the fee than parents who receive nothing.[30]

The smaller effects of cash are less robust to selective school transfers than the Voucher's effect. The Lee (2009) bounds on the Voucher's effect are UGX 19,204 to 22,904 for the amount paid and 54 to 62 percentage points for the share paid (Appendix Table A.6). For Early Cash and Fee-Day Cash, the lower bounds on the amount, UGX 903 and 369, and on the share, 5.6 and 4.3 percentage points, are not statistically distinguishable from zero. The Voucher and Fee-Day Cash arms retain almost identical shares of children, so the bounds on the earmarking contrast trim 0.08 percent of the Fee-Day Cash arm, less than one child, and stay close to the point estimate.

Earmarking rather than timing therefore accounts for almost the entire difference between the Voucher and Early Cash. Of the difference of UGX 18,151, earmarking accounts for UGX 18,038 ($p < 0.001$) and timing for UGX 113 ($p = 0.94$). Fees paid rise by only 13 to 14 percent of the value of the cash transfers, so most of the cash goes to other uses, even when it is delivered when fees are due.

In the model, these estimates suggest that parents more often fail to use cash for the fee when they have it than fail to keep it until fees are due. Fee-Day Cash raises payment over Early Cash only for parents who would fail to save the transfer until fee day. The Voucher raises payment over Fee-Day Cash for parents who have the cash on fee day but do not use it for the fee. The estimates are consistent with the weak ordering in Prediction 3, with the Voucher above both cash transfers and the two cash transfers statistically indistinguishable.

Parents expected timing, not earmarking, to raise their chance of clearing the fee. At baseline, they forecast a 60 percent chance of clearing the fee with Early Cash, 73 percent with Fee-Day Cash, and the same 73 percent with the Voucher (Section 4.2). In fact, Fee-Day

[30] An eight-day postponement of the term provides a second test of timing within the Fee-Day Cash arm (Section 2.5). We instrument the date on which a parent received the transfer with the randomized offer to choose it. The estimated effect on fees paid of receiving the transfer on February 10 rather than February 2 is −UGX 3,607, with a standard error of 3,786 (Appendix Table A.8). The compliers are parents who chose the later date when offered it, so the estimate applies to parents who want the delay.

Cash does not raise the share of children who clear the fee relative to Early Cash, and the Voucher raises it by a further 34.5 percentage points (Appendix Table A.7). We return to this gap between forecasts and outcomes in Section 6.

The absence of a timing effect contrasts with evidence from Bogotá (Barrera-Osorio et al., 2011). There, withholding a third of a monthly conditional transfer and paying it as a lump sum when students had to re-enroll raised re-enrollment by 4 percentage points, whereas the standard monthly transfer did not. That design turned part of a monthly stream into a lump sum available at the time of the decision. Both of our cash transfers are lump sums of the same size, so they differ only in when the money arrives. Our results suggest that, once parents receive a lump sum, receiving it several weeks before fees are due rather than when fees are due does not change what they pay toward the fee.

Earmarking has precedents in programs that tie transfers to schooling, either by labeling them or by paying fees directly. In Morocco, a cash transfer labeled as support for education but not conditioned on attendance raised the share of children in school by 7.4 percentage points when paid to fathers, with similar effects when paid to mothers (Benhassine et al., 2015). Its effect was 2 percentage points larger than that of the same transfer conditioned on attendance. The Voucher is the hard version of the label, since the earmark is enforced at the school rather than suggested.

In Malawi, the conditional transfer paid secondary-school fees directly to the school on confirmation of enrollment, while the unconditional transfer added the average fee to the cash for girls eligible for secondary school (Baird et al., 2011). The comparison between the two arms therefore combines conditionality with payment at source. The Voucher is likewise paid only if the child re-enrolls, but it does not depend on attendance and requires no monitoring of the household. Section 4 shows that parents are willing to pay for this restriction.

5.3 Enrollment and Attendance

Enrollment. The Voucher halves dropout during the term (Table 5, column 4). It reduces the share of children who drop out during the term or never enroll by 2.7 percentage points, from 5.1 to 2.4 percent ($q = 0.049$). For comparison, a labeled cash transfer in Morocco reduced dropout among children in the first four grades by 76 percent over two years (Benhassine et al., 2015). Free school uniforms in Kenya reduced girls' dropout by 3.1 percentage points over three years, from 18.8 percent (Duflo et al., 2015). Because the Voucher was paid only for children who re-enrolled at their baseline school, part of its effect may reflect that requirement.

The cash transfers reduce dropout by less, although the differences from the Voucher are not statistically significant ($p = 0.15$ and $p = 0.25$). Early Cash and Fee-Day Cash reduce dropout by 1.1 and 1.5 percentage points, and neither estimate is statistically significant. Nearly every child is enrolled in some school during the term, 98.4 percent in the control group, and no transfer detectably changes this share (column 3). The margin that the transfers can move is therefore whether children stay in school through the term.

Table 5: Effects on Fee Payment, Enrollment, and Attendance

	Fee Payments		Enrollment			Attendance	
	(1) Total Paid (1k UGX)	(2) Share Paid	(3) Enrolled (Any School)	(4) Dropped Out	(5) Transferred Schools	(6) All	(7) Did Not Transfer
T1: Early Cash	4.649***	0.116***	0.002	-0.011	-0.028	0.042**	0.024
	(1.417)	(0.028)	(0.008)	(0.013)	(0.019)	(0.019)	(0.015)
		[0.001]		[0.257]	[0.145]		[0.140]
T2: Fee-Day Cash	4.762***	0.111***	0.004	-0.015	-0.031*	0.052***	0.032**
	(1.428)	(0.028)	(0.007)	(0.013)	(0.019)	(0.018)	(0.015)
		[0.001]		[0.190]	[0.126]		[0.063]
T3: Voucher	22.800***	0.595***	0.009	-0.027**	-0.035*	0.068***	0.049***
	(1.260)	(0.026)	(0.007)	(0.012)	(0.019)	(0.018)	(0.015)
		[0.001]		[0.049]	[0.089]		[0.003]
Control Mean	20.123	0.374	0.984	0.051	0.119	0.696	0.772
Observations	1,776	1,776	1,960	1,960	1,960	22,251	20,180
T1 Effect (%)	23.1	31.0	0.2	-22.8	-23.4	6.0	3.1
T2 Effect (%)	23.7	29.6	0.4	-29.8	-26.1	7.4	4.2
T3 Effect (%)	113.3	159.1	0.9	-54.0	-29.4	9.8	6.4
p-value: T1 = T2	0.936	0.859	0.782	0.769	0.853	0.577	0.555
p-value: T2 = T3	0.000	0.000	0.384	0.248	0.819	0.348	0.209
p-value: T1 = T3	0.000	0.000	0.264	0.146	0.682	0.136	0.060

Notes: Intention-to-treat estimates from equation (3) with randomization-strata fixed effects, excluding the Choice group. Columns 1–2 use school fee records for Term 1 2026 for children who did not move to another school during the term, as reported at endline ($N = 1{,}776$), with missing records coded as zero. Columns 3–5 use the endline parent survey ($N = 1{,}960$) and control for the gender of the responding parent. Column 3 indicates enrollment in any school during the term, column 4 dropping out during the term or never enrolling, and column 5 attending a school other than the baseline school (a school transfer). Columns 6–7 use the weekly unannounced attendance checks at the baseline school, with week fixed effects. Column 6 includes all children and counts a child who moved schools as absent, and column 7 excludes children who moved. Appendix Table A.6 reports Lee bounds for columns 1, 2, and 7. T Effect (%) divides each coefficient by the control mean. Anderson (2008) sharpened q-values, in brackets, are computed across the five primary outcomes and three arms. Robust standard errors in parentheses, clustered by child in columns 6–7. $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

Moves to other schools are lower in every arm, which suggests that unpaid fees push some children to change schools. The share who move, 11.9 percent in the control group, is 2.8 to 3.5 percentage points lower, with q-values between 0.089 and 0.145 (column 5). The Voucher's re-enrollment requirement may account for part of its effect on moving, but the cash transfers, which carry no such requirement, have similar effects. In rural Uganda, informal fees also reduce poor children's enrollment in government schools (Sakaue, 2018).

Attendance. The transfers raise attendance, and Fee-Day Cash comes close to the Voucher (columns 6 and 7). Column 6 includes all children and counts a child who moved schools as absent from the baseline school, so it combines effects on attendance with effects on moving. In this sample, attendance rises by 4.2 percentage points under Early Cash, 5.2 under Fee-Day Cash, and 6.8 under the Voucher, from a control mean of 69.6 percent.

Among children who stayed at their baseline school, the Voucher and Fee-Day Cash raise attendance by 4.9 and 3.2 percentage points, and Early Cash by 2.4 points, from a control mean of 77.2 percent (column 7). The q-values for attendance refer to this sample. The Fee-Day Cash and Voucher estimates are significant at the 5 and 1 percent levels before adjustment for multiple testing ($q = 0.063$ and $q = 0.003$), and the Early Cash estimate is not significant. The Voucher's effect is equivalent to almost three additional days of school over the term. The effect of Fee-Day Cash is similar to the average effect of conditional cash transfers on primary-school attendance, 2.7 percentage points (García and Saavedra, 2017).

The Voucher's effect on attendance is robust to selective school transfers, and the effects of cash are less so. The Lee bounds are 4.0 to 7.7 percentage points for the Voucher, 2.7 to 6.5 for Fee-Day Cash, and 1.6 to 5.1 for Early Cash (Appendix Table A.6). The Voucher's lower bound is significant at the 5 percent level, the Fee-Day Cash lower bound only at the 10 percent level, and the Early Cash lower bound is not significant.

Most of the gain comes from children who remain enrolled through the term attending more often, rather than from fewer dropouts. We decompose the raw difference in attendance among children who stayed into the contributions of each enrollment status (**?**). In this descriptive decomposition, children enrolled for the full term account for 1.4 of the 2.4-point difference under Early Cash, 2.5 of 3.4 under Fee-Day Cash, and 3.1 of 4.7 under the Voucher (Appendix Figure A.6). The rest comes from fewer children dropping out or never enrolling.

We find no evidence that the transfers affected the attendance of control children in the same classrooms. Such spillovers have been found in other programs. A merit scholarship for girls in Kenya, for example, also raised teacher attendance (Kremer et al., 2009). In the daily class registers, the attendance of control children is statistically indistinguishable from that of classmates who were never in the study (Appendix Table A.9). The difference is 1.5 percentage points (standard error 1.3), on a classmate mean of 92 percent. This comparison would not detect effects common to the whole class.

Sanctions for unpaid fees. Attendance responds to cash much more than fee payment does, which is consistent with the sanctions that schools impose for unpaid fees. Fee-Day Cash raises fees paid by about a fifth as much as the Voucher, but its effect on attendance among children who stayed is two-thirds as large, and the two effects are not statistically distinguishable ($p = 0.21$). In the control group, 62 percent of children were sent home over unpaid

fees during the term, and 32 percent were sent home for five days or more (Table 6).

All three transfers reduce these sanctions. An index of the five sanctions documented at baseline (Section 2.1) falls by 0.41, 0.51, and 0.60 standard deviations under Early Cash, Fee-Day Cash, and the Voucher (column 1). Fee-Day Cash reduces the shares of children sent home and sent home for five days or more by 27.6 and 19.2 percentage points, and the Voucher by 33.0 and 20.0 points (columns 2 and 3). The Voucher also reduces the shares of children barred from exams and denied a report card by more than Early Cash does ($p = 0.053$ and $p = 0.049$), but not significantly more than Fee-Day Cash ($p = 0.21$ and $p = 0.17$).

Fee-Day Cash achieves most of the Voucher's reduction in sanctions with about a fifth of its effect on fees paid, which suggests that the returns to additional fee payment, in sanctions avoided, are nonlinear. Fee-Day Cash achieves 84 percent of the Voucher's reduction in the share of children sent home and 96 percent of its reduction in the share sent home for five days or more. Compared with Fee-Day Cash, the Voucher raises fees paid by another UGX 18,038 and the share of children who clear the fee by 34.5 percentage points (Appendix Table A.7). Yet it reduces the share sent home by only 5.4 more points ($p = 0.056$), and it does not detectably reduce the share sent home for five days or more (a difference of 0.8 points, $p = 0.71$).

The cash transfers also reduce the share sent home by more than their effects on the extensive margin of payment would suggest. They raise the share of children with any recorded payment by 8 to 9 percentage points and the share who clear the fee by 4 to 5 points, but reduce the share sent home by 19 to 28 points. This suggests that much of the reduction occurs among children whose parents pay part of the fee in either case, for whom a somewhat larger or earlier payment keeps the child in school. The model in Section 3 treats the fee as either paid or unpaid, whereas schools appear to respond to partial payment as well.

These comparisons are approximate. Payment is measured among children who stayed and sanctions among all enrolled children, and the cash effects on payment are less robust to selection than the Voucher's (Section 5.2). The comparison between Fee-Day Cash and the Voucher also attributes their differences to their effects on payment, although the two arms differ in when the money arrives and in the Voucher's re-enrollment requirement. Because the sanctions are reported by parents, the comparisons could also be affected by reporting. Attendance, which enumerators observed directly, shows the same pattern in weaker form. Fee-Day Cash achieves two-thirds of the Voucher's effect on attendance, compared with 84 and 96 percent of its effects on being sent home.

Parents report fewer children sent home under Fee-Day Cash than under Early Cash, even though the two transfers raise fees paid by similar amounts. Fee-Day Cash reduces the share of children sent home by 8.3 percentage points more than Early Cash ($p = 0.005$), and the share sent home for five days or more by 7.3 points more ($p = 0.002$). The two cash transfers do not differ significantly on the other sanctions, and attendance, which enumerators observed directly, does not differ significantly between them ($p = 0.56$).

Payment dates recorded in eleven schools show no difference between the cash arms on the first day of term and a small, imprecise difference a month later (Appendix Figure A.7). By

the first day of term, 23 to 24 percent of children in each cash arm had a recorded payment, compared with 5 percent in the control group. By March 10, 65 percent of Fee-Day Cash children had a recorded payment, compared with 58 percent of Early Cash children and 52 percent of control children ($p = 0.19$ for the difference between the cash arms). One interpretation is that Fee-Day Cash parents, who still held the cash when the term began, paid somewhat sooner in the first weeks of term, before children were sent home.

The transfers also reduce the pressure that headteachers put on parents (Appendix Table A.10). On a ten-point scale with a control mean of 5.7, pressure falls by 0.80 points under Early Cash and by 1.28 and 1.38 points under Fee-Day Cash and the Voucher. The difference between the two cash transfers is significant ($p = 0.03$).

Children in the Early Cash and Voucher arms report less fee-related stigma from classmates, but an index of stigma does not change significantly in any arm (Appendix Table A.11). In the control group, 52 percent of children report fee-related stigma from classmates and 54 percent from teachers or other adults. Early Cash and the Voucher reduce stigma from classmates by 6.2 and 5.3 percentage points (significant at the 5 and 10 percent levels), and the Voucher reduces stigma from teachers and other adults by 8.4 points.

Table 6: Effects on School Sanctions for Late or Unpaid Fees

	(1) Late-Fee Index (Std. Dev.)	(2) Sent Home	(3) Sent Home 5+ Days	(4) Late Start	(5) No Exams	(6) No Feeding	(7) No Report Card
T1: Early Cash	-0.407***	-0.193***	-0.119***	-0.074***	-0.042**	-0.113***	-0.092***
	(0.061)	(0.031)	(0.028)	(0.023)	(0.018)	(0.030)	(0.026)
T2: Fee-Day Cash	-0.506***	-0.276***	-0.192***	-0.084***	-0.052***	-0.121***	-0.106***
	(0.060)	(0.030)	(0.026)	(0.023)	(0.017)	(0.030)	(0.026)
T3: Voucher	-0.604***	-0.330***	-0.200***	-0.073***	-0.069***	-0.153***	-0.137***
	(0.059)	(0.029)	(0.026)	(0.023)	(0.016)	(0.030)	(0.025)
Control Mean	0.000	0.620	0.323	0.193	0.107	0.503	0.281
Observations	1,936	1,936	1,934	1,936	1,936	1,936	1,936
T1 Effect (%)		-31.1	-36.7	-38.4	-39.1	-22.5	-32.7
T2 Effect (%)		-44.5	-59.5	-43.3	-48.9	-24.1	-37.7
T3 Effect (%)		-53.3	-61.9	-37.6	-65.0	-30.5	-48.8
p-value: T1 = T2	0.085	0.005	0.002	0.642	0.490	0.777	0.552
p-value: T2 = T3	0.076	0.056	0.710	0.589	0.208	0.280	0.166
p-value: T1 = T3	0.000	0.000	0.000	0.942	0.053	0.171	0.049

Notes: Intention-to-treat estimates from equation (3) with strata fixed effects, excluding the Choice group. Outcomes are from the endline parent survey and refer to sanctions for late or unpaid fees during Term 1 2026. They are not defined for the 24 children who did not enroll in any school during the term. Column 3 indicates that the child was sent home for five or more school days in total and excludes two children whose reported number of days is invalid. The index in column 1 standardizes the sum of the five binary sanctions in columns 2 and 4–7 to control mean zero and unit standard deviation. T Effect (%) divides each coefficient by the control mean. Robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Comparison with other programs. Relative to the amount transferred, the Voucher's effects are at the high end of those found in randomized evaluations of other programs that reduce the cost of schooling. Following García and Saavedra (2017), who scale effects by the annual

transfer per child, we divide each effect by the program's average monthly transfer per child, spreading our one-time transfer over the term (Appendix Figure A.8 and Table A.12).

The Voucher raises attendance by about 2.0 percentage points per dollar of monthly transfer in the full sample and 1.5 among children who stayed. The comparable figures are 1.6 and 2.2 for the two programs in Mozambique, a cash transfer to parents conditioned on attendance and a voucher offered directly to girls (De Walque and Valente, 2023), and 1.1 for the conditional transfer in Malawi (Baird et al., 2011). For enrollment, which we measure as one minus dropout, the Voucher's ratio of 0.8 is similar to those of the more effective conditional cash transfer programs, and free school uniforms in Kenya have the largest ratio (Duflo et al., 2015). The confidence intervals of these ratios overlap widely.

Ratios of this kind favor small transfers when the effects of a transfer are concave in its size, and in a meta-analysis of conditional cash transfer programs, larger transfers do not have larger effects (García and Saavedra, 2017). In Cambodia, a modest conditional cash transfer raised school attendance by about 25 percentage points, and a somewhat larger transfer raised it no further (**?**). In Malawi, the minimum transfer appears to account for the entire effect of the conditional transfer on the number of terms enrolled, although in the unconditional arm enrollment rose with the amount offered to parents (Baird et al., 2011).

Where schools sanction unpaid balances, partial payment appears to avert most of the sanctions that the Voucher's larger payments avert, so larger transfers may add little. Our estimates cover a single term, whereas the comparison programs ran for longer, and short-run effects need not persist. Free school uniforms in Kenya reduced absenteeism by 37 percent, but eight years later there is no evidence of effects on primary school completion (Evans and Ngatia, 2021). The ratios also exclude administrative costs, which range from 4 to 63 cents per dollar transferred across conditional cash transfer programs (García and Saavedra, 2017).

5.4 Learning

Programs that raise enrollment and attendance often have small effects on learning. In a review of cash transfer programs, effects on test scores are "small at best" (Baird et al., 2014), and in Cambodia, scholarships that raised completed schooling by 0.6 grades had no significant effect on test scores (**?**). Table 7 reports effects on the pre-specified learning index and its English and math components. Enumerators administered the Uwezo and Beyond Basics assessments at home (Section 2.6), so the scores cover children who were barred from school examinations or who left school (Jones et al., 2014).

We detect no effects on learning after one term. The estimates on the learning index range from 0.000 to 0.059 standard deviations, and the 95 percent confidence intervals exclude effects larger than 0.15 standard deviations in every arm. The only individually significant estimate, a 0.096 standard-deviation effect of Early Cash on math, is significant at the 10 percent level before any adjustment for multiple testing. School examinations show the same pattern (Appendix Table A.13), although children with unpaid fees can be barred from sitting them, so treatment changes who is tested.

A null after one term is consistent with the size of the attendance gains. The Voucher adds almost three days of school over the term, and in Uganda, 57 percent of teachers were absent from the classroom during unannounced school visits (Bold et al., 2017). Comparable programs raised test scores, if at all, over longer horizons. The labeled transfer in Morocco had a modest and statistically insignificant effect on math scores after two years (Benhassine et al., 2015), and the conditional transfer in Malawi raised English reading comprehension by 0.14 standard deviations after two years (Baird et al., 2011).

Table 7: Effects on Learning

	(1) Learning Index	(2) English Index	(3) Math Index
T1: Early Cash	0.059	0.003	0.096*
	(0.044)	(0.043)	(0.052)
	[0.158]		
T2: Fee-Day Cash	0.015	0.009	0.030
	(0.045)	(0.044)	(0.051)
	[0.329]		
T3: Voucher	0.000	-0.006	0.020
	(0.044)	(0.044)	(0.051)
	[0.362]		
Control Mean	0.000	0.000	0.000
Observations	1,956	1,956	1,959
p-value: T1 = T2	0.327	0.903	0.209
p-value: T2 = T3	0.743	0.734	0.846
p-value: T1 = T3	0.180	0.818	0.140

Notes: Intention-to-treat estimates from equation (3) with strata fixed effects and the child's baseline score (ANCOVA). Outcomes are endline Uwezo and Beyond Basics assessments, z-scored within grade against the endline control distribution. Sample sizes differ because the endline assessment is incomplete for a few children. q-values as in Table 5. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.5 Household Outcomes

The Voucher prevents parents from using the transfer for anything other than the fee, so its larger effects on schooling could come at the expense of other needs. This is the standard argument for cash over in-kind transfers (Currie and Gahvari, 2008), and in models of self-control, the loss of flexibility is the cost of commitment (Amador et al., 2006). In Malawi, the conditional transfer raised schooling by more than the unconditional transfer, but only the unconditional transfer reduced teenage marriage and pregnancy, entirely among girls who left school (Baird et al., 2011). Table 8 compares the three transfers on food insecurity, disagreement within the household over school expenses, changes in livestock and durable assets, transfers to other households, and borrowing. Appendix Tables A.10 and A.14 report additional household outcomes.

On the outcomes in Table 8, we find no evidence that Voucher households fare worse than cash households. Food insecurity falls by 0.12 to 0.17 standard deviations in every arm, and the Voucher's effect is statistically indistinguishable from those of Early Cash and Fee-Day Cash ($p = 0.91$ and $p = 0.47$). The share of households that disagree over school expenses falls by about 8 percentage points in every arm, a quarter of the control mean of 33 percent. No arm detectably reduces livestock or durable assets, and the livestock index rises by 0.13 and 0.12 standard deviations under Early Cash and the Voucher, significant only at the 10 percent level.

One reason the Voucher need not leave households worse off is that it replaces fee payments that many parents would otherwise have made themselves. In the control group, 56 percent of children have a recorded payment toward the fee (Appendix Table A.7). For their parents, a transfer paid to the school frees money that would otherwise have gone to the fee, as an in-kind transfer does for households that would have bought the good anyway (Moffitt, 1989). The cash transfers leave parents with more money for other uses, some of which appears to have gone to other children's schooling and, under Early Cash, to other households.

Relative to the control group, the Voucher does not reduce the schooling of the child's siblings, but unlike cash, it does not improve it (Appendix Table A.15). Among siblings aged 6 to 17, 96 percent attend school in the control group, and no arm changes this share detectably. Fee-Day Cash reduces the share of siblings sent home over fees by 6.8 percentage points, from 77 percent. Both cash transfers raise the share of siblings who receive school meals by about 5 points, significant at the 10 percent level, whereas the Voucher does neither. These patterns suggest that parents used cash for the schooling costs of other children, which the Voucher does not allow.

Early Cash also raises net transfers to other households, by UGX 10,096. Control households receive UGX 6,804 on net, so Early Cash households become net givers. Fee-Day Cash and the Voucher produce no comparable outflow (UGX 2,600 and 1,188), although their differences from Early Cash are not statistically significant ($p = 0.13$ and $p = 0.10$). The outflow is consistent with claims on cash that is held for weeks before fees are due (Jakiela and Ozier, 2016), a motive that does not appear to drive demand for the restrictions (Section 4.4).

Both cash transfers raise borrowing, and much of the additional borrowing is for schooling. The share of households that took a loan since January rises by 8 to 9 percentage points under the cash transfers, from a control mean of 49 percent (column 6). The Voucher's effect, 2.5 points, is not significant. The share that borrowed for school fees or school expenses is 34 percent in the control group (column 7). It rises by 8.7 percentage points under Early Cash and 5.1 under Fee-Day Cash, significant at the 1 and 10 percent levels, and by 1.1 under the Voucher, less than under Early Cash ($p = 0.01$).

Cash recipients therefore appear to have financed part of their schooling costs with credit. This suggests that part of the cash went to other uses before school costs fell due, or that the transfer made it easier to borrow. The Voucher, which pays the fee directly, does not raise borrowing. Our follow-up covers a single term, so we cannot detect costs or benefits that arise later.

Table 8: Effects on Household Outcomes

	(1) Food Insecurity	(2) Any Fee Disagreement	(3) Δ Livestock (Index)	(4) Δ Durable Assets (Index)	(5) Net Transfers Out (1k UGX)	(6) Took Any Loan	(7) Borrowed for School
T1: Early Cash	-0.115*	-0.080***	0.133*	0.049	10.096**	0.083***	0.087***
	(0.062)	(0.028)	(0.079)	(0.067)	(4.924)	(0.032)	(0.031)
T2: Fee-Day Cash	-0.172***	-0.084***	0.035	0.024	2.600	0.089***	0.051*
	(0.065)	(0.028)	(0.069)	(0.068)	(4.933)	(0.032)	(0.030)
T3: Voucher	-0.123*	-0.078***	0.121*	-0.002	1.188	0.025	0.011
	(0.066)	(0.028)	(0.069)	(0.064)	(5.488)	(0.032)	(0.031)
Control Mean	0.000	0.327	0.000	0.000	-6.804	0.491	0.343
Observations	1,960	1,957	1,960	1,960	1,939	1,942	1,955
p-value: T1 = T2	0.380	0.901	0.246	0.718	0.130	0.832	0.234
p-value: T2 = T3	0.471	0.831	0.260	0.700	0.794	0.042	0.192
p-value: T1 = T3	0.911	0.929	0.886	0.439	0.101	0.068	0.013

Notes: Intention-to-treat estimates from equation (3) with strata fixed effects, excluding the Choice group. Outcomes are from the endline parent survey. Column 1 is the Food Insecurity Experience Scale score, standardized on the control group, with higher values indicating more food insecurity, and controls for its baseline value. Column 2 indicates any disagreement within the household over school expenses and controls for its baseline analog. Columns 3 and 4 are indices of the change between baseline and endline in livestock holdings (seven species) and in durable assets (bicycles, motorcycles, computers, radios, and refrigerators). Each item's change is winsorized at the 1st and 99th percentiles of nonzero changes and standardized on the control group, and the average across items is divided by its control-group standard deviation. Column 5 is money given to other households minus money received from them since January 2026, in thousands of UGX, winsorized, and controls for its baseline value. Column 6 indicates that the household took any loan since January 2026 and controls for whether it had an active loan at baseline. Column 7 indicates that the household borrowed for school fees or school expenses since January 2026 and has no baseline analog. Sample sizes vary with item nonresponse at endline and, for the baseline controls, at baseline. Robust standard errors in parentheses. $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

6 Anticipated and Realized Effects of Commitment

Demand for commitment shows that caregivers anticipate some difficulty following through on fee payment. Demand alone, however, does not reveal how well they understand that difficulty: a sophisticated caregiver correctly anticipates how her future behavior will respond to unrestricted cash, while a partially naïve caregiver recognizes a problem but underestimates its severity. We use caregivers' forecasts of fee payment under each instrument to distinguish these cases, then ask whether willingness to pay identifies those who benefit most from commitment, and whether experience changes demand.

6.1 Forecast Errors and Partial Naivete

We first ask whether caregivers correctly anticipate how the three instruments will affect their own fee payment. At baseline, caregivers report the probability that they would fully clear their child's Term 1 fees under four scenarios: no assistance, Early Cash, Fee-Day Cash, and the Voucher. For each instrument d, we define its anticipated effect, $\widetilde{\tau}_d$, as the within-caregiver difference between the forecast under d and the forecast under no assistance, averaged across caregivers. We compare this with the realized effect, τ_d, identified by random assignment relative to the control group. Following Augenblick and Rabin (2019) and Allcott et al. (2022), this comparison measures whether caregivers correctly predict the behavioral response to each instrument (see also Acland and Levy, 2015; Mahajan et al., 2026). If caregivers accurately anticipate their own behavior, anticipated and realized effects coincide on average.

Figure 6 presents the results. Panel A compares forecast and realized levels. Caregivers substantially overpredict fee clearing in every scenario: they predict a 42 percent probability of clearing without assistance, compared with 9.9 percent realized, and a 74 percent probability under the Voucher, compared with 50 percent realized. Level errors of this size do not identify naïveté, because realized clearing also depends on income, fees, and the school. The Voucher arm makes the point within the data: a 24-point error survives in the arm where clearing depends least on the caregiver's own behavior, so much of the level error concerns circumstances rather than behavior.[31] We therefore focus on whether caregivers correctly predict the *effects* of the instruments, which difference out forecast errors common to the scenarios.

Panel B shows that they do not. Caregivers expect Early Cash to increase full clearing by 17.7 percentage points (s.e. 0.7) and Fee-Day Cash by 29.8 points (0.7), compared with realized effects of 4.8 points (2.1) and 5.0 points (2.0). They expect the Voucher to increase clearing by 30.4 points (0.7), compared with a realized effect of 40.0 points (2.4). The corresponding forecast errors—anticipated minus realized effects—are 12.9 (2.2), 24.9 (2.3), and −9.6 (2.6) percentage points: caregivers overpredict what cash will do and underpredict

[31] This is why our test differs from settings in which levels themselves identify misprediction. In Allcott et al. (2022), borrowers forecast a familiar status-quo behavior over eight weeks and are nearly calibrated, predicting a 70 percent chance of reborrowing against a realized 74 percent; the level comparison is informative precisely because the environment component is small. Here the forecast concerns a term-long outcome priced by schools, and the environment component is first-order.

what the earmark will do.

Comparisons across instruments show where caregivers expect the problem to bind. They expect moving cash from six weeks before the fee date to the fee date to increase clearing by 12.1 percentage points (s.e. 0.6); experimentally, the difference is 0.1 points (2.2). Conversely, they expect the Voucher to outperform Fee-Day Cash by only 0.5 points (0.6) on the full-clearing margin, while experimentally the Voucher's advantage is 35.1 points (2.5). Caregivers thus expect the main difficulty to be preserving cash until fees are due. In fact, the far larger difficulty arises once cash is already in hand.

The Voucher–Fee-Day Cash comparison provides the sharpest test of naïveté. The two instruments deliver the same amount at the same time toward the same fee and differ only in whether the transfer remains fungible, so beliefs about fee-date resources enter both forecasts symmetrically and cancel. What remains is the caregiver's forecast of her own allocation of cash in hand. In the benchmark model, the Voucher settles the fee at source, so the forecast error in this comparison satisfies

$$\left(\widetilde{\tau}_V - \widetilde{\tau}_2\right) - \left(\tau_V - \tau_2\right) = \mu_2(\beta) - \mu_2(\tilde{\beta}), \tag{4}$$

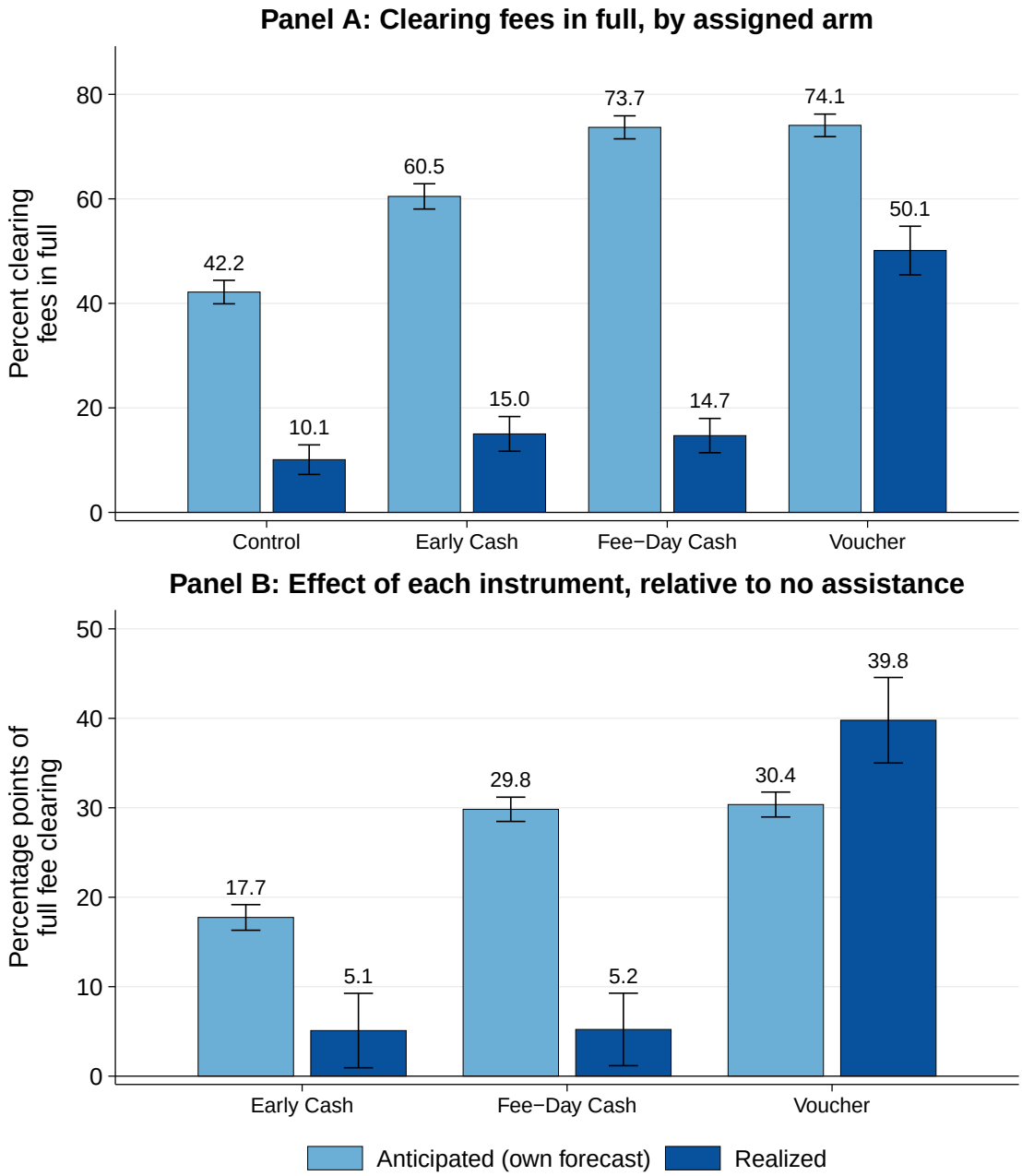
where $\mu_2(b)$ is the probability that a caregiver pays the fee out of cash in hand when future behavior is governed by present-focus parameter b. Because $\mu_2(b)$ is increasing, sophistication ($\tilde{\beta} = \beta$) implies a zero forecast error and partial naïveté ($\tilde{\beta} > \beta$) implies a negative one; no $\tilde{\beta} \leq \beta$ can produce a negative error, and the ordering $\beta \leq \tilde{\beta}$ rules out a positive one. The comparison is therefore a one-sided test that the data could have failed.

The estimated forecast error is -34.5 percentage points (bootstrap s.e. 2.7; one-sided $p < 0.001$). We therefore reject the sophisticated benchmark. Caregivers expect unrestricted cash on the fee date to work almost as well as the Voucher; in fact, the Voucher is substantially more effective. In the model, the negative error implies $\beta < \tilde{\beta}$: a caregiver who overrates her own follow-through expects to need the earmark less often than she does.

Two qualifications apply. First, the interpretation relies on the maintained model. Differencing removes forecast errors common to the scenarios—generic optimism, aspirational reporting—and no common error can make cash look too effective while the earmark looks insufficiently effective; but instrument-specific errors remain. A caregiver could, for example, systematically underestimate competing demands on cash at the fee date while correctly anticipating her own present focus. Errors of this kind still imply misprediction of her own future allocation, but need not map uniquely into $\tilde{\beta}$. Section 4.4 finds little evidence that demand for commitment reflects claims from other household members or kin, making present focus the leading interpretation.[32] Second, the experimental Voucher covers only part of the outstanding balance for most children, whereas equation (4) describes the benchmark case in which the payment is settled at source. The equation is therefore a sign prediction rather than an exact inversion of the experimental estimate, and the deviation works against the result: optimism about paying the remaining balance raises the anticipated Voucher effect

[32] Fedyk (2025) instead separates self-specific optimism by comparing forecasts of one's own and others' behavior. Our design identifies misprediction from the comparison of two instruments that hold fixed the timing and amount of resources.

Figure 6: Anticipated and Realized Full Fee Clearing



Notes: $N = 1{,}776$ non-transfer children; both panels use this common sample. Panel A reports the average baseline forecast (0–100 scale, elicited before treatment assignment) of fully clearing Term 1 fees under the scenario corresponding to each assigned arm, together with realized full clearing in that arm. Full clearing is defined as total fees paid at least equal to fees charged, excluding food, in school administrative records. The Voucher pays 85 percent of billed fees on average; full clearing, the event the belief elicitation concerns, occurs for 50 percent. Panel B reports anticipated effects, defined as within-caregiver differences in forecasts under each instrument relative to no assistance, and realized effects, defined as intention-to-treat differences relative to the control group, with randomization-strata fixed effects and heteroskedasticity-robust standard errors. Error bars are 95% confidence intervals. Forecast errors equal anticipated minus realized effects; their standard errors are based on 1,000 caregiver-level bootstrap replications.

and makes the measured error less negative.

The forecast evidence complements the demand evidence. Positive willingness to pay for commitment (Section 4) implies $\tilde{\beta} < 1$: caregivers anticipate some failure of future follow-through. The negative earmarking forecast error implies $\beta < \tilde{\beta}$: they underestimate that failure. Under the maintained model, the joint evidence implies

$$\beta < \tilde{\beta} < 1,$$

the defining ordering of partial naïveté.

6.2 Willingness to Pay and Heterogeneous Treatment Effects

We next ask whether caregivers who demand more commitment also benefit more from it. If willingness to pay accurately measures the severity of the underlying self-control problem, the effect of the Voucher relative to cash should rise with the Voucher premium.

Figure 7 shows the opposite. Fee payment under the Voucher is nearly constant across four groups of baseline Voucher premia, at approximately UGX 39,000–44,000. Under Fee-Day Cash, however, payment rises from roughly UGX 14,000 among caregivers at the lower WTP bound to UGX 24,000 among those at the upper bound. The Voucher's incremental effect therefore falls from UGX 27,855 to UGX 15,412 across these groups, a difference of UGX 12,443 (s.e. 3,561). In a linear specification, each additional UGX 1,000 of Voucher premium is associated with a UGX 338 smaller Voucher advantage ($p < 0.001$). Panel B shows no comparable relationship between the delay premium and the effect of Fee-Day relative to Early Cash. Appendix Table A.16 reports the underlying regressions and other schooling outcomes.

The gradient comes from behavior under cash, not the Voucher: caregivers with higher Voucher premia pay more when given Fee-Day Cash, leaving less scope for earmarking to increase payment. Willingness to pay therefore does not select the caregivers with the largest gains from commitment. If anything, allocating the Voucher on the basis of willingness to pay would select caregivers with smaller incremental effects.

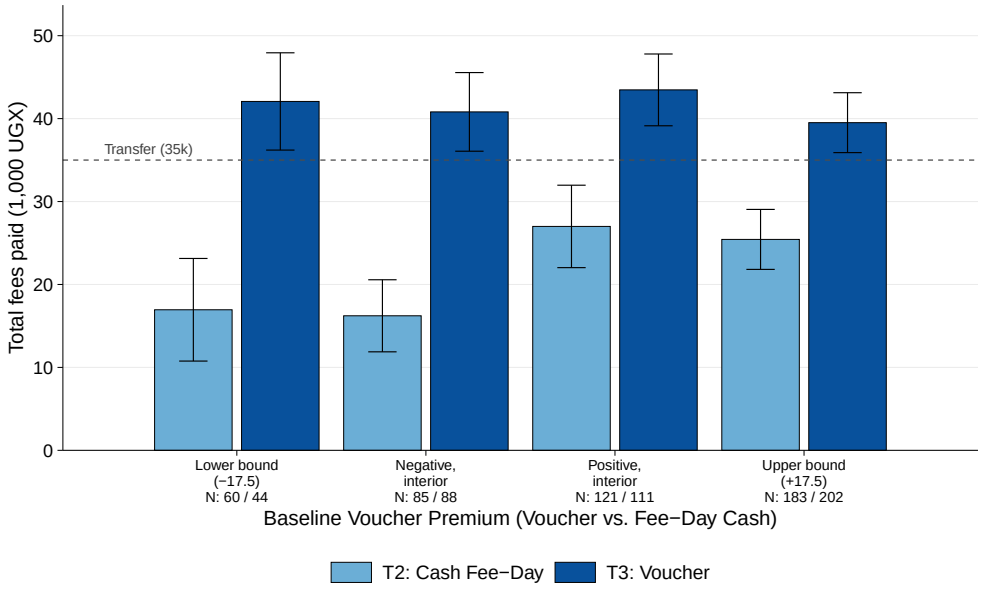
This disconnect between demand and effects appears in other commitment settings: hypertensive patients in Bai et al. (2021) overestimate the behavioral effect of commitment contracts, and demand for commitment correlates only weakly with subsequent behavior elsewhere (Carrera et al., 2022; John, 2020). Here, random assignment lets us observe demand and the causal return to the same instrument for the same population.

6.3 Preference Updating

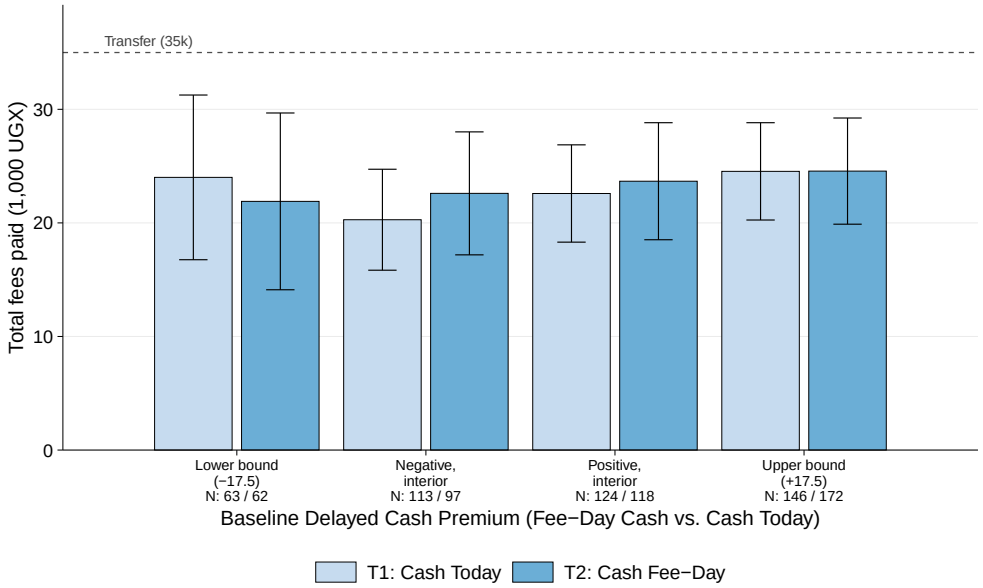
At endline, we repeat the Voucher-versus-cash task for a future school term. Because baseline demand is elicited before random assignment and control caregivers repeat the task without receiving a transfer, treatment differences in endline demand identify the effect of experiencing each instrument.

Figure 7: Total Fees Paid, by Baseline WTP Premium

Panel A: Voucher vs. Fee-Day Cash, by Voucher Premium



Panel B: Fee-Day Cash vs. Early Cash, by Delayed Cash Premium



Notes: The outcome is total Term 1 fees paid, in 1,000 UGX, from school administrative records; children with no recorded payment are coded as zero. Caregivers are grouped by their baseline premium into the lower censoring point (−17,500 UGX), negative interior values, positive interior values, and the upper censoring point (+17,500 UGX). Panel A compares the Voucher with Fee-Day Cash; Panel B compares Fee-Day Cash with Early Cash. Light bars are raw reference-arm means with 95% confidence intervals; dark bars add the group-specific treatment effect from a regression saturated in group × arm with randomization-strata fixed effects. Cell sizes are reported below each group. The dashed horizontal line marks the UGX 35,000 transfer amount.

Figure 8 shows substantial updating. Control caregivers do not revise: their mean Voucher premium changes from UGX 5,907 to UGX 5,823 (−83, s.e. 792). Voucher recipients increase their premium by UGX 3,524 (729), while Early Cash and Fee-Day Cash recipients reduce it by UGX 2,687 (759) and UGX 1,719 (735).

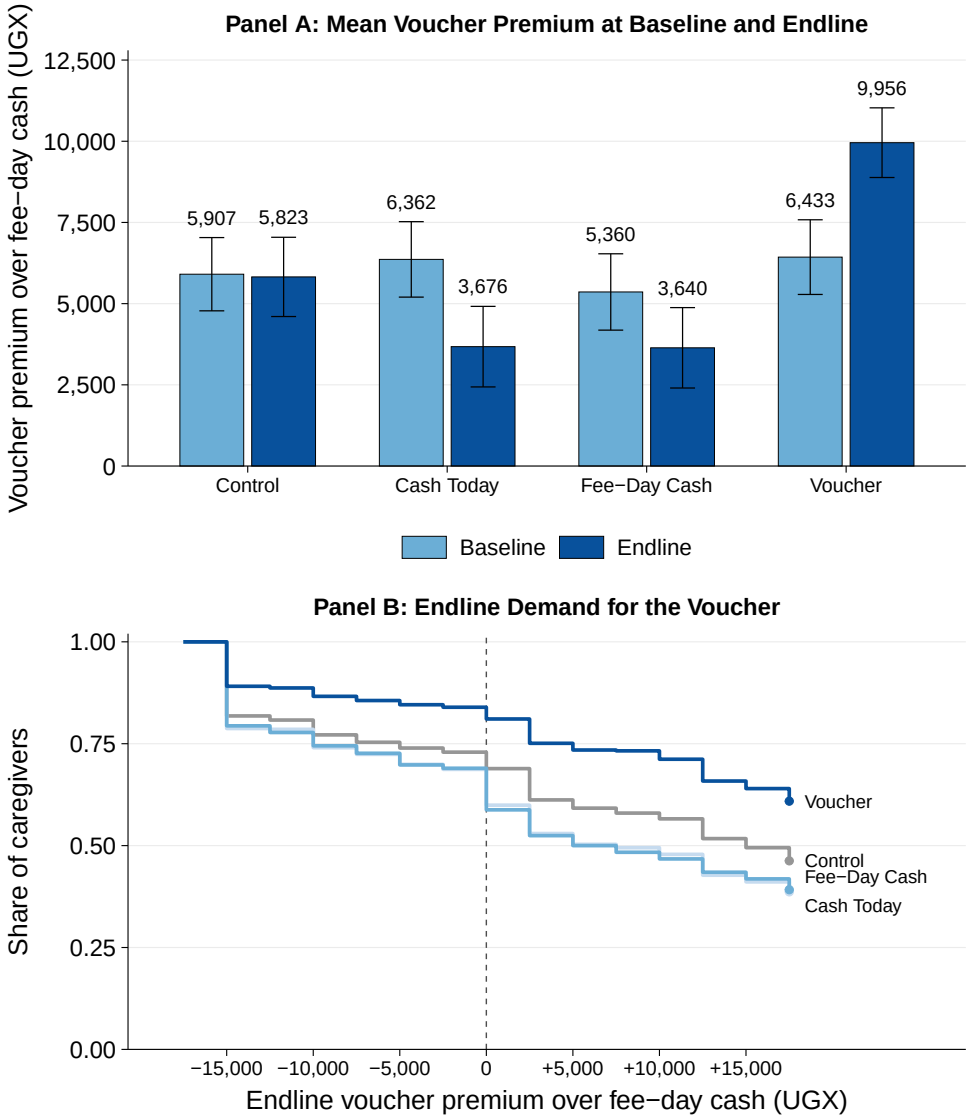
The experimental comparisons yield the same pattern. Among control caregivers, 68.9 percent choose the Voucher at endline. Voucher assignment raises this share by 12.1 percentage points, while Early Cash and Fee-Day Cash reduce it by 8.8 and 9.7 points. In the preferred ANCOVA Tobit specification, Voucher assignment raises latent valuation by UGX 10,778, 26 percent of the control mean, while Early Cash and Fee-Day Cash reduce it by UGX 5,484 and UGX 4,836; the corresponding marginal effects on the observed, censored premium are 4,211, $-2,143$, and $-1,890$. The two cash effects are statistically indistinguishable ($p = 0.74$), while each differs from the Voucher effect at conventional levels.

The distribution shifts in the same direction: 61 percent of Voucher recipients are at the upper bound of the elicitation, compared with 46 percent of control caregivers and 39 percent in either cash arm. Appendix Table A.18 shows similar results across alternative specifications and samples. Experimenter demand is unlikely to explain the pattern: control caregivers do not revise, the treatment arms move in opposite directions despite receiving the same endline elicitation, and a desire to stay consistent with baseline answers would attenuate rather than generate the differences.

These results show that experience changes valuations, but not what caregivers learn. In other settings, repeated experience with clear feedback moves beliefs about one's own self-control toward accuracy (Kaur et al., 2015; Le Yaouanq and Schwardmann, 2022). Because each caregiver experiences only her assigned instrument and we do not re-elicit the counterfactual clearing beliefs at endline, the experiment does not distinguish learning about β from learning about the environment. Nor does one term of experience eliminate demand: caregivers in the cash arms still value earmarking at roughly UGX 3,600–3,700, and 59–60 percent continue to choose the Voucher at par.

Overall, caregivers anticipate a need for commitment but imperfectly predict its returns: they overpredict what cash will accomplish, underpredict the effect of earmarking, and their valuations do not select those who benefit most. Experience moves demand in the direction of the instrument received. Willingness to pay is therefore informative about the value caregivers place on commitment, but it is not a sufficient statistic for the gains commitment produces.

Figure 8: Experience Shifts Demand for the Voucher



Notes: $N = 1{,}960$. The Voucher premium is defined as in Figure 3; the endline premium comes from the re-elicitation for an upcoming term. Panel A reports baseline and endline means by treatment arm. Panel B reports the share of each arm with an endline premium of at least each value on the elicitation grid.

7 Conclusion

Why do households fail to complete investments they appear to value? This paper shows that insufficient resources are only part of the answer. When an investment requires a lumpy payment at a fixed date, households must also preserve resources until the deadline and then convert those resources into the required expenditure. Our experiment separates these margins by providing identical resources while varying their timing and fungibility.

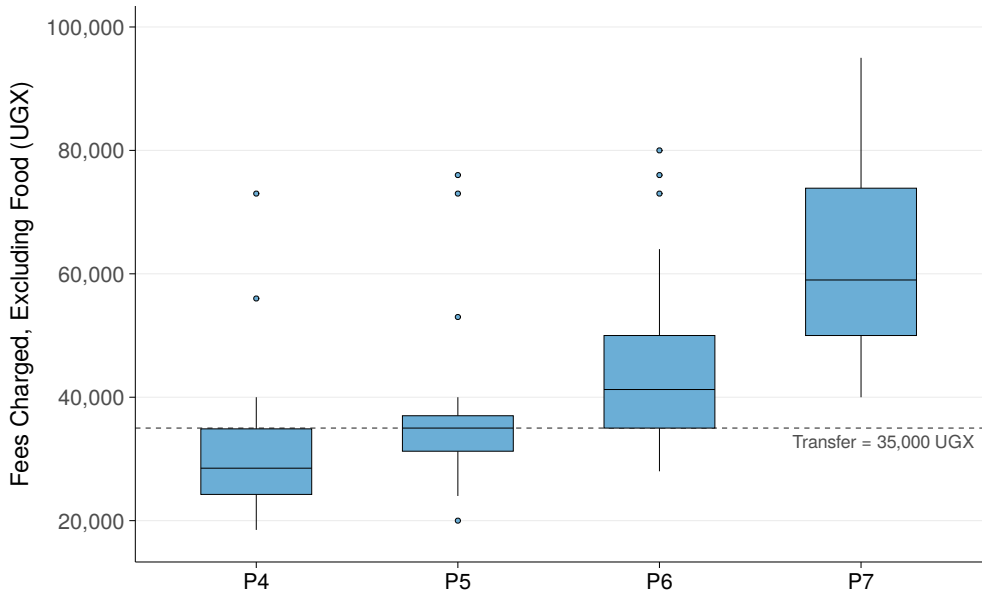
The results point to the importance of the second margin. Cash raises fee payment modestly, and delivering it precisely when fees are due does no more than delivering it several weeks earlier. Paying the same transfer directly to the school, by contrast, raises the share of fees paid by 52 percentage points and reduces dropout by more than half. The difference is not that cash is spent primarily outside education: caregivers devote most of it to schooling, but spread it across inputs and children. The voucher instead completes the particular payment required for the focal child to remain fully engaged in school. For lumpy expenditures, relaxing a household's budget constraint need not be enough to ensure that the investment is completed.

Caregivers themselves place substantial value on restrictions that help them make these payments. Most are willing to give up cash both to delay a transfer until fees are due and to have it earmarked for the school. This costly demand is consistent with caregivers anticipating limits to their own ability to follow through. But demand is only imperfectly related to effectiveness. Caregivers value earmarking more than delay, correctly ranking the two restrictions, yet substantially overestimate what delaying cash will accomplish. Across caregivers, those willing to pay the most for earmarking do not benefit most from receiving it relative to cash. Willingness to pay therefore reveals demand for commitment without providing a sufficient statistic for the gains from commitment.

These findings have a broader implication for how we think about investments that require discrete, time-sensitive payments. The conventional resource constraint asks whether a household can afford the investment. Our results point to a separate question: whether available resources are converted into the transaction needed to complete it. Restrictions on timing or use can address this problem, but their effectiveness depends on where in the decision process the investment breaks down. In our setting, preserving liquidity until the deadline is not enough. Completing the payment is.

A Additional Tables and Figures

Figure A.1: Term fees by grade in the study schools

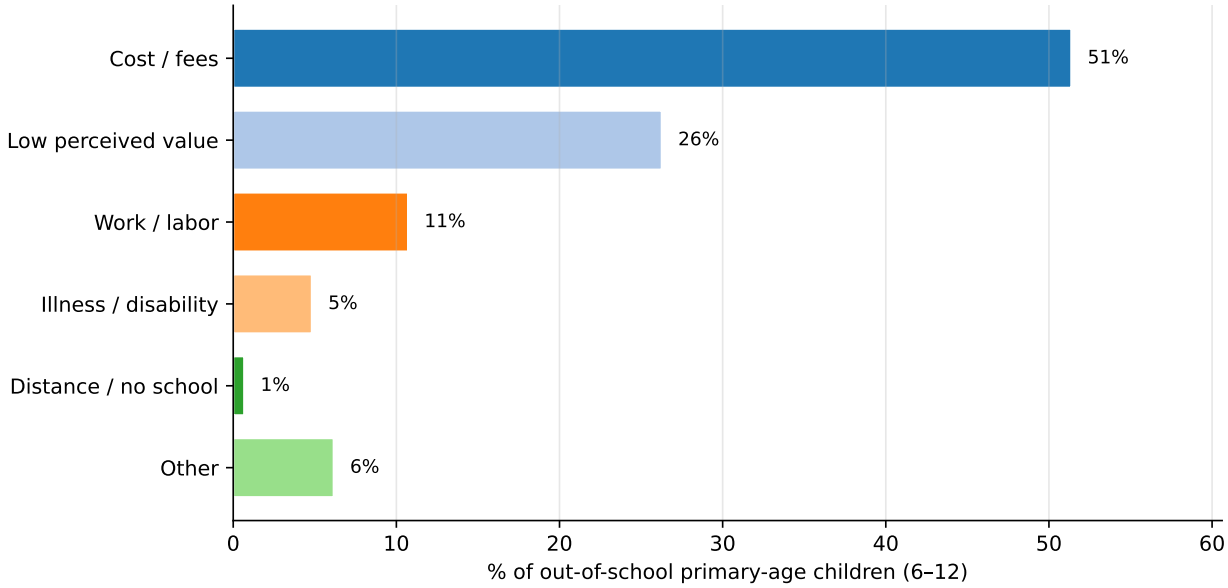


Notes: Term 1 2026 fees charged, excluding food, by grade in the 30 study schools, with one observation per school and grade. Boxes show the median and interquartile range, whiskers extend to the most extreme value within 1.5 times the interquartile range, and points are schools beyond the whiskers. The dashed line marks the UGX 35,000 transfer.

Table A.1: Saving, forecast payment, and willingness to pay by region

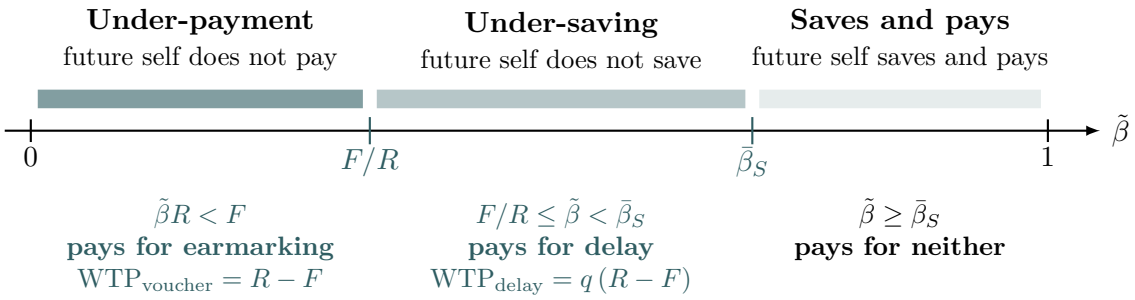
	Under-payment $\tilde{\beta}R < F$	Under-saving $F/R \leq \tilde{\beta} < \bar{\beta}_S$	Saves and pays $\tilde{\beta} \geq \bar{\beta}_S$
Saving under Early Cash	0	0	$F - y_L$
$\tilde{\mu}_1, \tilde{\mu}_2, \tilde{\mu}_V$	$0, 0, 1$	$1{-}q,\ 1,\ 1$	$1, 1, 1$
$\text{WTP}_{\text{voucher}}$	$R - F > 0$	0	0
$\text{WTP}_{\text{delay}}$	0	$q(R - F) > 0$	0

Figure A.2: Reason for Non-Enrollment



Notes: Reasons for being out of school among children aged 6–12 in the UNPS 2019/20, among the 86 previously enrolled children with a reported reason; children never enrolled are not asked. Survey weights applied. Categories group the original response options; "Cost / fees" is "too expensive."

Figure A.3: Willingness to pay by perceived present focus $\tilde{\beta}$



Notes: The figure shows the three regions of Proposition 1, which assumes linear utility, drawn for $y_L < F(1-q)$. A lower $\tilde{\beta}$ means that the parent believes they are more present-focused. With concave utility the same parent can pay for both restrictions (Appendix B.6).

Table A.2: Willingness to Pay and Forecast Gains, with Controls

	(1)	(2)	(3)	(4)
Panel A: Earmarking premium				
Forecast gain from earmarking (per 10 pp)	481***	504***	425***	579***
	(122)	(122)	(124)	(130)
Forecast gain from delay (per 10 pp)				424***
				(123)
Mean of dependent variable	6,000	5,996	5,996	5,996
Panel B: Delay premium				
Forecast gain from delay (per 10 pp)	821***	811***	796***	789***
	(113)	(114)	(116)	(121)
Forecast gain from earmarking (per 10 pp)				−21
				(133)
Mean of dependent variable	4,569	4,569	4,569	4,569
Controls	No	Yes	Yes	Yes
Strata fixed effects	No	No	Yes	Yes
Observations	2,000	1,995	1,995	1,995

Notes: OLS regressions of the earmarking premium (Panel A) and the delay premium (Panel B), in UGX, on the parent's forecast gains, measured in units of 10 percentage points. The forecast gain from earmarking is the forecast probability of fully clearing Term 1 fees under the Voucher minus that under Fee-Day Cash; the forecast gain from delay is the forecast under Fee-Day Cash minus that under Early Cash. Column 1 corresponds to the slopes in Figure 5. Controls are an indicator for Primary 4, the asset index, the child's baseline numeracy and literacy scores, the parent's expected return to schooling, and the school's Term 1 fee (per UGX 10,000). Strata are the randomization strata. Robust standard errors in parentheses. $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

Table A.3: Heterogeneity in Willingness to Pay: Wealth, Credit, Liquidity, and Seasonality

		Earmarking Premium *(Voucher vs. Fee-Day Cash)*		**Delay Premium** *(Fee-Day Cash vs. Early Cash)*	
	Mean	Marg. effect	(s.e.)	Marg. effect	(s.e.)
Panel A: Wealth					
Asset index (z-score)	0.06	−816***	(272)	−685**	(277)
Income above median (0/1)	0.51	636	(569)	−418	(579)
Panel B: Access to Credit					
Has an active loan (0/1)	0.64	−1,012*	(589)	−117	(603)
Rejected for a loan (0/1)	0.26	43	(650)	−626	(667)
Fast loan (0/1)	0.21	−237	(703)	−1,037	(720)
Ease of obtaining a loan (z-score)	0.00	−338	(283)	−726**	(288)
Panel C: Liquidity and Savings					
Liquidity index (z-score)	0.00	−834***	(283)	−675**	(288)
Has any savings (0/1)	0.76	−2,584***	(643)	−549	(682)
Member of a savings group (0/1)	0.58	−1,944***	(570)	−450	(585)
Has crops in storage (0/1)	0.43	−189	(574)	−888	(584)
Food insecurity index (z-score)	0.00	−153	(287)	313	(291)
Panel D: Seasonality					
Main income from agriculture (0/1)	0.64	141	(593)	−148	(603)
Investment opportunity in February (0/1)	0.24	1,467**	(653)	−1,693**	(692)
February is the lowest-income month (0/1)	0.43	1,259**	(571)	−141	(584)
February is the highest-expense month (0/1)	0.74	−1,111*	(640)	−772	(657)

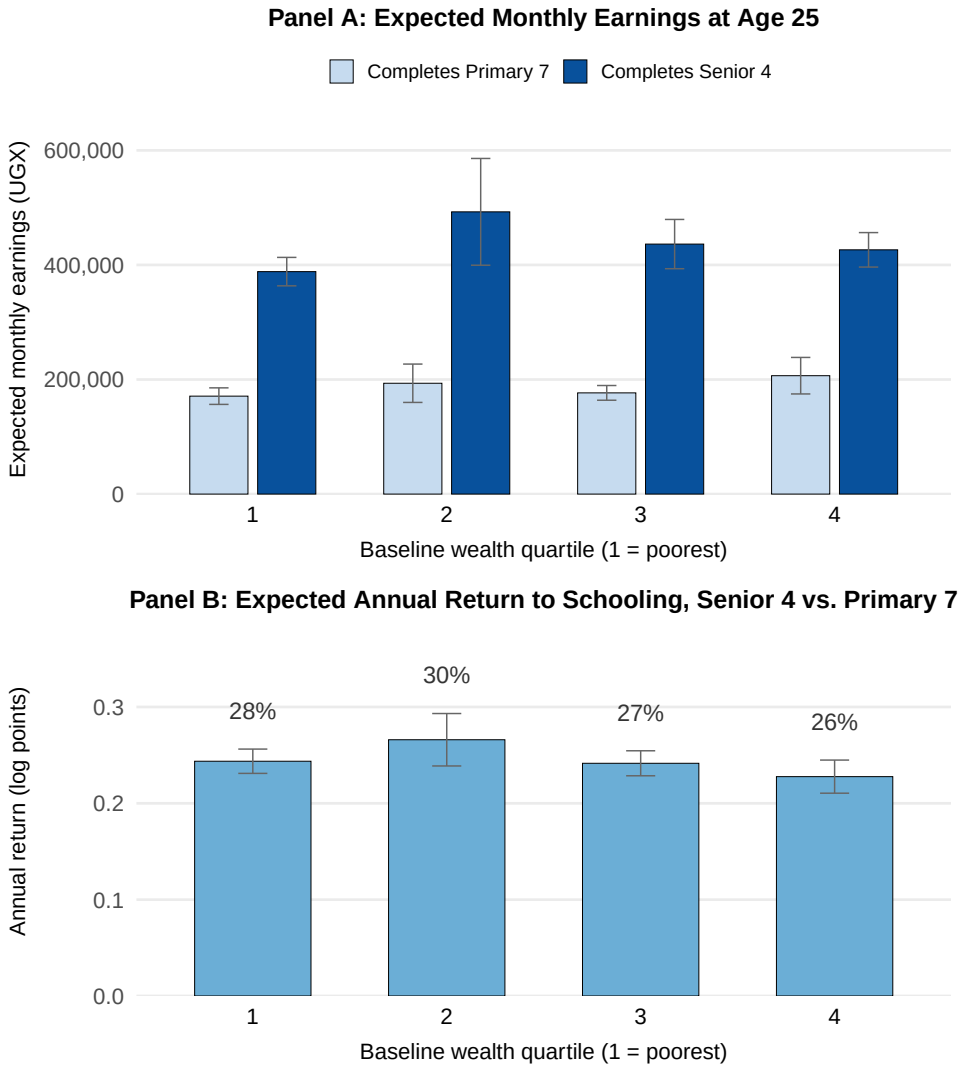
Notes: $N = 2{,}000$ parents. Average marginal effects from separate two-limit Tobit models with no other controls, as in Table 3. For binary variables, the effect is the difference between one and zero. Standard errors in parentheses. $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

Table A.4: Heterogeneity in Willingness to Pay: Parents' Beliefs about Schooling and Fee Payment

			Earmarking Premium *(Voucher vs. Fee-Day Cash)*		**Delay Premium** *(Fee-Day Cash vs. Early Cash)*	
	Mean	(s.d.)	Marg. effect	(s.e.)	Marg. effect	(s.e.)
Panel A: Beliefs about the Child's Schooling						
Expected return to schooling (z-score)	0.00	(1.00)	9	(283)	48	(293)
Expects Division 1 on the PLE (0/1)	0.32	(0.46)	−150	(613)	377	(621)
Expects above-average PLE score (0/1)	0.39	(0.49)	−875	(570)	−217	(581)
Expects child to complete Primary 7 (0/1)	0.98	(0.13)	76	(2,137)	−3,346*	(1,998)
Expects child to complete Senior 4 (0/1)	0.85	(0.36)	−141	(786)	−114	(802)
Panel B: Forecasts of Clearing Fees (0–100)						
With no transfer	43.04	(24.47)	−26.6**	(11.5)	−23.9**	(11.7)
With Early Cash	60.48	(26.14)	−67.6***	(10.6)	−68.2***	(10.8)
With Fee-Day Cash	72.52	(23.54)	−34.5***	(12.1)	17.5	(12.3)
With the Voucher	72.95	(23.92)	10.1	(11.8)	−12.3	(12.1)

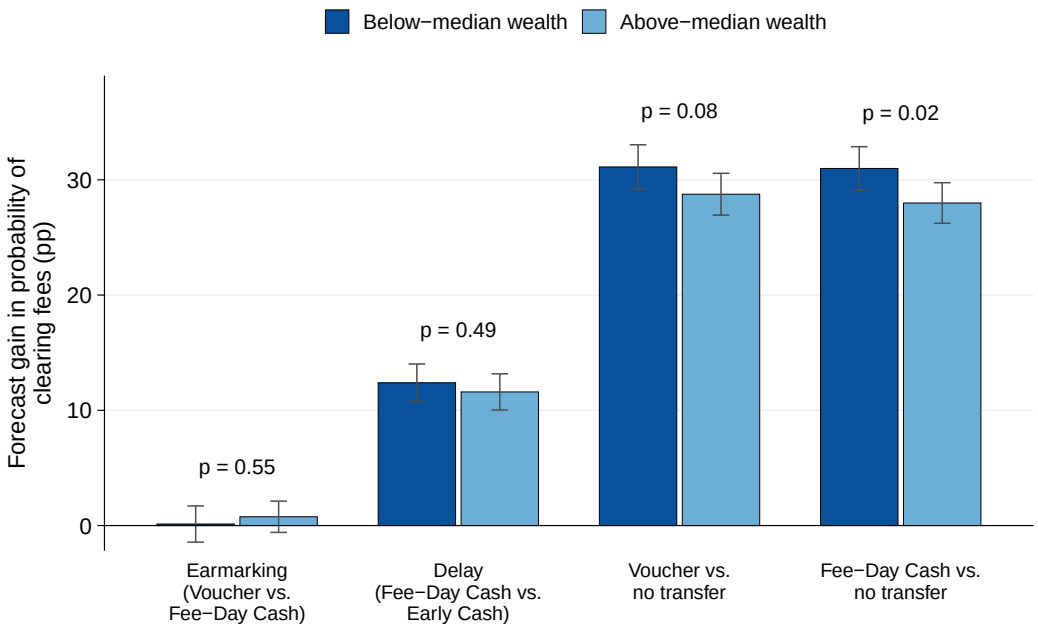
Notes: $N = 2{,}000$ parents. Average marginal effects from separate two-limit Tobit models with no other controls, as in Table 3. For binary variables, the effect is the difference between one and zero. The expected return to schooling follows Attanasio and Kaufmann (2014). Panel B reports the parent's baseline forecasts of the probability of fully clearing Term 1 fees under each scenario (Section 2.4); effects are per percentage point. Standard errors in parentheses. $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

Figure A.4: Expected Returns to Schooling by Baseline Wealth



Notes: $N = 1{,}996$ parents with complete earnings expectations, by quartile of the baseline asset index. Panel A reports the midpoint of the parent's expected range of the child's monthly earnings at age 25, if employed, under two scenarios: the child completes only Primary 7 or completes Senior 4. Panel B reports the expected log earnings difference between the two scenarios divided by the four years that separate them, following Attanasio and Kaufmann (2014); labels give the corresponding percent return. Error bars are 95% confidence intervals.

Figure A.5: Forecast Gains by Baseline Wealth



Notes: $N = 2{,}000$ parents, split at the median of the baseline asset index. Each bar is the mean difference, in percentage points, between two of the parent's baseline forecasts of the probability of fully clearing Term 1 fees (Section 2.4). Error bars are 95% confidence intervals. The p-values test equality of the means across the two wealth groups.

Table A.5: Who Has a Positive Premium

	Positive Earmarking Premium		**Positive Delay Premium**	
	Coef.	(s.e.)	Coef.	(s.e.)
Panel A: Wealth and Liquidity				
Asset index (z-score)	-0.030^{***}	(0.010)	-0.031^{***}	(0.011)
Liquidity index (z-score)	-0.019^{*}	(0.010)	-0.016	(0.011)
Forecast: clears fees with no transfer (0–100)	-0.0011^{***}	(0.0004)	-0.0007	(0.0004)
Panel B: Child Achievement and Expected Returns				
Learning index (z-score)	0.013	(0.011)	0.021^{*}	(0.011)
Numeracy score (z-score)	0.012	(0.010)	0.028^{***}	(0.011)
Literacy score (z-score)	0.011	(0.010)	0.009	(0.011)
Expected return to schooling (z-score)	-0.007	(0.010)	0.011	(0.010)

Notes: $N = 2{,}000$ parents. Linear probability models of an indicator for a strictly positive earmarking or delay premium on the row variable, with the same controls as in Table 3. Robust standard errors in parentheses. $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

Table A.6: Lee (2009) Bounds for Outcomes Observed at the Baseline School

	Fee Payments		Attendance
	(1) Total Paid (1k UGX)	(2) Share Paid	(3) Share of Weeks Present
Panel A: Treatment arms vs. Control (treatment arm trimmed)			
T1: Early Cash — trimming share 3.43%			
ITT (did-not-transfer sample)	4.649***	0.116***	0.025
	(1.417)	(0.028)	(0.016)
Lee lower bound	0.903	0.056	0.016
	(2.419)	(0.040)	(0.017)
Lee upper bound	4.825	0.129	0.051
	(1.986)	(0.035)	(0.021)
Imbens–Manski 95% CI	[-3.088, 8.102]	[-0.010, 0.187]	[-0.012, 0.085]
T2: Fee-Day Cash — trimming share 3.88%			
ITT (did-not-transfer sample)	4.762***	0.111***	0.033**
	(1.428)	(0.028)	(0.016)
Lee lower bound	0.369	0.043	0.027
	(2.411)	(0.041)	(0.016)
Lee upper bound	4.942	0.123	0.065
	(2.108)	(0.038)	(0.020)
Imbens–Manski 95% CI	[-3.601, 8.414]	[-0.024, 0.185]	[0.000, 0.098]
T3: Voucher — trimming share 3.80%			
ITT (did-not-transfer sample)	22.800***	0.595***	0.050***
	(1.260)	(0.026)	(0.015)
Lee lower bound	19.204	0.541	0.040
	(1.943)	(0.036)	(0.016)
Lee upper bound	22.904	0.620	0.077
	(1.734)	(0.034)	(0.019)
Imbens–Manski 95% CI	[16.005, 25.759]	[0.483, 0.676]	[0.014, 0.109]
Panel B: Earmarking margin (T2 trimmed)			
T3: Voucher vs. T2: Fee-Day Cash — trimming share 0.08%			
ITT (did-not-transfer sample)	18.038***	0.484***	0.017
	(1.242)	(0.027)	(0.014)
Lee lower bound	17.980	0.486	0.012
	(1.771)	(0.033)	(0.016)
Lee upper bound	18.127	0.488	0.013
	(1.838)	(0.033)	(0.015)
Imbens–Manski 95% CI	[14.577, 21.658]	[0.422, 0.552]	[-0.018, 0.043]
Share not transferring: Control 0.881; T1 0.912; T2 0.916; T3 0.916			
Observations (did not transfer)	1,776	1,776	1,776
Observations (randomized)	1,960	1,960	1,960

Notes: Fee records (columns 1–2) and spot checks at the baseline school (column 3) are compared among children who did not move to another school, and the share who move is 2.8 to 3.5 percentage points lower in every treatment arm (Table 5, column 5). Each comparison trims the arm with more children who stayed by its excess share, from the tail of its outcome distribution that minimizes the difference for the lower bound and from the tail that maximizes it for the upper bound (Lee, 2009). Panel A trims each treated arm relative to the control group and assumes that no child moves schools because of treatment. Panel B trims Fee-Day Cash relative to the Voucher and assumes that every child who stays under the Voucher would also stay under Fee-Day Cash. ITT rows use strata fixed effects, whereas the bounds compare trimmed means without them, so the bounds need not contain the ITT estimate. Column 3 is the share of weeks present, with dropouts counted as absent, estimated at the child level, so its ITT differs slightly from Table 5, column 7. Standard errors of the bounds are bootstrapped (1,000 replications, resampling within arm), and the Imbens–Manski intervals cover the effect with 95 percent probability (**?**). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.7: Effects on Fee Payment: Any Payment and Full Clearing

	(1) Total Fees Paid (1k UGX)	(2) Share of Fees Paid	(3) Any Fees Paid (binary)	(4) Cleared Fees (binary)
T1: Early Cash	4.649***	0.116***	0.086***	0.045**
	(1.417)	(0.028)	(0.030)	(0.021)
T2: Fee-Day Cash	4.762***	0.111***	0.079***	0.039*
	(1.428)	(0.028)	(0.030)	(0.021)
T3: Voucher	22.800***	0.595***	0.429***	0.384***
	(1.260)	(0.026)	(0.024)	(0.024)
Control Mean	20.123	0.374	0.557	0.119
Observations	1,776	1,776	1,776	1,776
T1 Effect (%)	23.1	31.0	15.4	38.0
T2 Effect (%)	23.7	29.6	14.2	32.7
T3 Effect (%)	113.3	159.1	76.9	321.6
p-value: T1 = T2	0.936	0.859	0.824	0.765
p-value: T2 = T3	0.000	0.000	0.000	0.000
p-value: T1 = T3	0.000	0.000	0.000	0.000

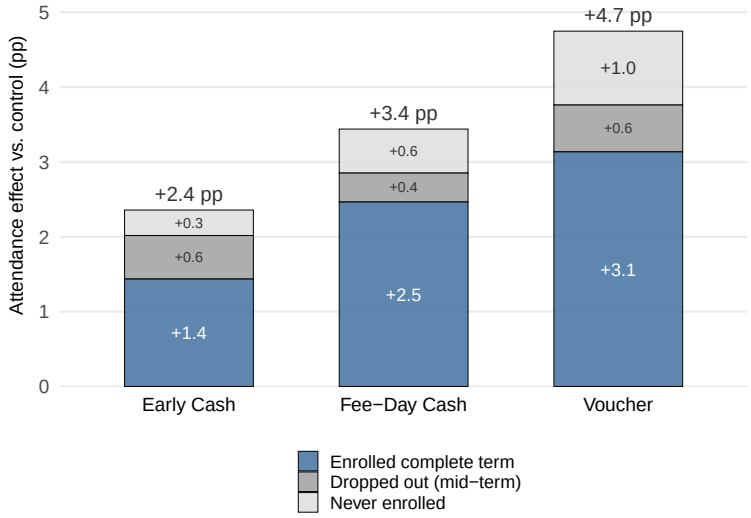
Notes: Intention-to-treat estimates from equation (3) with strata fixed effects, for children who did not move to another school during the term ($N = 1{,}776$). Columns 1–2 repeat Table 5. Column 3 indicates any recorded payment toward the Term 1 fee, and column 4 that recorded payments cover the full fee excluding the food charge. Payments include in-kind contributions valued at the school's unit prices, and missing records are coded as zero. T Effect (%) divides each coefficient by the control mean. Robust standard errors in parentheses. $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

Table A.8: The February Timing Sub-Experiment: IV Estimates Within Fee-Day Cash

	Fee Payments		Enrollment		Attendance
	(1) Total Paid (1k UGX)	(2) Share Paid	(3) Dropped Out	(4) Transferred Schools	(5) Did Not Transfer
Paid on February 10 (IV)	-3.607	-0.043	0.012	-0.010	0.021
	(3.786)	(0.076)	(0.034)	(0.047)	(0.035)
February-2 Mean	24.367	0.477	0.032	0.090	0.805
Observations	446	446	487	487	5,101
First stage: offered choice	0.671	0.671	0.632	0.632	0.671
First-stage F	208.0	208.0	185.5	185.5	275.2

Notes: Fee-Day Cash arm only. Being paid on February 10 is instrumented with random assignment to the binding date-choice offer (Section 2.5); two-stage least squares with strata fixed effects and robust standard errors, clustered by child in the attendance panel. The first-stage coefficient is the offer's effect on being paid February 10; the first-stage F is Kleibergen–Paap. Outcomes and samples match Table 5, except that three children who are the only Fee-Day Cash child in their randomization stratum are dropped.
$^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

Figure A.6: Decomposition of the Attendance Effect by Enrollment Status



Notes: Children who did not move to another school during the term ($N = 1{,}776$). Each bar splits the difference in mean attendance between a treatment arm and the control group into the contributions of three enrollment states, following **?** with midpoint weights. The first is higher attendance among children enrolled for the full term. The second is fewer children dropping out during the term, valued at the attendance gap between full-term children and dropouts, plus higher attendance among dropouts before they left. The third is fewer children who never enrolled, valued in the same way. The components sum to the raw difference in means, which differs slightly from the regression estimates in Table 5, column 7. Enrollment states are themselves affected by treatment, so the decomposition is descriptive.

Table A.9: Attendance of Control Children Relative to Classmates Outside the Study

	(1) Rate	(2) Weekly	(3) Daily	(4) Daily
In study, assigned control	0.0154	0.0167	0.0163	0.0159
	(0.0127)	(0.0136)	(0.0136)	(0.0135)
Classmate mean	0.920	0.921	0.923	0.923
Observations	3,353	37,771	165,092	165,092

Notes: Daily class registers for Term 1 2026. The sample consists of study children assigned to the control group and their classmates in the same Primary 5 or Primary 7 class who were not in the study. Study children are those in the attendance sample of Table 5, column 7, excluding children who never enrolled. The coefficient is the difference in attendance between control children and classmates outside the study. Column 1 uses each child's attendance rate over the term with class fixed effects, column 2 a weekly panel with class and week fixed effects, column 3 a daily panel with class and day fixed effects, and column 4 a daily panel with class-by-day fixed effects. Standard errors clustered by class in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure A.7: Timing of Fee Payments

Notes: Share of children with any recorded payment toward the Term 1 fee, in cash or in kind, on or before each date, in the 11 schools that recorded the date of each payment, among children who did not move to another school ($N = 611$). The Voucher is recorded when the school is credited, on February 10 for 95 percent of Voucher children, and is not shown. In a regression with strata fixed effects, the share with a payment recorded by March 10 is 52.0 percent in the control group and is higher by 5.2 percentage points (standard error 5.1) under Early Cash and by 11.9 percentage points (5.0) under Fee-Day Cash. The difference between the two cash arms is 6.7 percentage points (5.1; $p = 0.19$).

Table A.10: Effects on Fee Pressure and Household Disagreement

	(1) Fee Pressure from Headteacher (1–10)	(2) Any Fee Disagreement	(3) Disagreement Difficulty (1–10)
T1: Early Cash	-0.804***	-0.080***	-0.670***
	(0.217)	(0.028)	(0.191)
T2: Fee-Day Cash	-1.275***	-0.084***	-0.613***
	(0.210)	(0.028)	(0.191)
T3: Voucher	-1.379***	-0.078***	-0.738***
	(0.210)	(0.028)	(0.187)
Control Mean	5.673	0.327	3.507
Observations	1,960	1,957	1,960
T1 Effect (%)	-14.2	-24.6	-19.1
T2 Effect (%)	-22.5	-25.6	-17.5
T3 Effect (%)	-24.3	-23.9	-21.0
p-value: T1 = T2	0.027	0.901	0.749
p-value: T2 = T3	0.618	0.831	0.478
p-value: T1 = T3	0.007	0.929	0.702

Notes: Intention-to-treat estimates from equation (3) with strata fixed effects; outcomes from the endline parent survey. Column 1 is reported pressure from the headteacher over fees (1–10; no baseline analog, so unadjusted). Columns 2–3 are any disagreement in the household over school expenses and the difficulty of reaching agreement (1–10), each controlling for its baseline analog (ANCOVA). T Effect (%) divides each coefficient by the control mean. Robust standard errors in parentheses. $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

Table A.11: Effects on Peer and Teacher Stigma

	(1) Social Consequences Index (Std. Dev.)	(2) Peer Stigma	(3) Adult (Teacher) Stigma
T1: Early Cash	-0.100	-0.062**	-0.038
	(0.064)	(0.032)	(0.032)
T2: Fee-Day Cash	-0.033	-0.002	-0.028
	(0.064)	(0.032)	(0.032)
T3: Voucher	-0.094	-0.053*	-0.084***
	(0.065)	(0.032)	(0.032)
Control Mean	0.000	0.524	0.539
Observations	1,907	1,907	1,907
T1 Effect (%)		-11.9	-7.1
T2 Effect (%)		-0.3	-5.3
T3 Effect (%)		-10.0	-15.7
p-value: T1 = T2	0.303	0.058	0.758
p-value: T2 = T3	0.354	0.115	0.083
p-value: T1 = T3	0.924	0.759	0.153

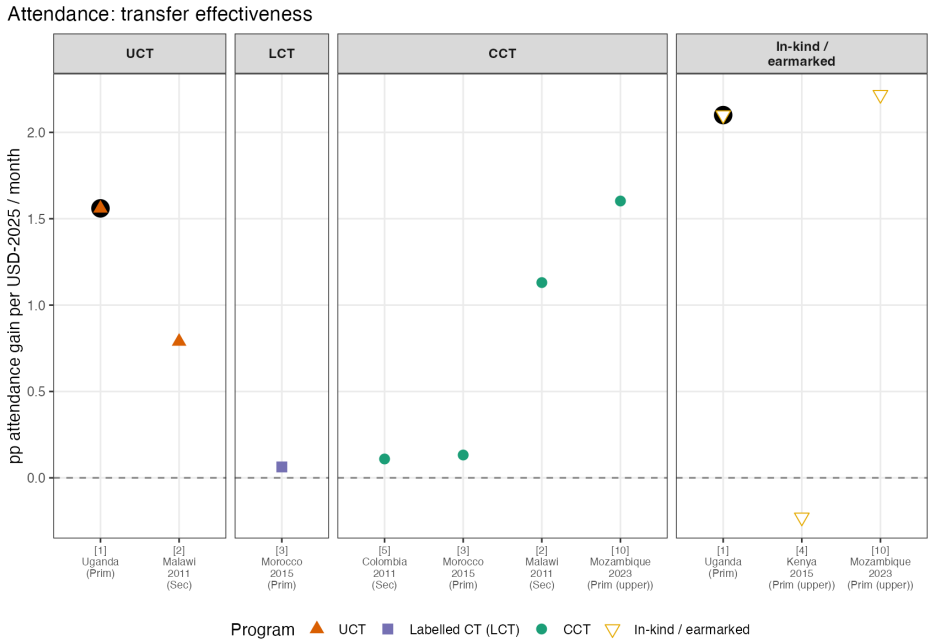
Notes: Intention-to-treat estimates from equation (3) with strata fixed effects; outcomes from the endline child survey. Peer and adult (teacher) stigma are indicators for child-reported fee-related mistreatment or shame, so coefficients are percentage-point effects; the index standardizes their sum. Stigma measures are missing for 53 children. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.12: Comparison Studies: Estimates Underlying Figure A.8

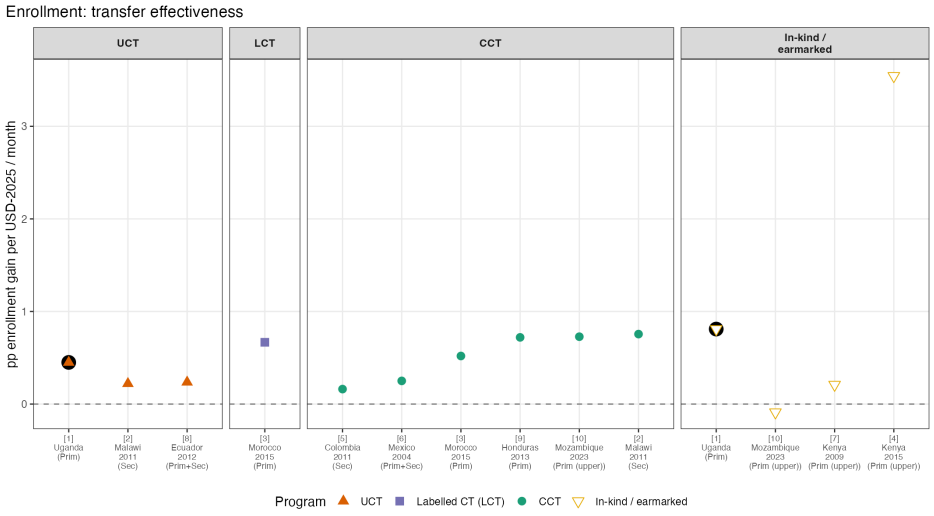
						Enrollment			Attendance		
Study	Country	Arm	Level	Follow-up (years)	Transfer ($/mo.)	Effect (pp)	Control mean	pp per $-mo.	Effect (pp)	Control mean	pp per $-mo.
This study	Uganda	Voucher	Prim	0.25	3.33	2.7 (1.2)	95	0.81	6.8 (1.8)	70	2.04
This study	Uganda	Fee-Day Cash	Prim	0.25	3.33	1.5 (1.3)	95	0.45	5.2 (1.8)	70	1.56
Baird et al. (2011)	Malawi	CCT	Sec	2	14.95	11.3 (3.9)	70	0.76	16.9 (8.5)	69	1.13
Baird et al. (2011)	Malawi	UCT	Sec	2	14.95	3.3 (3.9)	70	0.22	11.8 (10.2)	69	0.79
Benhassine et al. (2015)	Morocco	Labeled cash	Prim	2	11.09	7.4 (1.6)	74	0.67	0.7 (1.0)	96	0.06
Benhassine et al. (2015)	Morocco	CCT	Prim	2	10.59	5.5 (1.2)	74	0.52	1.4 (0.7)	96	0.13
Duflo et al. (2015)	Kenya	Free uniforms	Prim	7	0.87	3.1 (1.2)	81	3.54	-0.2 (0.6)	94	-0.23
Barrera-Osorio et al. (2011)	Colombia	CCT (basic)	Sec	1	24.72	4.0 (1.1)	70	0.16	2.7 (0.7)	79	0.11
Schultz (2004)	Mexico	CCT (sec. entry)	Sec	3	44.34	11.1 (3.6)	58	0.25	–	–	–
Kremer et al. (2009)	Kenya	Merit scholarship	Prim	2	2.86	3.2 (1.8)	87	1.12	–	–	–
Edmonds and Schady (2012)	Ecuador	UCT (BDH)	Prim+Sec	2	26.24	6.2 (2.0)	49	0.24	–	–	–
Galiani and McEwan (2013)	Honduras	CCT (PRAF II)	Prim	2–3	14.02	10.1 (3.6)	65	0.72	–	–	–
De Walque and Valente (2023)	Mozambique	Parent cash CCT	Prim (upper)	1	3.74	2.7 (1.5)	95	0.73	6.0 (2.2)	65	1.60
De Walque and Valente (2023)	Mozambique	Girl voucher	Prim (upper)	1	3.74	-0.3 (1.5)	95	-0.09	8.3 (2.2)	65	2.22

Notes: One row per treatment arm; intention-to-treat effects in percentage points with standard errors in parentheses, control means in percent. Transfer is the program's average monthly transfer per child in 2025 US dollars at base-year market rates; "This study" spreads UGX 35,000 over the three-month term. Enrollment for this study is one minus dropout (Table 5, column 4), the share of children enrolled at the end of the term. Attendance for this study is the full-sample estimate in Table 5, column 6. With the estimates among children who did not move (column 7), the attendance ratios are 1.47 for the Voucher and 0.96 for Fee-Day Cash. Standard errors reconstructed from published t-statistics or significance levels are flagged in the comparison database; for Schultz (2004), the transfer is the junior-secondary first-year grant at the extracted margin; for Benhassine et al. (2015), the CCT standard error is that of the difference between the CCT and the labeled transfer. Dashes indicate that the outcome is not reported in a comparable metric.

Figure A.8: Effects of Cost-of-Schooling Interventions per Dollar Transferred



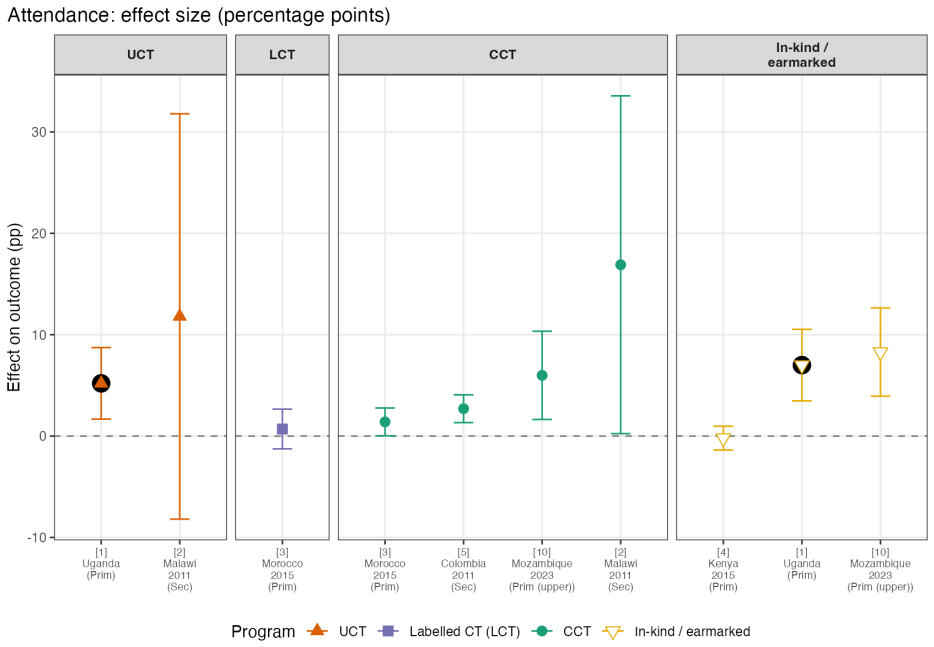
(a) Attendance, percentage points per 2025 US dollar per month



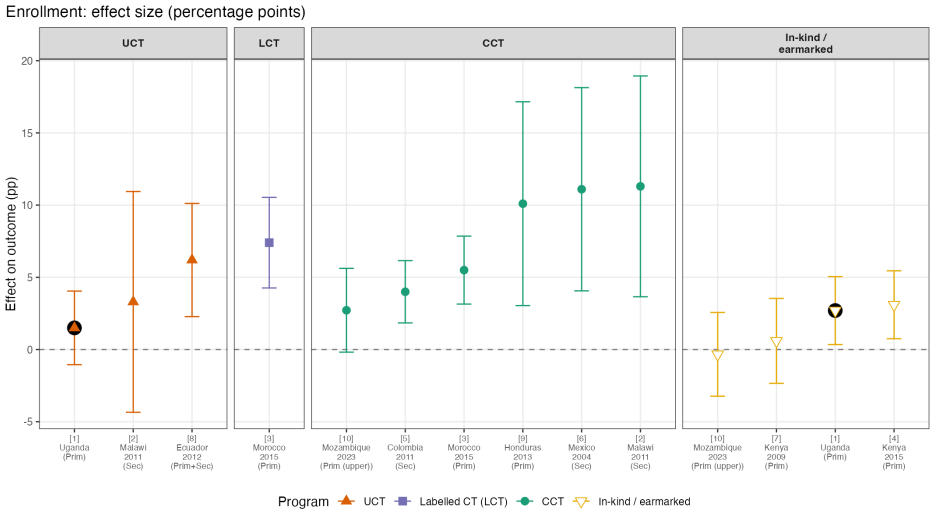
(b) Enrollment, percentage points per 2025 US dollar per month

Notes: Each marker is one arm of a randomized evaluation of a program that reduces the cost of schooling in a low- or middle-income country. The ratio is the intention-to-treat effect divided by the program's average monthly transfer per child in 2025 US dollars, a monthly version of the transfer-effectiveness measure of García and Saavedra (2017), who divide by the annual transfer. "This study" spreads the one-time UGX 35,000 over the three-month term. Enrollment for this study is one minus dropout, the share of children enrolled at the end of the term. Whiskers are 95% confidence intervals. Our outcomes are measured after one term, whereas the comparison programs follow children for one to seven years. The ratios exclude administrative costs, which differ across delivery mechanisms, and mechanically favor smaller transfers when effects are concave in transfer size, so they are benchmarks for magnitude rather than estimates of cost-effectiveness. Underlying estimates are in Table A.12, unscaled effects in Figure A.9, and ratios at purchasing-power parity in Figure A.10.

Figure A.9: Effects of Cost-of-Schooling Interventions in Percentage Points



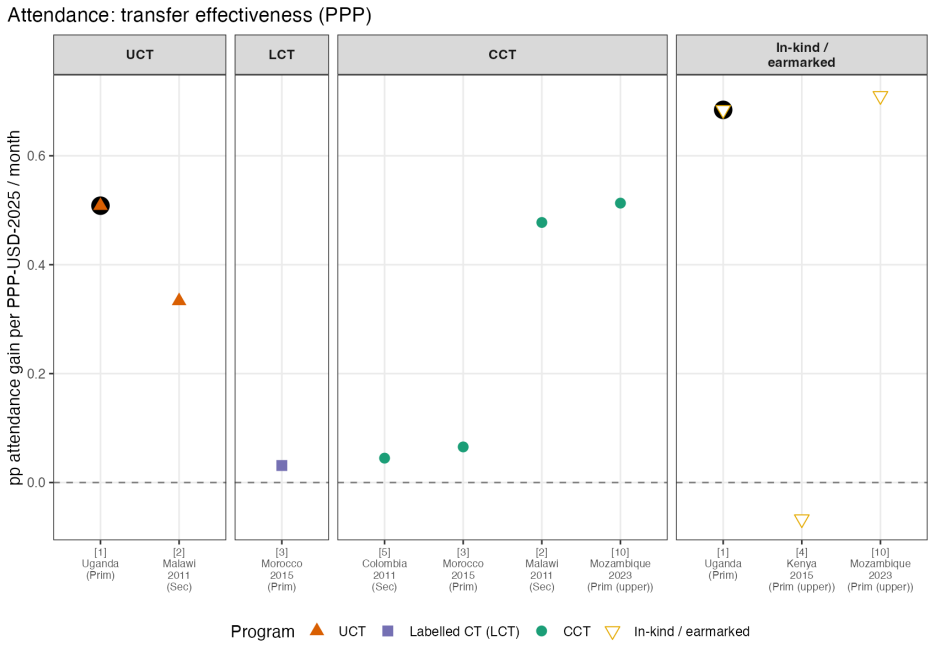
(a) Attendance, percentage-point effect



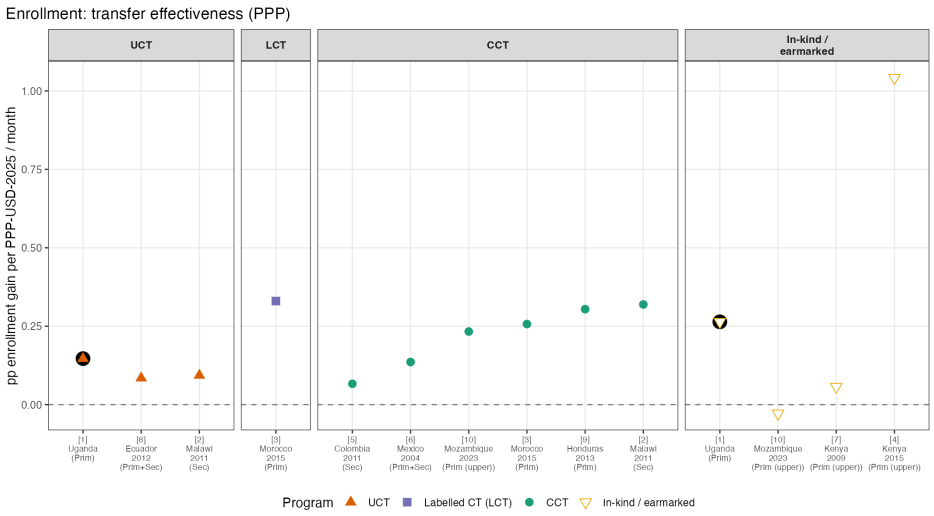
(b) Enrollment, percentage-point effect

Notes: Unscaled version of Figure A.8: intention-to-treat effects in percentage points, with 95% confidence intervals. Control means differ widely across studies (49 to 96 percent), so effects are not directly comparable across rows; Table A.12 reports each arm's control mean.

Figure A.10: Effects per Dollar Transferred at Purchasing-Power Parity



(a) Attendance, percentage points per 2025 PPP dollar per month



(b) Enrollment, percentage points per 2025 PPP dollar per month

Notes: As Figure A.8, with transfers converted at purchasing-power parity (World Bank GDP conversion factors) and inflated to 2025 with the US CPI-U. Every ratio shrinks, but the ordering of arms is essentially unchanged.

Table A.13: Effects on School Examination Scores

Panel A: Mid-Term Assessment (T1 2026)					
	Total Marks	English	Math	Science	Social Studies
T1: Cash Today	-0.087	-0.012	-0.011	-0.072	-0.057
	(0.078)	(0.080)	(0.075)	(0.079)	(0.082)
T2: Cash Fee-Day	-0.052	0.009	0.001	0.053	-0.028
	(0.079)	(0.082)	(0.076)	(0.079)	(0.083)
T3: Voucher	-0.078	-0.036	-0.002	0.021	-0.100
	(0.077)	(0.083)	(0.078)	(0.078)	(0.080)
Control Mean	0.000	0.000	0.000	0.000	0.000
Observations	869	860	860	857	843
p-value: T1 = T2	0.645	0.803	0.884	0.110	0.727
p-value: T2 = T3	0.727	0.601	0.972	0.684	0.377
p-value: T1 = T3	0.903	0.762	0.915	0.220	0.589
Panel B: End-of-Term Assessment (T1 2026)					
	Total Marks	English	Math	Science	Social Studies
T1: Cash Today	-0.058	-0.037	-0.039	-0.053	-0.033
	(0.062)	(0.068)	(0.059)	(0.059)	(0.065)
T2: Cash Fee-Day	-0.032	0.035	-0.040	-0.019	-0.013
	(0.060)	(0.075)	(0.057)	(0.057)	(0.060)
T3: Voucher	-0.020	-0.006	-0.003	-0.020	-0.011
	(0.060)	(0.064)	(0.059)	(0.057)	(0.062)
Control Mean	0.000	0.000	0.000	0.000	0.000
Observations	1,570	1,569	1,565	1,566	1,557
p-value: T1 = T2	0.657	0.371	0.978	0.513	0.751
p-value: T2 = T3	0.837	0.586	0.511	0.986	0.967
p-value: T1 = T3	0.517	0.644	0.530	0.517	0.732

Notes: Intention-to-treat estimates from equation (3) with strata fixed effects. Panel A uses mid-term and Panel B end-of-term school examinations, standardized to control mean zero and unit standard deviation. Exam samples are selected, since children with unpaid fees can be barred from sitting them (Section 5.4). Robust standard errors in parentheses. $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

Table A.14: Effects on Household Schooling Spending, Transfers Out, and Food Security

	(1) Total HH Schooling Spending (UGX)	(2) Net Transfers Out (UGX)	(3) Food Insecurity (FIES)
T1: Early Cash	35236**	10096**	-0.115*
	(16506)	(4924)	(0.062)
T2: Fee-Day Cash	15842	2600	-0.172***
	(15818)	(4933)	(0.065)
T3: Voucher	38095**	1188	-0.123*
	(16251)	(5488)	(0.066)
Control Mean	288000	-6804	0.000
Observations	1,960	1,939	1,960
T1 Effect (%)	12.2		
T2 Effect (%)	5.5		
T3 Effect (%)	13.2		
p-value: T1 = T2	0.230	0.130	0.380
p-value: T2 = T3	0.159	0.794	0.471
p-value: T1 = T3	0.863	0.101	0.911

Notes: Intention-to-treat estimates from equation (3) with strata fixed effects; outcomes from the endline parent survey. Column 1 is total household schooling spending across all children, adding UGX 35,000 for Voucher-arm children enrolled at the baseline school, since payments at source do not appear in out-of-pocket reports. Column 2 is net transfers out of the household (money given minus money received, winsorized, ANCOVA). Column 3 is the FIES food-insecurity z-score (ANCOVA). Robust standard errors in parentheses. $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

Table A.15: Effects on the Schooling of Siblings

	(1) Share Attending School	(2) Share Sent Home over Fees	(3) Share w/ School Feeding
T1: Early Cash	-0.018	-0.017	0.049*
	(0.012)	(0.025)	(0.029)
T2: Fee-Day Cash	-0.014	-0.068***	0.051*
	(0.011)	(0.026)	(0.029)
T3: Voucher	-0.018	-0.020	-0.012
	(0.011)	(0.024)	(0.029)
Control Mean	0.961	0.771	0.545
Observations	1,802	1,766	1,767
T1 Effect (%)	-1.9	-2.2	9.0
T2 Effect (%)	-1.5	-8.8	9.4
T3 Effect (%)	-1.9	-2.6	-2.2
p-value: T1 = T2	0.754	0.056	0.948
p-value: T2 = T3	0.747	0.065	0.030
p-value: T1 = T3	1.000	0.907	0.033

Notes: Intention-to-treat estimates from equation (3) with strata fixed effects, excluding the Choice group, with one observation per household. Outcomes are from the endline parent survey's roster of the other children in the household, restricted to those aged 6 to 17. Column 1 is the share of these children attending school in 2026, among the 1,802 households with at least one. Columns 2 and 3 are the shares sent home over unpaid fees since January 2026 and receiving school meals, which were asked only about children attending school. The roster was collected only at endline, so there is no baseline control. T Effect (%) divides each coefficient by the control mean. Robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.16: Heterogeneous Treatment Effects by Baseline Willingness-to-Pay Premium

	Fee Payments		School Participation			Attendance	
	(1) Total Paid (1k UGX)	(2) Share Paid	(3) Enrollment	(4) Dropout	(5) Transfers	(6) All	(7) Did Not Transfer
Panel A: Interaction with Voucher Premium (1k UGX)							
T1: Cash Today	3.259**	0.099***	0.003	-0.014	-0.019	0.035	0.019
	(1.606)	(0.032)	(0.009)	(0.016)	(0.023)	(0.021)	(0.018)
T2: Cash Fee-Day	3.726**	0.097***	0.003	-0.021	-0.031	0.043**	0.023
	(1.556)	(0.030)	(0.008)	(0.015)	(0.021)	(0.021)	(0.017)
T3: Voucher	23.310***	0.626***	0.007	-0.031**	-0.032	0.070***	0.053***
	(1.370)	(0.029)	(0.008)	(0.015)	(0.022)	(0.021)	(0.017)
T1 × Voucher Premium	0.206*	0.002	0.000	0.000	-0.001	0.001	0.001
	(0.112)	(0.002)	(0.001)	(0.001)	(0.002)	(0.002)	(0.001)
T2 × Voucher Premium	0.186*	0.003	0.000	0.001	0.000	0.002	0.002
	(0.106)	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
T3 × Voucher Premium	-0.073	-0.005**	0.000	0.001	0.000	-0.001	-0.001
	(0.096)	(0.002)	(0.000)	(0.001)	(0.002)	(0.001)	(0.001)
Equality of interaction coefficients (p-value)							
T1 = T2	0.848	0.912	0.542	0.641	0.366	0.578	0.334
T2 = T3	0.003	0.000	0.854	0.825	0.782	0.089	0.041
T1 = T3	0.002	0.001	0.418	0.774	0.540	0.276	0.319
All three = 0	0.003	0.001	0.840	0.861	0.796	0.339	0.219
Panel B: Interaction with Delayed Cash Premium (1k UGX)							
T1: Cash Today	4.474***	0.116***	-0.003	-0.009	-0.026	0.035*	0.017
	(1.489)	(0.030)	(0.008)	(0.014)	(0.020)	(0.020)	(0.016)
T2: Cash Fee-Day	4.445***	0.104***	-0.002	-0.007	-0.029	0.048**	0.030*
	(1.451)	(0.029)	(0.008)	(0.014)	(0.020)	(0.020)	(0.016)
T3: Voucher	22.863***	0.606***	0.008	-0.026**	-0.025	0.063***	0.049***
	(1.270)	(0.027)	(0.006)	(0.013)	(0.021)	(0.019)	(0.015)
T1 × Delayed Cash Premium	0.043	0.000	0.001*	-0.001	0.000	0.002	0.002
	(0.110)	(0.002)	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)
T2 × Delayed Cash Premium	0.064	0.001	0.001**	-0.002	-0.001	0.001	0.000
	(0.108)	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
T3 × Delayed Cash Premium	-0.010	-0.002	0.000	0.000	-0.002	0.001	0.000
	(0.093)	(0.002)	(0.000)	(0.001)	(0.002)	(0.001)	(0.001)
Control Mean	20.123	0.374	0.984	0.051	0.119	0.696	0.772
Observations	1,776	1,776	1,960	1,960	1,960	22,251	20,180
Equality of interaction coefficients (p-value)							
T1 = T2	0.846	0.556	0.837	0.366	0.954	0.604	0.301
T2 = T3	0.423	0.087	0.061	0.155	0.213	0.836	0.786
T1 = T3	0.588	0.301	0.150	0.704	0.218	0.744	0.179
All three = 0	0.853	0.371	0.081	0.424	0.445	0.751	0.483

Notes: This table reproduces Table 5 with each treatment indicator interacted with the parent's baseline premium, per 1,000 UGX: the voucher premium in Panel A, the delayed-cash premium in Panel B. Each regression includes the premium main effect; premia are not demeaned, so treatment coefficients apply at indifference. Both panels share the estimation sample. Lower rows report pairwise and joint tests of the interaction coefficients. Robust standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table A.17: Effects of Experience on Voucher Demand

	(1) Full Sample	(2) Same Parent	(3) Term 1 Only	(4) Same Parent, Term 1
T1: Early Cash	-2,229***	-2,574***	-1,960	-2,204
	(858)	(885)	(1,692)	(1,735)
T2: Fee-Day Cash	-2,128**	-2,528***	-2,672*	-2,348
	(856)	(879)	(1,601)	(1,617)
T3: Voucher	4,019***	3,762***	3,247**	3,659**
	(810)	(838)	(1,627)	(1,654)
Control Mean	5,823	6,223	6,225	6,402
Observations	1,960	1,855	566	546

Notes: The sample excludes the Choice group ($N = 1{,}960$). Column 1 is a linear probability model of choosing the UGX 35,000 voucher over equal Fee-Day Cash in the first round of the endline task. Column 2 is a two-limit Tobit of the endline indifference point, censored at UGX 17,500 and 52,500; the Marg. Effect rows give the implied effect on the observed indifference point. Columns 3–4 are linear probability models of the endline indifference point lying at the ceiling and floor of the grid. Each column controls for its baseline analog (ANCOVA) and strata fixed effects; ITT estimates without baseline controls are similar (Appendix Table A.18), since the control-group baseline–endline correlation of the indifference point is 0.127. T Effect (%) divides each coefficient by the control mean. Robust standard errors in parentheses. $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

Table A.18: Preference Updating: Alternative Estimators and Samples

	(1) Endline WTP OLS, ITT	(2) Endline WTP OLS, ANCOVA	(3) Δ WTP OLS, ITT	(4) Endline WTP Tobit, Same Parent	(5) Endline WTP Tobit, Term 1
T1: Early Cash	-2163**	-2229***	-2540**	-6278***	-3718
	(868)	(858)	(1093)	(2029)	(3460)
T2: Fee-Day Cash	-2234**	-2128**	-1628	-5723***	-5694*
	(869)	(856)	(1082)	(2012)	(3341)
T3: Voucher	4105***	4019***	3613***	10033***	9277**
	(817)	(810)	(1062)	(2155)	(3739)
Control Mean	40823	40823	-83	41223	41178
p-value: T1 = T2	0.935	0.906	0.393	0.776	0.562
p-value: T2 = T3	0.000	0.000	0.000	0.000	0.000
p-value: T1 = T3	0.000	0.000	0.000	0.000	0.001
Observations	1,960	1,960	1,960	1,855	584

Notes: The dependent variable is the endline indifference point (columns 1–2, 4–5) or its change from baseline (column 3). Columns 1–3 are OLS with strata fixed effects; columns 4–5 are two-limit Tobits restricted to same-respondent households ($N = 1{,}855$) and to the Term 1 2027 version of the task ($N = 584$). With a control-group baseline–endline correlation of 0.127, ANCOVA is more efficient than the change score. Robust standard errors in parentheses. $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

Table A.19: The Baseline Belief Gap and the Voucher Premium, Baseline and Endline

	(1) Baseline Voucher Premium	(2) Endline Voucher Premium	(3) Endline Voucher Premium
Perceived Earmarking Gain (per 10 pp)	477***	117	12.907
	(122)	(130)	(128)
Baseline Voucher Premium			0.218***
			(0.024)
Observations	1,960	1,960	1,960

Notes: $N = 1{,}960$. The perceived earmarking gain is the baseline belief gap (Voucher minus Fee-Day Cash clearing probability), per 10 percentage points; outcomes are the baseline and endline voucher premia, defined as in Figure 3. OLS with robust standard errors. $^{*}\, p < 0.10$, $^{**}\, p < 0.05$, $^{***}\, p < 0.01$.

B Model Derivations

This appendix collects the formal results behind Section 3: the decision rules (Lemmas B.1 and B.2), the derivation of the value functions and of the willingness-to-pay expressions in equation (2), the proofs of Propositions 1 and 2, and four extensions. Sections B.1 to B.3 maintain the assumptions of Section 3.1: $R > F > 0$, $s \geq 0$, $C = F$, $\beta, \tilde{\beta} \in (0, 1]$, and $y_L < F \leq y_H$. Decision rules are stated for the forecast parameter $\tilde{\beta}$, which the $t = 0$ self uses to predict their later selves; realized behavior obtains on replacing $\tilde{\beta}$ with the true β in each self's own trade-off (see the proof of Proposition 2).

B.1 Decision rules

Lemma B.1 (Fee-day payment)**.** *Under either cash arm, the fee-day self holding cash ℓ pays ($a = 1$) if and only if $\ell \geq F$ and $\tilde{\beta}R \geq F$. Under the Voucher, the fee is paid directly to the school regardless of ℓ and $\tilde{\beta}$.*

Proof. Paying yields $(\ell - F) + \tilde{\beta}R$ and not paying yields ℓ, so payment is preferred if and only if $\tilde{\beta}R \geq F$, and it is feasible if and only if $\ell \geq F$ given the no-borrowing constraint. Under the Voucher the program pays the school directly upon enrollment, so no fee-day choice remains. □

Lemma B.2 (Saving)**.** *Under Early Cash, optimal saving is a corner solution, $s^* \in \{0, F - y_L\}$. If $\tilde{\beta}R \geq F$, then $s^* = F - y_L$ if and only if $\tilde{\beta} \geq \bar{\beta}_S$, and $s^* = 0$ otherwise; if $\tilde{\beta}R < F$, then $s^* = 0$. Under Fee-Day Cash and the Voucher, $s^* = 0$.*

Proof. The period-1 self maximizes $U_1 = c_1 + \tilde{\beta}\,\mathbb{E}[c_2 + c_3]$. Moving one shilling from $t = 1$ to $t = 2$ costs one unit of current consumption and returns $\tilde{\beta}$ in expectation, a net loss of $1 - \tilde{\beta} \geq 0$, unless the shilling changes the fee-day payment set. Under Fee-Day Cash the low-state budget is $y_L + F \geq F$ and under the Voucher the fee is paid directly to the school, so saving never changes payment and $s^* = 0$. Under Early Cash, cash on hand on fee day is $\ell = y_2 + s$, so saving changes payment only in the low state and only if $s \geq F - y_L$; the only candidate other than $s = 0$ is therefore $s = F - y_L$, since any smaller s is a pure loss and any larger s adds one. Suppose $\tilde{\beta}R \geq F$, so the fee-day self pays whenever cash on hand covers the fee (Lemma B.1). Comparing the two candidates,

$$U_1(F - y_L) - U_1(0) = -(F - y_L) + \tilde{\beta}\big[(F - y_L) + q(R - F)\big],$$

which is nonnegative if and only if $\tilde{\beta} \geq \bar{\beta}_S \equiv (F - y_L)/[(F - y_L) + q(R - F)]$, equivalently

$\tilde{\beta} q(R-F) \geq (1-\tilde{\beta})(F-y_L)$. If instead $\tilde{\beta} R < F$, the fee-day self never pays, saving cannot change payment, and $s^* = 0$. □

Lemmas B.1 and B.2 together deliver the payment probabilities in equation (1): under the Voucher the fee is always paid; under Fee-Day Cash it is paid if and only if the payment condition holds, since the fee-day budget always covers F; and under Early Cash it is paid if and only if the payment condition holds and either the high state realizes or the period-1 self saved.

B.2 Value functions and willingness to pay

Under a cash arm $d \in \{1, 2\}$, realized total consumption equals total resources plus the net return when the fee is paid, $c_1 + c_2 + c_3 = y_1 + C + y_2 + (R - F)\, a$. Saving only reallocates consumption across dates, which the $t = 0$ self values equally, so

$$V_0^{(d)} = \beta\big[y_1 + \mathbb{E}y_2 + C + (R-F)\,\tilde{\mu}_d\big], \qquad d \in \{1, 2\}. \tag{B.1}$$

Under the Voucher, the parent's own resources are untouched, the fee is paid directly to the school, and the return accrues with forecast probability $\tilde{\mu}_V$, so $V_0^{(V)} = \beta\big[y_1 + \mathbb{E}y_2 + \tilde{\mu}_V R\big]$. Using $C = F$, this can be written as $V_0^{(V)} = \beta\big[y_1 + \mathbb{E}y_2 + C + (R-F)\tilde{\mu}_V - F(1-\tilde{\mu}_V)\big]$: relative to a cash arm, an unused Voucher forfeits the transfer, whereas unspent cash is kept.

The willingness to pay for transfer d over d' is the amount of fee-day cash x that makes the $t = 0$ self indifferent, $V_0^{(d)} = V_0^{(d')} + \beta x$, that is, $x = (V_0^{(d)} - V_0^{(d')})/\beta$. Applying this to the two elicitation tasks gives equation (2):

$$\begin{aligned} \mathrm{WTP}_{\text{voucher}} &= \frac{V_0^{(V)} - V_0^{(2)}}{\beta} = (R-F)(\tilde{\mu}_V - \tilde{\mu}_2) - F(1-\tilde{\mu}_V), \\ \mathrm{WTP}_{\text{delay}} &= \frac{V_0^{(2)} - V_0^{(1)}}{\beta} = (R-F)(\tilde{\mu}_2 - \tilde{\mu}_1), \end{aligned}$$

where the forfeiture term in the earmarking premium vanishes because $\tilde{\mu}_V \equiv 1$. After dividing by β, willingness to pay depends on preferences only through the forecast payment probabilities.

B.3 Proofs of the propositions and comparative statics

Proof of Proposition 1. Substitute the payment probabilities (1) into (2) region by region. If $\tilde{\beta} R < F$, then $\tilde{\mu}_1 = \tilde{\mu}_2 = 0$, so $\mathrm{WTP}_{\text{voucher}} = R - F > 0$ and $\mathrm{WTP}_{\text{delay}} = 0$. If $F/R \leq \tilde{\beta} < \bar{\beta}_S$,

then $\tilde{\mu}_2 = 1$ and $\tilde{\mu}_1 = 1 - q$, so $\text{WTP}_{\text{voucher}} = 0$ and $\text{WTP}_{\text{delay}} = q(R - F) > 0$. If $\tilde{\beta} \geq \bar{\beta}_S$, then $\tilde{\mu}_1 = \tilde{\mu}_2 = 1$ and both premia are zero. The supports are disjoint by inspection. The three regions partition $(0, 1]$ in this order provided $F/R < \bar{\beta}_S$, which holds if and only if

$$\frac{F}{R} < \frac{F - y_L}{(F - y_L) + q(R - F)} \iff qF(R - F) < (R - F)(F - y_L) \iff y_L < F(1 - q).$$

Finally, both supports lie strictly below one, since $\tilde{\beta}R < F$ implies $\tilde{\beta} < F/R < 1$ and the under-saving region requires $\tilde{\beta} < \bar{\beta}_S \leq 1$; positive willingness to pay for either restriction therefore implies $\tilde{\beta} < 1$. □

Proof of Proposition 2. In realized behavior each self applies the true β to its own trade-off and uses $\tilde{\beta}$ to forecast later selves. The fee-day self pays if and only if cash on hand covers the fee and $\beta R \geq F$, so $\mu_2 = \mathbf{1}\{\beta R \geq F\}$. Under Early Cash the period-1 self saves $F - y_L$ if and only if $\beta \geq \bar{\beta}_S$ and $\tilde{\beta}R \geq F$, so $\mu_1 = \mathbf{1}\{\beta R \geq F\}\big(1 - q\,\mathbf{1}\{\beta < \bar{\beta}_S \text{ or } \tilde{\beta}R < F\}\big)$. Since $\mu_V = \tilde{\mu}_V = 1$, $e_d = \mu_d - \tilde{\mu}_d$. Let $m_d(b)$ denote the forecast probability in equation (1) evaluated at b, so that $\tilde{\mu}_d = m_d(\tilde{\beta})$; both m_1 and m_2 are weakly increasing step functions. If $\beta \leq \tilde{\beta}$, then $\tilde{\beta}R < F$ implies $\beta R < F$, so $\mu_d = m_d(\beta)$ and $e_d = m_d(\beta) - m_d(\tilde{\beta}) \leq 0$. If $\tilde{\beta} < \beta$, then $\mu_2 = m_2(\beta) \geq m_2(\tilde{\beta})$, and $\mu_1 \geq m_1(\tilde{\beta})$: the forecast is zero when $\tilde{\beta}R < F$, and $\mu_1 = m_1(\beta) \geq m_1(\tilde{\beta})$ when $\tilde{\beta}R \geq F$. Hence $e_d \geq 0$. It follows that $e_d = 0$ when $\tilde{\beta} = \beta$, that $e_d < 0$ only if $\beta < \tilde{\beta}$, and that $e_d > 0$ only if $\tilde{\beta} < \beta$. For timing, $m_2(b) - m_1(b) = q\,\mathbf{1}\{F/R \leq b < \bar{\beta}_S\}$ is not monotone in b. With $\beta < \tilde{\beta}$, the timing error equals $-q$ when $F/R \leq \beta < \bar{\beta}_S \leq \tilde{\beta}$ and q when $\beta < F/R \leq \tilde{\beta} < \bar{\beta}_S$; with $\tilde{\beta} < \beta$, it equals $-q$ when $\tilde{\beta} < F/R \leq \beta < \bar{\beta}_S$ and q when $F/R \leq \tilde{\beta} < \bar{\beta}_S \leq \beta$. □

Demand and effects. The realized effect of earmarking on payment is $\mu_V - \mu_2 = \mathbf{1}\{\beta R < F\}$, and by Proposition 1 a parent pays for earmarking if and only if $\tilde{\beta}R < F$. If $\beta \leq \tilde{\beta}$ for every parent, then $\tilde{\beta}R < F$ implies $\beta R < F$, so the effect equals one for every parent who pays and is at most one for the others. The effect among parents who pay can be smaller than among those who do not only if $\tilde{\beta}R < F \leq \beta R$ for some parents who pay. For delay, the realized effect is $\mu_2 - \mu_1 = q\,\mathbf{1}\{\beta R \geq F\}\,\mathbf{1}\{\beta < \bar{\beta}_S \text{ or } \tilde{\beta}R < F\}$, which is weakly positive, so realized payment probabilities are ordered $\mu_V \geq \mu_2 \geq \mu_1$ for every parent.

The comparative statics used in the text follow directly. On its support, $\text{WTP}_{\text{delay}} = q(R - F)$ is strictly increasing in q. The saving cutoff $\bar{\beta}_S$ is increasing in $F - y_L$ and decreasing in q, so a lower low-state income widens the region in which delay is demanded, while a more likely shortfall raises the premium conditional on demanding but narrows the region itself. The earmarking premium involves neither q nor y_L on its support. It rises with the return

R on its support, while the support itself, $\tilde{\beta} < F/R$, shrinks as R rises, so the relationship between the return and average demand for earmarking is ambiguous.

Appendix Figure B.1 displays the partition of the forecast present-focus parameter, and Appendix Table B.1 collects saving, forecast payment, and willingness to pay in each region.

B.4 Long-run discounting

The main text sets $\delta = 1$, following the convention of O'Donoghue and Rabin (1999, 2001) for short-horizon problems. This subsection restores $\delta < 1$ and shows that the characterization survives, with one substitution: the maintained assumption $R > F$ becomes $\delta R > F$, so that paying the fee is worthwhile from the ex ante perspective.

Preferences. Each self now discounts exponentially as well: $U_2 = c_2 + \beta\delta c_3$, $U_1 = c_1 + \beta\delta\,\mathbb{E}c_2 + \beta\delta^2\,\mathbb{E}c_3$, and $V_0^{(d)} = \beta\big[\delta\,\mathbb{E}c_1 + \delta^2\,\mathbb{E}c_2 + \delta^3\,\mathbb{E}c_3\big]$. Beliefs concern β only; δ is known and common to forecast and behavior.

Decision rules. The fee-day self pays if and only if $\ell \geq F$ and $\tilde{\beta}\delta R \geq F$, so the payment threshold rises from F/R to $F/(\delta R)$. The saving problem remains a corner (the argument of Lemma B.2 is unchanged), and the cutoff generalizes to

$$\bar{\beta}_S(\delta) \equiv \frac{F - y_L}{\delta\big[(F - y_L) + q(\delta R - F)\big]},$$

which collapses to $\bar{\beta}_S$ at $\delta = 1$. Because $\delta < 1$ both shrinks the forecast benefit of saving and adds an exponential cost to postponing consumption, $\bar{\beta}_S(\delta) > \bar{\beta}_S$, and $\bar{\beta}_S(\delta)$ can exceed one, in which case no parent saves.

Willingness to pay. A fee-day shilling is worth $\beta\delta^2$ to the $t = 0$ self, so premia are $x = (V_0^{(d)} - V_0^{(d')})/(\beta\delta^2)$. Differencing the value functions region by region gives, for $\tilde{\beta} < F/(\delta R)$,

$$\mathrm{WTP}_{\text{voucher}} = \delta R - F > 0, \qquad \mathrm{WTP}_{\text{delay}} = -\frac{F(1-\delta)}{\delta} < 0;$$

for $F/(\delta R) \leq \tilde{\beta} < \bar{\beta}_S(\delta)$,

$$\mathrm{WTP}_{\text{voucher}} = 0, \qquad \mathrm{WTP}_{\text{delay}} = q\,(\delta R - F) - \frac{F(1-\delta)}{\delta};$$

and for $\tilde{\beta} \geq \bar{\beta}_S(\delta)$,

$$\mathrm{WTP}_{\text{voucher}} = 0, \qquad \mathrm{WTP}_{\text{delay}} = -\frac{y_L(1-\delta)}{\delta} \leq 0.$$

The structure of Proposition 1 is intact: earmarking is valued exactly in the under-payment region, and delay is most valuable in the under-saving region. Each delay premium now nets the commitment benefit against the pure cost of waiting, $F(1-\delta)/\delta$ on the amount held back (or $y_L(1-\delta)/\delta$ on the saver's own set-aside). In the middle region the delay premium is positive if and only if $q(\delta R - F) > F(1-\delta)/\delta$, that is, if the risk of a shortfall outweighs the cost of waiting, which holds automatically at $\delta = 1$. The middle region is non-empty, $F/(\delta R) < \bar{\beta}_S(\delta)$, if and only if

$$y_L \;<\; F - \frac{qF(\delta R - F)}{R - F},$$

which nests $y_L < F(1-q)$ at $\delta = 1$ and is weaker for $\delta < 1$.

Remark 1. Positive willingness to pay for either restriction still requires $\tilde{\beta} < 1$. The earmarking premium is positive only for $\tilde{\beta} < F/(\delta R) < 1$. For delay, if $\bar{\beta}_S(\delta) \leq 1$ a parent with $\tilde{\beta} = 1$ saves on their own and $\text{WTP}_{\text{delay}} = -y_L(1-\delta)/\delta \leq 0$; if $\bar{\beta}_S(\delta) > 1$ they do not save, but $\bar{\beta}_S(\delta) > 1$ is equivalent to $q(\delta R - F) < (1-\delta)(F - y_L)/\delta \leq F(1-\delta)/\delta$, so their delay premium is again negative. Exponential discounting alone therefore cannot generate demand for either restriction.

The extension also rationalizes a feature of the data that the $\delta = 1$ model cannot: with $\delta = 1$ every premium is zero or positive, whereas with $\delta < 1$ parents who expect to save and pay on their own strictly prefer earlier cash. The negative delay premia in Section 4.1, revealed by 38 percent of parents, are consistent with this region, without changing where demand for commitment comes from. Negative earmarking premia do not arise here; Section B.7 shows how they arise when the Voucher covers only part of the fee.

B.5 Costly borrowing

The main text rules out borrowing. This subsection lets a parent cover a fee-day shortfall by borrowing at a gross interest rate $\rho > 1$. The earmarking premium is unchanged. The delay premium remains positive only for $\tilde{\beta} < 1$, equals the expected interest for parents who are willing to borrow, and converges to zero as $\rho \to 1$.

Borrowing. Borrowing is possible only on fee day. A parent who holds cash $\ell < F$ can borrow the shortfall $F - \ell$ and pay the fee. The loan is repaid with interest before the schooling return accrues, so its cost falls on the fee-day self: paying with a shortfall yields $-\rho(F - \ell) + \tilde{\beta}R$, against ℓ from not paying.[33] Let $\kappa \equiv (\rho - 1)(F - y_L)$ denote the interest

[33] With linear utility only differences in consumption matter, so consumption may be negative. No self

on the full low-state shortfall.

Decision rules. A fee-day self with $\ell \geq F$ pays if and only if $\tilde{\beta}R \geq F$, as before. A fee-day self with $\ell < F$ borrows and pays if and only if $\tilde{\beta}R \geq \ell + \rho(F - \ell)$, that is, if and only if

$$\ell \;\geq\; \ell^*(\tilde{\beta}) \equiv \frac{\rho F - \tilde{\beta}R}{\rho - 1}.$$

The threshold satisfies $\ell^* \leq F$ if and only if $\tilde{\beta}R \geq F$, and $\ell^* \leq y_L$ if and only if $\tilde{\beta}R \geq F + \kappa$. Under Fee-Day Cash and the Voucher cash on hand covers the fee, so borrowing is never used and payment is as in equation (1). Under Early Cash a parent with $\tilde{\beta}R \geq F$ chooses among three levels of saving: nothing; $\max\{\ell^* - y_L, 0\}$, which leaves a low-state shortfall that the fee-day self is just willing to borrow; and $F - y_L$, which avoids borrowing. Each shilling saved beyond $\ell^* - y_L$ replaces a shilling borrowed in the low state and is worth $\tilde{\beta}[1 + q(\rho - 1)]$ to the period-1 self, so saving the full shortfall is preferred if and only if

$$\tilde{\beta} \;\geq\; \bar{\beta}_B \equiv \frac{1}{1 + q(\rho - 1)}.$$

When $\ell^* > y_L$, saving $\ell^* - y_L$ rather than nothing changes period-1 utility by $(1 - \tilde{\beta})\big[\tilde{\beta}qR - (\ell^* - y_L)\big]$, which is nonnegative if and only if

$$\tilde{\beta} \;\geq\; \bar{\beta}_P \equiv \bar{\beta}_B \, \frac{F + \kappa}{R}.$$

Willingness to pay. Premia are differences in $t = 0$ values, as in Section B.2. The earmarking premium compares the Voucher with Fee-Day Cash, neither of which involves borrowing, so $\text{WTP}_{\text{voucher}} = (R - F)\,\mathbf{1}\{\tilde{\beta}R < F\}$, as in Proposition 1. If $\kappa \geq R - F$, the interest on the shortfall exceeds the net return, no parent borrows, and the delay premium is also as in Proposition 1. If $\kappa < R - F$,

$$\text{WTP}_{\text{delay}} = \begin{cases} q\,(R - F) & \text{if } F/R \leq \tilde{\beta} < \bar{\beta}_P, \\ q\,\min\{\tilde{\beta}R - F,\ \kappa\} & \text{if } \max\{F/R, \bar{\beta}_P\} \leq \tilde{\beta} < \bar{\beta}_B, \\ 0 & \text{otherwise.} \end{cases}$$

In the first region the parent neither saves nor borrows, and the fee goes unpaid in the low

borrows to consume, because $\rho > 1$. If repayment instead fell due when the return accrues, the fee-day self would discount the cost of the loan along with the return, and borrowing would itself relax the payment condition for a parent with little cash on hand.

state, as in the main text. In the second the fee is paid in both states and the premium is the expected interest: $q\kappa$ if the parent borrows the full shortfall, and $q(\tilde{\beta}R - F)$ if they save part of it and borrow the rest. The first region is non-empty if and only if $y_L < F(1-q)$, the condition in Proposition 1.

Remark 2. Both premia are zero at $\tilde{\beta} = 1$, because $\bar{\beta}_B < 1$ whenever $\rho > 1$. As $\rho \to 1$, $\kappa \to 0$, $\bar{\beta}_P \to F/R$, and $\bar{\beta}_B \to 1$, so the delay premium converges to zero for every $\tilde{\beta}$: when credit is free, the timing of the transfer does not matter. A positive delay premium therefore requires both $\tilde{\beta} < 1$ and costly credit, whereas a positive earmarking premium requires only $\tilde{\beta} < 1$.

B.6 Concave utility

The main text assumes linear utility. This subsection shows that the central implication of Proposition 1, that a positive premium for either restriction requires $\tilde{\beta} < 1$, holds for any increasing utility function, and describes which results are specific to the linear case.

Preferences. Each self has per-period utility $u(c)$, with u strictly increasing, and discounts as in Section B.4: $U_2 = u(c_2) + \beta\delta\, u(c_3)$, $U_1 = u(c_1) + \beta\delta\, \mathbb{E}u(c_2) + \beta\delta^2\, \mathbb{E}u(c_3)$, and $V_0^{(d)} = \beta\big[\delta\, \mathbb{E}u(c_1) + \delta^2\, \mathbb{E}u(c_2) + \delta^3\, \mathbb{E}u(c_3)\big]$, where $c_3 = y_3 + aR$ and y_3 is income at $t = 3$. Fee-day income y_2 has an arbitrary distribution. Without a linear numeraire we define the premia as in the experiment. Let $V_0^{(2)}(X)$ denote the $t = 0$ value of Fee-Day Cash of amount X, and let X_E and X_D solve $V_0^{(2)}(X_E) = V_0^{(V)}$ and $V_0^{(2)}(X_D) = V_0^{(1)}$, where the Voucher and Early Cash are of amount C. Then $\mathrm{WTP}_{\mathrm{voucher}} \equiv X_E - C$ and $\mathrm{WTP}_{\mathrm{delay}} \equiv C - X_D$.

Proposition 3 (No demand without perceived present focus)**.** *Suppose* $\tilde{\beta} = 1$. *For any strictly increasing* u, *any* $\delta \in (0, 1]$, *and any distribution of* y_2, $\mathrm{WTP}_{\mathrm{voucher}} \leq 0$ *and* $\mathrm{WTP}_{\mathrm{delay}} \leq 0$.

Proof. When $\tilde{\beta} = 1$ the $t = 0$ self expects each later self t to maximize $\sum_{\tau \geq t} \delta^{\tau - t} u(c_\tau)$, which ranks continuation plans as V_0 does. The outcome the parent forecasts under a transfer therefore maximizes V_0 over the plans that are feasible under that transfer. Every plan that is feasible under Fee-Day Cash of amount X is feasible under Early Cash of the same amount, by saving the transfer until fee day. Every plan that is feasible under the Voucher is feasible under Fee-Day Cash of the same amount, by paying it to the school, because the two carry the same transaction costs (Section 2.4). Hence $V_0^{(1)} \geq V_0^{(2)}(C) \geq V_0^{(V)}$. Because u is strictly increasing, $V_0^{(2)}(X)$ is strictly increasing in X when $\tilde{\beta} = 1$, so $X_E \leq C \leq X_D$. □

The proposition does not use concavity, the absence of borrowing, or the two-point distri-

bution of income. It uses only that cash can be saved at no cost and that paying the school in cash costs the same as the Voucher. A positive premium therefore implies either $\tilde{\beta} < 1$ or a failure of one of these two conditions, for example cash that cannot be held safely or that others can claim. Section 4.4 examines these alternatives.

What is specific to linear utility. When u is strictly concave, a fee-day self holding cash $\ell \geq F$ pays if and only if

$$u(\ell) - u(\ell - F) \;\leq\; \tilde{\beta}\,\delta\,\big[u(y_3 + R) - u(y_3)\big].$$

The left-hand side falls with ℓ, so the payment condition is easier to meet when cash on hand is higher, and a parent may pay in the high state and not in the low state. Saving is interior. The closed forms of Proposition 1, and the result that no parent pays for both restrictions, then no longer hold exactly. A parent who expects to pay only in the high state under Fee-Day Cash values the Voucher, under which the fee is also paid in the low state, and can value delay at the same time. The delay premium also no longer depends on fee payment alone: delay limits consumption at $t = 1$, which raises the premium when $\tilde{\beta} < 1$, and it prevents the parent from smoothing consumption when y_1 is low, which lowers it. The earmarking premium can be negative, because in a state with little cash on hand even the $t = 0$ self may prefer cash to a paid fee. This is the cost of lost flexibility that limits demand for commitment under uncertainty (Amador et al., 2006; Laibson, 2015; Carrera et al., 2022). For example, with $u = \log$, $\delta = 1$, $F = C = 1$, $R = 10$, $q = 0.4$, $y_H = 2$, $y_L = 0.5$, $y_1 = 1.5$, and $y_3 = 1$, a parent with $\tilde{\beta} = 0.3$ has an earmarking premium of 0.12 and a delay premium of 0.48, and a parent with $\tilde{\beta} = 1$ has neither.

B.7 A Voucher that covers part of the fee

The main text assumes that the transfer covers the fee. In the experiment it covered 85 percent of the term fee on average, and the full fee for 37 percent of children. This subsection sets $C < F$. The premia in equation (2) carry over with a forfeiture term, the earmarking premium is positive on a narrower range of $\tilde{\beta}$, and the delay premium is unchanged.

Setup. All transfers are of amount $C < F$. Under the Voucher, C is paid to the school and the parent owes the balance $F - C$. The return R accrues only if the full fee is paid; if the balance goes unpaid, the amount paid by the Voucher stays with the school and yields no return. We assume $y_L \geq F - C$, so that a transfer received on fee day covers the low-state shortfall.

Decision rules. Under the Voucher the fee-day self pays the balance if and only if $\tilde{\beta}R \geq F - C$,

since the parent's own cash always covers it. Under Fee-Day Cash cash on hand is at least $y_L + C \geq F$, and the fee-day self pays if and only if $\tilde{\beta}R \geq F$. Under Early Cash the period-1 self must save $F - y_L$ for the fee to be paid in the low state, and Lemma B.2 applies unchanged. Forecast payment is therefore

$$\tilde{\mu}_V = \mathbf{1}\{\tilde{\beta}R \geq F - C\},$$

with $\tilde{\mu}_1$ and $\tilde{\mu}_2$ as in equation (1).

Willingness to pay. Under a cash transfer, $V_0^{(d)} = \beta\big[y_1 + \mathbb{E}y_2 + C + (R - F)\tilde{\mu}_d\big]$, as in equation (B.1). Under the Voucher the parent pays $F - C$ and receives R when the fee is paid in full, and receives nothing otherwise, so $V_0^{(V)} = \beta\big[y_1 + \mathbb{E}y_2 + (C + R - F)\tilde{\mu}_V\big]$. Differencing,

$$\text{WTP}_{\text{voucher}} = (R - F)(\tilde{\mu}_V - \tilde{\mu}_2) - C\,(1 - \tilde{\mu}_V) = \begin{cases} -C & \text{if } \tilde{\beta} < (F - C)/R, \\ R - F & \text{if } (F - C)/R \leq \tilde{\beta} < F/R, \\ 0 & \text{if } \tilde{\beta} \geq F/R, \end{cases}$$

and $\text{WTP}_{\text{delay}} = (R - F)(\tilde{\mu}_2 - \tilde{\mu}_1)$ is as in Proposition 1.

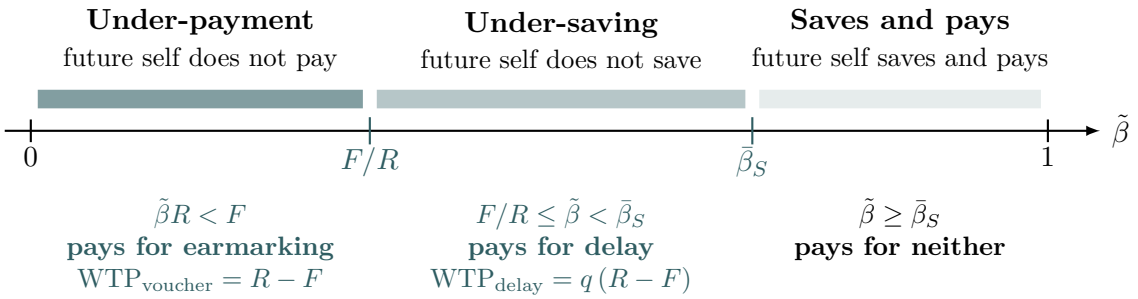
The earmarking premium is still positive where the fee-day self would not pay the fee out of cash, provided that self would pay the balance. A partial Voucher lowers the out-of-pocket cost of paying the fee from F to $F - C$, so it changes behavior for parents with $(F - C)/R \leq \tilde{\beta} < F/R$. Parents below this range expect to leave the balance unpaid and the Voucher's payment to be forfeited, and they value the Voucher at C less than cash. The negative earmarking premia in Section 4.1, revealed by 31 percent of parents, are consistent with this region. As $C \to F$ the region disappears and Proposition 1 obtains.

Realized payment under the Voucher is $\mu_V = \mathbf{1}\{\beta R \geq F - C\}$, which is below one for a sufficiently present-focused parent. Proposition 2 continues to hold for parents with β and $\tilde{\beta}$ both at least $(F - C)/R$, for whom $\mu_V = \tilde{\mu}_V = 1$.

Remark 3. (i) If $y_L < F - C$, a transfer received on fee day leaves a low-state shortfall of $g \equiv F - C - y_L$, which the parent must save out of y_1. The period-1 self saves it if and only if $\tilde{\beta} \geq g/[g + q(R - F)]$ under Fee-Day Cash and $\tilde{\beta} \geq g/[g + q(R - F + C)]$ under the Voucher, where the larger gain under the Voucher reflects the payment that saving keeps from being forfeited; the cutoff under Early Cash remains $\bar{\beta}_S$. The premia are given by the same expressions in $\tilde{\mu}_d$. (ii) If the return is instead proportional to the amount paid, so that

each shilling paid toward the fee yields R/F, the Voucher secures a return of RC/F and the earmarking premium is $(C/F)(R-F)\,\mathbf{1}\{\tilde{\beta}R < F\}$, with no negative region.

Figure B.1: Willingness to pay by perceived present focus $\tilde{\beta}$



Notes: The figure shows the three regions of Proposition 1, which assumes linear utility, drawn for $y_L < F(1-q)$. A lower $\tilde{\beta}$ means that the parent believes they are more present-focused. With concave utility the same parent can pay for both restrictions (Appendix B.6).

Table B.1: Saving, forecast payment, and willingness to pay by region

	Under-payment $\tilde{\beta}R < F$	Under-saving $F/R \le \tilde{\beta} < \bar{\beta}_S$	Saves and pays $\tilde{\beta} \ge \bar{\beta}_S$
Saving under Early Cash	0	0	$F - y_L$
$\tilde{\mu}_1, \tilde{\mu}_2, \tilde{\mu}_V$	$0, 0, 1$	$1{-}q,\ 1,\ 1$	$1, 1, 1$
$\text{WTP}_{\text{voucher}}$	$R - F > 0$	0	0
$\text{WTP}_{\text{delay}}$	0	$q(R-F) > 0$	0

C Preference and Belief Elicitation

Figure C.1 shows the sequence of offers in each task.

The elicited premia are censored by construction at the bounds of the elicitation grid, so we estimate them with two-limit Tobit models. The latent premium is

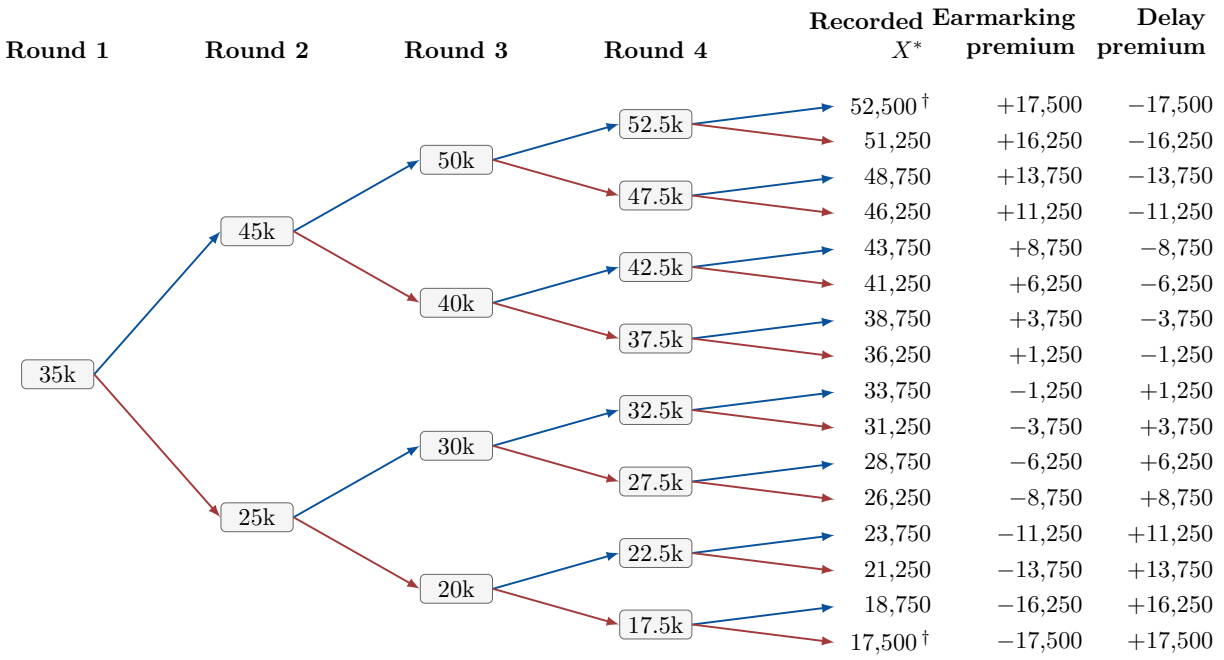
$$\text{WTP}_i^* = Z_i'\pi + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \tag{C.1}$$

observed as

$$\text{WTP}_i = \begin{cases} -17{,}500 & \text{if } \text{WTP}_i^* \le -17{,}500, \\ \text{WTP}_i^* & \text{if } -17{,}500 < \text{WTP}_i^* < 17{,}500, \\ 17{,}500 & \text{if } \text{WTP}_i^* \ge 17{,}500, \end{cases}$$

separately for the voucher and delay premia. With only a constant in Z_i, the Tobit recovers

Figure C.1: The adaptive elicitation procedure



→ chooses the fixed option: next Fee-Day Cash offer rises
→ chooses Fee-Day Cash: next Fee-Day Cash offer falls
Fixed option: the 35,000 Voucher (earmarking task) or 35,000 in Early Cash (timing task).
†Same choice in all four rounds; recorded at the grid bound.

Notes: Both tasks use the same procedure. Each box is the Fee-Day Cash amount X offered in that round against the fixed option, which is the UGX 35,000 Voucher in the earmarking task and UGX 35,000 in Early Cash in the timing task. The first offer is 35,000; the offer then rises by 10,000, 5,000, and 2,500 after each choice of the fixed option and falls by the same steps after each choice of Fee-Day Cash. The four choices bracket the switch point within a 2,500 interval, and the recorded indifference point X^* is its midpoint. The earmarking premium is $X^* - 35{,}000$ and the delay premium is $35{,}000 - X^*$.

the latent means reported in Table 2; the heterogeneity estimates in Table 3 add baseline covariates to Z_i.

(to be complete)

D Implementation Details

This appendix describes the field protocols behind the three instruments and the administrative data collection.

Cash delivery. Both cash arms were delivered by mobile money to the phone number and provider the caregiver confirmed at the close of the baseline interview. Early Cash transfers were initiated within 24 hours of the interview; Fee-Day Cash transfers were preceded by a

verification call in the days before the start of term. When a transfer failed because a line was inactive or an account was blocked, field staff visited the household to update the number and, where mobile money was not feasible, disbursed in person. Each cash transfer included a UGX 1,000 top-up that offsets the mobile-money withdrawal fee, so the full UGX 35,000 was available in cash.

Voucher redemption. A voucher was disbursed only after the research team verified, against the school's class registers, that the child had enrolled at the school attended in 2025. Payments were made by mobile money to the school, accompanied by a written list of beneficiary pupils and amounts. Schools issued receipts and recorded the payments in their fee ledgers, which the team collected and reconciled against the beneficiary lists. A voucher for a child who did not re-enroll was forfeited rather than converted to cash.

Elicitation scripts. Before the preference tasks, enumerators read a script stating that cash could be used for any purpose; that the voucher could be used only at the school the child attended in 2025 and required the caregiver to return to the school on a separate day to enroll the child; and that each cash transfer included the UGX 1,000 top-up. Caregivers were told that answering truthfully was in their best interest, and payouts for the Choice group followed the Becker–DeGroot–Marschak procedure described in Section 2.4.

Replacements. Households that could not be located, whose child had moved away, or who declined to participate were replaced by the next pre-randomized name in the same school–grade–gender cell, following the pre-analysis plan.

Fee schedules and in-kind payments. The team collected each school's fee schedule by grade at baseline. Where fees were paid in kind, in maize or beans, school records state quantities, which we value at the unit prices posted by the school, so cash and in-kind payments are expressed in common units.

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